

RISP: A Mathematical Deep Dive

Retention and Invariance in Decision-Focused Strategy Pools for Non-Stationary Markets

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Abstract

This document is the exhaustive, ground-up mathematical companion to the RISP research paper (*When the Regime Returns*) and its explanatory report. It develops, from first principles, every tool the framework’s theory uses—decision-focused learning and the SPO+ surrogate, concentration of measure, distances between distributions and Pinsker’s inequality, exponentiated-gradient dynamics, and invariant risk across environments—then assembles them into the four results that organize the contribution: the episode-transfer lemma (the invariance gap), the eviction-rate lemma (the forgetting deficit), the post-reactivation regret decomposition with its predicted-and-confirmed super-additive interaction, and the break-even-dormancy pricing of retention against carrying costs. The framework’s central claim is that financial non-stationarity is two-layered—regimes *recur* (so dormant expertise is worth retaining, and any reward-driven capacity allocator provably fails to retain it) and recurrences *differ* (so retained expertise must be trained for across-episode invariance, or it faithfully preserves the last episode’s accidents)—and that the two layers are governed by independent design choices whose product, but not either factor alone, recovers a hand-designed oracle assignment without hand design. Every theoretical statement is given with its full proof or an explicitly labeled sketch; every assumption carries an audit of its idealizations; every experimental number in the evidence part traces to a released result file; and the document reports, at full prominence, the three places where our pre-registered expectations were refuted by our own experiments (the real-data structure gate, the competition-batching story, and the high-signal regime where the entire mechanism inverts). **Prerequisites:** probability at the level of a first graduate course, linear algebra, and convexity basics; everything else is developed here.

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Part I

Mathematical Foundations

Before describing the RISP mechanism itself, we assemble the mathematical toolkit its theory rests on. Four ingredients recur throughout. First, *decision-focused learning*: RISP never scores a forecast by how close it is to the truth, but by the portfolio decision it induces, so we need the exact geometry of the decision layer, the induced regret, and a trainable surrogate (SPO+). Second, *concentration of measure*: every retention and transfer claim ultimately reduces to “a quantity estimated from 5–7 episodes is close to its mean,” and we must be honest about what such small samples can and cannot certify. Third, *distances between distributions*: the regime-conditional guarantees we import are stated in divergence (KL/total-variation) terms, and a small but load-bearing “Pinsker bridge” converts them into statements about bounded decision regret. Fourth, *online learning on the simplex*: the gate that routes episodes to niches is an exponentiated-gradient update, and we must understand both what its classical regret theory gives us and—just as importantly—what it does *not* give us once losses become endogenous. The final section formalizes why pooled empirical risk minimization fails across non-stationary episodes and why an across-episode variance penalty detects exactly the failure mode, positioning the RISP objective inside the invariant-risk literature.

1 Decision-Focused Learning from First Principles

1.1 Prediction Versus Decision: the Predict-then-Optimize Pipeline

Classical supervised learning evaluates a forecast $\hat{y}$ of an unknown quantity y by a *prediction loss* such as the squared error $\|\hat{y} - y\|_2^2$. But in RISP—as in a large class of operational problems—the forecast is not the product. The forecast is an *intermediate input* to an optimization problem whose solution is the actual decision, and the decision is what earns or loses money. This is the **predict-then-optimize** pipeline [1, 2, 3]:

$$x \xrightarrow{\text{model}} \hat{y} = f_\theta(x) \xrightarrow{\text{optimizer}} z^*(\hat{y}) \in \arg \max_{z \in Z} F(z, \hat{y}) \xrightarrow{\text{nature reveals } y} \text{realized value } F(z^*(\hat{y}), y). \quad (1)$$

Here x are observable features (market state), $y \in \mathbb{R}^n$ is the unknown parameter vector (next-period asset returns), $Z \subseteq \mathbb{R}^n$ is a known feasible set of decisions (portfolios), and $F : Z \times \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective.

Definition 1.1 (Linear-objective decision layer). A decision layer is **linear-objective** if $F(z, y) = y^\top z$, i.e. the unknown parameter enters the objective linearly, while all problem structure (combinatorics, budgets, bounds) is carried by the feasible set Z , which does not depend on y . We write

$$z^*(y) \in \arg \max_{z \in Z} y^\top z, \quad V(y) := \max_{z \in Z} y^\top z, \quad (2)$$

and call the map $y \mapsto z^*(y)$ the **decision oracle** and V the **support function** of Z .

Remark 1.1. Linearity of the objective in y is a genuine restriction, but it covers shortest paths, assignment, knapsack relaxations, and—our case—linear portfolio construction. It buys us two things: the optimal value $V(y)$ is a maximum of linear functions of y and hence *convex* in y , and the SPO+ surrogate of Section 1.3 exists in closed form.

RISP’s decision layer is a cardinality-constrained, long-only allocation.

Definition 1.2 (Cardinality-constrained long-only portfolio). Fix the universe size $n \in \mathcal{N}$, a per-asset cap $w_{\max} > 0$, a gross budget $W > 0$, and a cardinality $k \in \{1, \dots, n\}$. The feasible set is

$$Z = \left\{ z \in [0, w_{\max}]^n : \sum_{i=1}^n z_i \leq W, \ \|z\|_0 \leq k \right\}, \quad (3)$$

where $\|z\|_0 = |\{i : z_i \neq 0\}|$ counts nonzero positions. The objective is the portfolio return $F(z, y) = y^\top z$ for a return vector $y \in \mathbb{R}^n$.

Note that Z is *nonconvex* (the ℓ_0 constraint carves the box into $\binom{n}{k}$ -many faces), so generic convex-optimization machinery does not apply. Fortunately the problem admits an exact, closed-form, sort-based solution.

Theorem 1.1 (Exact decision oracle). Assume $W \geq k w_{\max}$ (the budget never binds before the cap and the cardinality do). For $y \in \mathbb{R}^n$ let

$$P(y) = \{i : y_i > 0\}, \quad S^*(y) = \text{the } \min\{k, |P(y)|\} \text{ indices of } P(y) \text{ with largest } y_i,$$

with ties broken toward the smaller index. Then

$$z^*(y)_i = w_{\max} \mathbf{1}\{i \in S^*(y)\} \quad (4)$$

is an optimal solution of $\max_{z \in Z} y^\top z$, with optimal value $V(y) = w_{\max} \sum_{i \in S^*(y)} y_i \geq 0$.

Proof. Feasibility. The vector in (4) has entries in $\{0, w_{\max}\} \subseteq [0, w_{\max}]$, support size $|S^*(y)| \leq k$, and gross exposure $|S^*(y)| w_{\max} \leq k w_{\max} \leq W$. So $z^*(y) \in Z$.

Optimality. Let $z \in Z$ be arbitrary with support $T = \{i : z_i > 0\}$, so $|T| \leq k$. Using $0 \leq z_i \leq w_{\max}$ and writing $(u)_+ = \max\{u, 0\}$,

$$y^\top z = \sum_{i \in T} y_i z_i \leq \sum_{i \in T} (y_i)_+ z_i \leq w_{\max} \sum_{i \in T} (y_i)_+.$$

The first inequality holds termwise: if $y_i \leq 0$ then $y_i z_i \leq 0 = (y_i)_+ z_i$; if $y_i > 0$ the two sides agree. Now $\sum_{i \in T} (y_i)_+$ is a sum of at most k of the nonnegative numbers $\{(y_i)_+\}_{i=1}^n$, hence is at most the sum of the k largest of them, which equals $\sum_{i \in S^*(y)} y_i$ (indices outside $P(y)$ contribute $(y_i)_+ = 0$ and are never needed). Therefore $y^\top z \leq w_{\max} \sum_{i \in S^*(y)} y_i = y^\top z^*(y)$. Finally $V(y) \geq 0$ because $z = 0 \in Z$. $\square$

Key Point

The oracle is a **sort**: rank the forecast coordinates, keep the top k strictly positive ones, allocate the cap $w_{\max}$ to each. It costs $O(n \log n)$, is exact (no relaxation gap despite the nonconvex ℓ_0 constraint), and will be called *twice per gradient step* by the SPO+ subgradient of Section 1.3. The standing assumption $W \geq k w_{\max}$ is maintained throughout the document; when it fails the oracle becomes a fractional-knapsack rule and only the boundary case changes.

Throughout, we fix the tie-breaking rule above so that $z^*(\cdot)$ is a well-defined *function* (not a set-valued map); all statements below are for this selection.

1.2 Decision Regret and Its Geometry

Definition 1.3 (Decision regret). For a forecast $\hat{y} \in \mathbb{R}^n$ and realized returns $y \in \mathbb{R}^n$, the **decision regret** (also: SPO loss [1]) is

$$\rho(\hat{y}; y) = F(z^*(y), y) - F(z^*(\hat{y}), y) = y^\top z^*(y) - y^\top z^*(\hat{y}). \quad (5)$$

It is the return given up by acting on the forecast instead of acting on the truth.

Proposition 1.1 (Range of the regret). *Suppose $\|y\|_\infty \leq y_{\max}$. Then for every $\hat{y} \in \mathbb{R}^n$,*

$$0 \leq \rho(\hat{y}; y) \leq B, \quad B := 2k w_{\max} y_{\max}. \quad (6)$$

Proof. Lower bound. $z^*(\hat{y}) \in Z$, and $z^*(y)$ maximizes $y^\top z$ over Z (Theorem 1.1), so $y^\top z^*(y) \geq y^\top z^*(\hat{y})$.

Upper bound. For any $z \in Z$ with support T , $|T| \leq k$ and $0 \leq z_i \leq w_{\max}$, so

$$|y^\top z| \leq \sum_{i \in T} |y_i| z_i \leq k w_{\max} y_{\max}.$$

Hence $y^\top z^*(y) \leq k w_{\max} y_{\max}$ and $y^\top z^*(\hat{y}) \geq -k w_{\max} y_{\max}$; subtracting gives $\rho \leq 2k w_{\max} y_{\max}$. $\square$

Remark 1.2. The constant B is attained only in a worst case where the forecast selects k assets that each lose $y_{\max}$ while the best portfolio gains $y_{\max}$ on each of k assets. With daily equity returns this is astronomically pessimistic; we return to this honestly in Remark 3.1 when B enters the Pinsker bridge.

The next observation is small but conceptually central: the decision layer destroys all information about the *scale* of the forecast.

Proposition 1.2 (Positive scale invariance; regret is a ranking loss). *For every $c > 0$ and $\hat{y} \in \mathbb{R}^n$:*

1. $z^*(c\hat{y}) = z^*(\hat{y})$, and consequently $\rho(c\hat{y}; y) = \rho(\hat{y}; y)$.
2. More generally, $\rho(\hat{y}; y)$ depends on $\hat{y}$ only through the sign pattern $(\text{sign}(\hat{y}_i))_i$ and the relative order of the positive coordinates of $\hat{y}$.

Proof. (1) Multiplication by $c > 0$ preserves the positivity set, $P(c\hat{y}) = P(\hat{y})$, and preserves all pairwise comparisons $c\hat{y}_i > c\hat{y}_j \iff \hat{y}_i > \hat{y}_j$ (including ties, which are broken by the same index rule). Hence $S^*(c\hat{y}) = S^*(\hat{y})$ and, by (4), $z^*(c\hat{y}) = z^*(\hat{y})$. The regret (5) depends on $\hat{y}$ only through $z^*(\hat{y})$.

(2) Inspecting Theorem 1.1: the selected set $S^*(\hat{y})$ is determined by which coordinates are strictly positive and by the ordering among those, and nothing else. Since $\rho(\cdot; y)$ factors through $S^*(\hat{y})$, the claim follows. $\square$

Intuition

The optimizer asks the forecast exactly one question: “*which k assets do you like, and do you like them at all?*” Calibration, magnitude, even the gap between the best and second-best asset are invisible to the decision. A loss that trains the forecaster should therefore reward correct *rankings*, not small residuals—this is the entire *raison d’être* of decision-focused learning [1, 4].

Example 1.1 (Zero MSE correlation with regret: a 4-asset case). Let $n = 4$, $k = 2$, $w_{\max} = 0.5$, $W = 1$, and realized returns

$$y = (0.05, 0.03, -0.02, 0.01).$$

The oracle selects the two largest positive coordinates: $S^*(y) = \{1, 2\}$, $z^*(y) = (0.5, 0.5, 0, 0)$, and the optimal value is $V(y) = 0.5(0.05 + 0.03) = 0.04$.

Forecast A (high MSE, zero regret). Take $\hat{y}^A = 10y = (0.5, 0.3, -0.2, 0.1)$, a wildly miscalibrated forecast. Its squared error is

$$\text{MSE}(\hat{y}^A) = \frac{1}{4}(0.45^2 + 0.27^2 + 0.18^2 + 0.09^2) = \frac{1}{4}(0.2025 + 0.0729 + 0.0324 + 0.0081) \approx 0.0790.$$

But by Proposition 1.2, $z^*(\hat{y}^A) = z^*(y)$, so $\rho(\hat{y}^A; y) = 0$: the decision is *perfect*.

Forecast B (tiny MSE, large regret). Take $\hat{y}^B = (0.025, 0.030, -0.020, 0.031)$. Its squared error is

$$\text{MSE}(\hat{y}^B) = \frac{1}{4}(0.025^2 + 0^2 + 0^2 + 0.021^2) = \frac{1}{4}(0.000625 + 0.000441) \approx 0.000267,$$

about 296 times smaller than forecast A's. Yet the induced ranking puts assets 4 and 2 on top: $S^*(\hat{y}^B) = \{4, 2\}$, $z^*(\hat{y}^B) = (0, 0.5, 0, 0.5)$, realized value $y^\top z^*(\hat{y}^B) = 0.5(0.03 + 0.01) = 0.02$, and

$$\rho(\hat{y}^B; y) = 0.04 - 0.02 = 0.02,$$

i.e. *half* the achievable return is forfeited. MSE ranks B far above A; the decision ranks A strictly above B.

Important

Example 1.1 is not a pathological corner case. In a cross-section of assets the return differences that determine the top- k set are routinely smaller than the irreducible noise level, so an MSE-optimal forecast spends its capacity matching magnitudes that the decision layer ignores, while leaving the decision-relevant orderings to chance. This is the failure that the SPO family of losses is designed to repair.

1.3 The SPO Loss, Its Pathologies, and the SPO+ Surrogate

The natural training loss is the regret itself, viewed as a function of the prediction: $\ell_{\text{SPO}}(\hat{y}; y) := \rho(\hat{y}; y)$. It is, unfortunately, almost unusable directly.

Example 1.2 (Discontinuity and nonconvexity of the SPO loss). Let $n = 2$, $k = 1$, $w_{\max} = 1$, $W \geq 1$, $y = (0.04, 0.02)$, and consider the forecast family $\hat{y}(t) = (t, 0.02)$ for $t \in \mathbb{R}$. The oracle picks asset 1 iff $t > 0.02$ (at $t = 0.02$ the tie-break also picks asset 1), and asset 2 for $0 < t < 0.02$. Hence

$$\ell_{\text{SPO}}(\hat{y}(t); y) = \begin{cases} 0, & t \geq 0.02 \quad (\text{decision } z = e_1, \text{ value } 0.04), \\ 0.02, & t < 0.02 \quad (\text{decision } z = e_2, \text{ value } 0.02), \end{cases}$$

a step function: **discontinuous** at $t = 0.02$, with derivative 0 everywhere else. Any nonconstant function that is piecewise constant is nonconvex (a finite convex function on $\mathbb{R}$ is continuous, and a continuous piecewise-constant function is constant), so ℓ_{SPO} is also **nonconvex** in $\hat{y}$. Gradient-based training receives either no signal (zero gradient) or an undefined one (at the jump).

The remedy of [1] is a convex surrogate, SPO+. Their derivation is stated for cost *minimization*; RISP maximizes return, so we derive the maximization form explicitly via the substitution $c = -y$.

Derivation. In the minimization convention one has cost vectors c , decisions $w^*(c) \in \arg \min_{w \in Z} c^\top w$, and the SPO+ loss

$$\ell_{\text{SPO}+}^{\min}(\hat{c}; c) = \max_{w \in Z} \{(c - 2\hat{c})^\top w\} + 2\hat{c}^\top w^*(c) - c^\top w^*(c), \quad (7)$$

obtained from the SPO loss by a duality relaxation and the (provably tightest) scaling $\hat{c} \mapsto 2\hat{c}$ [1]. Substituting $c = -y$ and $\hat{c} = -\hat{y}$, and noting that $\arg \min_w (-y)^\top w = \arg \max_z y^\top z$ so $w^*(c) = z^*(y)$, the three terms of (7) become $\max_z (2\hat{y} - y)^\top z$, $-2\hat{y}^\top z^*(y)$, and $+y^\top z^*(y)$. We adopt the result as our definition.

Definition 1.4 (SPO+ loss, maximization form). For forecast $\hat{y}$ and realized returns y ,

$$\begin{aligned}\ell_{\text{SPO}+}(\hat{y}; y) &= \max_{z \in Z} (2\hat{y} - y)^\top z - 2\hat{y}^\top z^*(y) + y^\top z^*(y). \\ &= \underbrace{V(2\hat{y} - y)}\end{aligned}\quad (8)$$

Both extremal problems are instances of the same oracle (Theorem 1.1), applied to the *perturbed* vector $2\hat{y} - y$ and to the true vector y .

A reassuring sanity check: at the truth, the surrogate vanishes,

$$\ell_{\text{SPO}+}(y; y) = V(y) - 2y^\top z^*(y) + y^\top z^*(y) = V(y) - y^\top z^*(y) = 0.$$

Proposition 1.3 (Convexity in the prediction). *For every fixed y , the map $\hat{y} \mapsto \ell_{\text{SPO}+}(\hat{y}; y)$ is convex on $\mathbb{R}^n$ (indeed, piecewise affine).*

Proof. The first term is $\hat{y} \mapsto V(2\hat{y} - y) = \max_{z \in Z} \{2z^\top \hat{y} - z^\top y\}$, a pointwise maximum of affine functions of $\hat{y}$, hence convex (a supremum of convex functions is convex, and affine functions are convex). The second term $-2\hat{y}^\top z^*(y)$ is affine in $\hat{y}$ ($z^*(y)$ is a constant once y is fixed), and the third term is a constant. A sum of convex functions is convex. Since Z has finitely many extreme points (each coordinate of an optimal z is 0 or $w_{\max}$ on one of finitely many supports), the maximum is over finitely many affine functions, so the loss is piecewise affine. $\square$

Proposition 1.4 (Majorization: SPO+ upper-bounds the decision regret). *For all $\hat{y}, y \in \mathbb{R}^n$,*

$$\rho(\hat{y}; y) \leq \ell_{\text{SPO}+}(\hat{y}; y). \quad (9)$$

Proof. Lower-bound the maximum in (8) by the particular feasible point $z = z^*(\hat{y})$:

$$\ell_{\text{SPO}+}(\hat{y}; y) \geq (2\hat{y} - y)^\top z^*(\hat{y}) - 2\hat{y}^\top z^*(y) + y^\top z^*(y) = 2\hat{y}^\top z^*(\hat{y}) - y^\top z^*(\hat{y}) - 2\hat{y}^\top z^*(y) + y^\top z^*(y).$$

Now $z^*(\hat{y})$ maximizes $\hat{y}^\top z$ over Z , so $\hat{y}^\top z^*(\hat{y}) \geq \hat{y}^\top z^*(y)$, i.e. $2\hat{y}^\top z^*(\hat{y}) - 2\hat{y}^\top z^*(y) \geq 0$. Dropping this nonnegative quantity,

$$\ell_{\text{SPO}+}(\hat{y}; y) \geq y^\top z^*(y) - y^\top z^*(\hat{y}) = \rho(\hat{y}; y). \quad \square$$

Combining Propositions 1.1 and 1.4: minimizing the convex surrogate pushes down the true (discontinuous) regret. The surrogate is moreover statistically faithful in the following sense.

Theorem 1.2 (Fisher consistency of SPO+; [1], stated without proof). *Suppose the conditional distribution of y given x is continuous, centrally symmetric about its mean $\bar{y}(x) = \mathbb{E}[y \mid x]$, and assigns the optimal decision uniquely for almost every realization. Then the minimizer of the conditional SPO+ risk $\mathbb{E}[\ell_{\text{SPO}+}(\hat{y}; y) \mid x]$ over $\hat{y} \in \mathbb{R}^n$ is $\hat{y} = \bar{y}(x)$; in particular, minimizing SPO+ risk also minimizes the true SPO (regret) risk.*

Subgradient. Convexity (Proposition 1.3) gives the loss a nonempty subdifferential everywhere; the support-function structure makes one element of it available in closed form. We first record the elementary lemma usually attributed to Danskin in this finite setting.

Lemma 1.1 (Subgradient of a support function). *Let $h(u) = \max_{z \in Z} u^\top z$ with Z compact, and let $\bar{z} \in \arg \max_{z \in Z} u^\top z$. Then $\bar{z} \in \partial h(u)$, i.e. $h(u') \geq h(u) + \bar{z}^\top (u' - u)$ for all u' .*

Proof. $h(u') \geq u'^\top \bar{z} = u^\top \bar{z} + (u' - u)^\top \bar{z} = h(u) + \bar{z}^\top (u' - u)$, where the first step holds because $\bar{z}$ is feasible for the maximization defining $h(u')$ and the last because $\bar{z}$ attains $h(u)$. $\square$

Proposition 1.5 (SPO+ subgradient and the linear-model chain rule). *Fix y and let $\hat{y} \in \mathbb{R}^n$. Then*

$$g(\hat{y}; y) := 2 \left(z^*(2\hat{y} - y) - z^*(y) \right) \in \partial_{\hat{y}} \ell_{\text{SPO}+}(\hat{y}; y), \quad (10)$$

so a subgradient-descent step moves the forecast in the direction

$$-g(\hat{y}; y) = 2 \left(z^*(y) - z^*(2\hat{y} - y) \right), \quad (11)$$

i.e. toward the assets the truth selects and away from the assets the perturbed forecast selects. If the forecast is produced by a linear model $\hat{y} = \Theta x$ with $\Theta \in \mathbb{R}^{n \times d}$ and features $x \in \mathbb{R}^d$, then by the affine chain rule for subgradients,

$$g(\Theta x; y) x^\top \in \partial_{\Theta} \ell_{\text{SPO}+}(\Theta x; y) \subseteq \mathbb{R}^{n \times d}, \quad (12)$$

and for a batch $\{(x_s, y_s)\}_{s=1}^m$ the subgradient of the average loss is $\frac{1}{m} \sum_s g(\Theta x_s; y_s) x_s^\top$.

Proof. Write $\ell_{\text{SPO}+}(\hat{y}; y) = h(2\hat{y} - y) - 2\hat{y}^\top z^*(y) + \text{const}$, with h the support function of Z . By Lemma 1.1, $z^*(2\hat{y} - y) \in \partial h(2\hat{y} - y)$ (our oracle applied to the vector $2\hat{y} - y$ is exactly an argmax for h). The map $\hat{y} \mapsto 2\hat{y} - y$ is affine with Jacobian $2I$, so by the affine chain rule $2z^*(2\hat{y} - y)$ is a subgradient of $\hat{y} \mapsto h(2\hat{y} - y)$. The term $-2\hat{y}^\top z^*(y)$ is differentiable with gradient $-2z^*(y)$. Subdifferentials of sums of convex functions (one of them differentiable) add, giving (10). The chain rule (12) is the same affine-composition rule applied to $\Theta \mapsto \Theta x$: for any Θ' , $\ell(\Theta' x) \geq \ell(\Theta x) + g^\top(\Theta' - \Theta)x = \ell(\Theta x) + \langle g x^\top, \Theta' - \Theta \rangle_F$. $\square$

Remark 1.3 (Sign conventions). In the cost-minimization convention of [1] the same object reads $2(w^*(c) - w^*(2\hat{c} - c))$. Under $c = -y$ the two expressions map onto each other; the expression $2(z^*(y) - z^*(2\hat{y} - y))$ that appears in informal descriptions of RISP is the *descent direction* (11) of the maximization form, not the subgradient itself. Both conventions produce the identical parameter update; only the bookkeeping differs.

Intuition

The update (11) has a “contrastive” reading. The oracle is run on the truth y and on the reflection $2\hat{y} - y$ of the truth through the forecast (the point such that $\hat{y}$ is the midpoint of y and $2\hat{y} - y$; the reflection exaggerates the forecast’s deviation from the truth). Assets selected by the truth but not by the exaggerated forecast get their predicted returns pushed *up* by $2w_{\max}\eta$ per step; assets selected only by the exaggerated forecast get pushed *down* by the same amount; assets on which both agree (or both pass) receive no gradient. The loss is exactly zero gradient when the exaggerated forecast already makes the right selection—training stops adjusting magnitudes the moment the ranking is robustly correct, which is precisely the scale-invariance of Proposition 1.2 made operational.

Example 1.3 (One SPO+ gradient step on four assets). Continue with $n = 4$, $k = 2$, $w_{\max} = 0.5$, $W = 1$, true returns $y = (0.05, 0.03, -0.02, 0.01)$, and current forecast

$$\hat{y} = (0.02, 0.01, 0.04, 0.00).$$

Step 1: the two oracle calls. The truth selects $S^*(y) = \{1, 2\}$, so $z^*(y) = (0.5, 0.5, 0, 0)$. The perturbed vector is

$$2\hat{y} - y = (0.04 - 0.05, 0.02 - 0.03, 0.08 + 0.02, 0 - 0.01) = (-0.01, -0.01, 0.10, -0.01),$$

whose only positive coordinate is the third, so $S^*(2\hat{y} - y) = \{3\}$ and $z^*(2\hat{y} - y) = (0, 0, 0.5, 0)$.

Step 2: *loss value*. By (8),

$$\ell_{\text{SPO}+} = \underbrace{0.10 \times 0.5}_{V(2\hat{y}-y)=0.05} - \underbrace{2(0.02+0.01)(0.5)}_{2\hat{y}^\top z^*(y)=0.03} + \underbrace{0.04}_{y^\top z^*(y)} = 0.06.$$

For comparison, the true regret of $\hat{y}$: the forecast selects its top two positive coordinates $S^*(\hat{y}) = \{3, 1\}$, realized value $0.5(-0.02 + 0.05) = 0.015$, so $\rho(\hat{y}; y) = 0.04 - 0.015 = 0.025 \leq 0.06$, consistent with Proposition 1.4.

Step 3: *subgradient and update*. By (10),

$$g = 2((0, 0, 0.5, 0) - (0.5, 0.5, 0, 0)) = (-1, -1, 1, 0).$$

A descent step with rate $\eta = 0.02$ gives

$$\hat{y}' = \hat{y} - \eta g = (0.02 + 0.02, 0.01 + 0.02, 0.04 - 0.02, 0.00) = (0.04, 0.03, 0.02, 0.00).$$

The new forecast ranks the assets $1 \succ 2 \succ 3 \succ 4$, selects $S^*(\hat{y}') = \{1, 2\}$, and achieves $\rho(\hat{y}'; y) = 0$: one step flipped the decision from losing 0.025 to optimal. Note what the step did *not* do: it made the forecast magnitudes *less* accurate for asset 3 in MSE terms en route to fixing the ranking.

Key Point

Summary of the decision-focused toolkit used by RISP:

- the decision layer is an $O(n \log n)$ sort oracle (Theorem 1.1);
- the quantity that matters is the regret $\rho \in [0, B]$, a bounded *ranking* loss (Propositions 1.1, 1.2);
- ρ itself is untrainable (Example 1.2), so each niche head is trained on the convex majorant $\ell_{\text{SPO}+}$ (Definition 1.4), whose subgradient costs two oracle calls (Proposition 1.5).

Every “episode risk” appearing later in the document is an average of $\ell_{\text{SPO}+}$ (for training) or of ρ (for evaluation) over the days of an episode.

2 Concentration of Measure

RISP’s retention and transfer arguments repeatedly take the form: “the average regret of niche head r over its E_r stored episodes is small; therefore its regret on a fresh episode of the same regime is small with high probability.” Since E_r is tiny (5–7 in the companion experiments), we need the sharpest available finite-sample inequalities—and an honest accounting of how weak even those are at such sample sizes.

2.1 Markov, Chebyshev, and Cantelli

Theorem 2.1 (Markov’s inequality). *Let $X \geq 0$ be a random variable with $\mathbb{E}[X] < \infty$. For every $t > 0$,*

$$\Pr(X \geq t) \leq \frac{\mathbb{E}[X]}{t}. \quad (13)$$

Proof. Since $X \geq 0$, the pointwise inequality $t \mathbf{1}\{X \geq t\} \leq X$ holds on every outcome: if $X \geq t$ the left side is $t \leq X$; otherwise the left side is $0 \leq X$. Taking expectations and using monotonicity and linearity, $t \Pr(X \geq t) = \mathbb{E}[t \mathbf{1}\{X \geq t\}] \leq \mathbb{E}[X]$. Divide by t . $\square$

Corollary 2.1 (Chebyshev’s inequality). *Let X have mean μ and variance $\sigma^2 < \infty$. For every $t > 0$,*

$$\Pr(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}. \quad (14)$$

Proof. Apply Theorem 2.1 to the nonnegative variable $(X - \mu)^2$ at level t^2 : $\Pr(|X - \mu| \geq t) = \Pr((X - \mu)^2 \geq t^2) \leq \mathbb{E}[(X - \mu)^2]/t^2 = \sigma^2/t^2$. $\square$

Chebyshev is two-sided. When only one direction of deviation is harmful—a niche head performing *worse* on a fresh episode than on its stored ones—we can do strictly better.

Theorem 2.2 (Cantelli’s one-sided inequality). *Let X have mean μ and variance $\sigma^2 < \infty$. For every $t > 0$,*

$$\Pr(X - \mu \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}. \quad (15)$$

Proof. Let $Y = X - \mu$, so $\mathbb{E}[Y] = 0$ and $\mathbb{E}[Y^2] = \sigma^2$. Fix an arbitrary auxiliary constant $u > 0$. On the event $\{Y \geq t\}$ we have $Y + u \geq t + u > 0$, hence $(Y + u)^2 \geq (t + u)^2$ there, and $(Y + u)^2 \geq 0$ everywhere. Therefore

$$\Pr(Y \geq t) = \Pr(Y + u \geq t + u) \leq \Pr((Y + u)^2 \geq (t + u)^2) \leq \frac{\mathbb{E}[(Y + u)^2]}{(t + u)^2} = \frac{\sigma^2 + u^2}{(t + u)^2},$$

where the middle step is Markov (Theorem 2.1) applied to the nonnegative variable $(Y + u)^2$, and the last step expands $\mathbb{E}[(Y + u)^2] = \mathbb{E}[Y^2] + 2u\mathbb{E}[Y] + u^2 = \sigma^2 + u^2$.

The bound holds for every $u > 0$, so we minimize the right-hand side. Let $\varphi(u) = (\sigma^2 + u^2)/(t + u)^2$. Then

$$\varphi'(u) = \frac{2u(t + u)^2 - 2(t + u)(\sigma^2 + u^2)}{(t + u)^4} = \frac{2[u(t + u) - (\sigma^2 + u^2)]}{(t + u)^3} = \frac{2(ut - \sigma^2)}{(t + u)^3},$$

which vanishes at $u^* = \sigma^2/t$, is negative for $u < u^*$ and positive for $u > u^*$; so u^* is the global minimizer on $(0, \infty)$. Substituting,

$$\varphi(u^*) = \frac{\sigma^2 + \sigma^4/t^2}{(t + \sigma^2/t)^2} = \frac{\sigma^2(t^2 + \sigma^2)/t^2}{(t^2 + \sigma^2)^2/t^2} = \frac{\sigma^2}{\sigma^2 + t^2}. \quad \square$$

Remark 2.1 (Why one-sidedness matters for the episode-transfer lemma). In the transfer argument of Part II, the certified quantity is the average regret of a niche head over its stored episodes, and the failure event is one-directional: the fresh-episode regret exceeding that average by more than a tolerance t . A fresh episode on which the head does *better* than its certificate is a windfall, not a failure. Cantelli prices exactly the harmful tail: at $t = \sigma$, Chebyshev gives the vacuous bound 1 split across two tails (effectively 1/2 per side by symmetry heuristics, but with no valid one-sided guarantee better than 1), while Cantelli gives $\sigma^2/(2\sigma^2) = 1/2$ *as a theorem*, and at $t = 2\sigma$ gives 1/5 versus Chebyshev’s two-sided 1/4. Moreover Cantelli requires no boundedness and no symmetry—only a variance—which suits regret distributions that are skewed by construction (they live in $[0, B]$ and pile up near 0 for a well-adapted head).

2.2 Hoeffding’s Inequality

Variance-based bounds decay polynomially in t . For *bounded* variables—and the regret is bounded in $[0, B]$ by Proposition 1.1—exponential tails are available. The engine is a bound on the moment generating function.

Lemma 2.1 (Hoeffding's lemma; proof sketch). *Let $X \in [a, b]$ almost surely, with $\mathbb{E}[X] = 0$. Then for all $\lambda \in \mathbb{R}$,*

$$\mathbb{E}[e^{\lambda X}] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right). \quad (16)$$

Proof sketch. Define the cumulant generating function $\psi(\lambda) = \log \mathbb{E}[e^{\lambda X}]$, which is finite for all λ (boundedness) and twice differentiable. Direct computation gives $\psi(0) = 0$ and $\psi'(0) = \mathbb{E}[X] = 0$. The second derivative is

$$\psi''(\lambda) = \mathbb{E}_\lambda[X^2] - (\mathbb{E}_\lambda[X])^2 = \text{Var}_\lambda(X),$$

the variance of X under the exponentially tilted law $d\text{Pr}_\lambda \propto e^{\lambda X} d\text{Pr}$ (this is the step we do not carry out in full detail; it is a one-line differentiation under the expectation, legitimate by dominated convergence on a bounded variable). The tilted law is still supported in $[a, b]$, and any random variable $V \in [a, b]$ satisfies

$$\text{Var}(V) \leq \mathbb{E}\left[\left(V - \frac{a+b}{2}\right)^2\right] \leq \left(\frac{b-a}{2}\right)^2,$$

where the first inequality holds because the variance minimizes the expected squared deviation over all centering constants, and the second because $|V - (a+b)/2| \leq (b-a)/2$ pointwise. Hence $\psi''(\lambda) \leq (b-a)^2/4$ for all λ , and Taylor's theorem with the exact remainder gives, for some ξ between 0 and λ ,

$$\psi(\lambda) = \psi(0) + \lambda \psi'(0) + \frac{\lambda^2}{2} \psi''(\xi) \leq \frac{\lambda^2(b-a)^2}{8}.$$

Exponentiate. □

Theorem 2.3 (Hoeffding's inequality). *Let $X_1, \dots, X_n$ be independent with $X_i \in [a_i, b_i]$ almost surely, and let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, $\mu = \mathbb{E}[\bar{X}]$. Then for every $t > 0$,*

$$\Pr(\bar{X} - \mu \geq t) \leq \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right), \quad (17)$$

and the same bound holds for $\Pr(\mu - \bar{X} \geq t)$; a union bound gives the two-sided version with an extra factor 2. In particular, if $X_i \in [0, B]$ for all i , then with probability at least $1 - \delta$,

$$|\bar{X} - \mu| \leq B \sqrt{\frac{\ln(2/\delta)}{2n}}. \quad (18)$$

Proof. Let $S = \sum_i (X_i - \mathbb{E}X_i)$, so $\{\bar{X} - \mu \geq t\} = \{S \geq nt\}$. For any $\lambda > 0$, by Markov's inequality applied to the nonnegative variable $e^{\lambda S}$ (the Chernoff step) and independence,

$$\Pr(S \geq nt) \leq e^{-\lambda nt} \mathbb{E}[e^{\lambda S}] = e^{-\lambda nt} \prod_{i=1}^n \mathbb{E}[e^{\lambda(X_i - \mathbb{E}X_i)}] \leq \exp\left(-\lambda nt + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right),$$

using Lemma 2.1 for each centered $X_i - \mathbb{E}X_i \in [a_i - \mathbb{E}X_i, b_i - \mathbb{E}X_i]$, an interval of the same width $b_i - a_i$. The exponent $-\lambda nt + \frac{\lambda^2}{8} \sum_i (b_i - a_i)^2$ is a convex parabola in λ , minimized at $\lambda^* = 4nt / \sum_i (b_i - a_i)^2 > 0$, with minimum value $-2n^2 t^2 / \sum_i (b_i - a_i)^2$. This proves (17); the lower tail follows by applying the result to $-X_i$. For (18), set $a_i = 0$, $b_i = B$, equate the two-sided bound $2 \exp(-2nt^2/B^2)$ to δ , and solve for t . □

2.3 Empirical Bernstein: Paying the Observed Variance, Not the Range

Hoeffding’s radius (18) scales with the worst-case range B , no matter how concentrated the data actually are. When the sample is tiny this is painful, because B for the decision regret is a gross worst-case constant (Proposition 1.1). Bernstein-type inequalities replace B by the standard deviation, but the classical Bernstein bound needs the *true* variance, which is unknown. The empirical-Bernstein inequality of Maurer and Pontil [5] closes the loop by using the *sample* variance, at the price of a lower-order B/n term.

Theorem 2.4 (Empirical Bernstein inequality [5]; stated without proof). *Let $X_1, \dots, X_n$ ($n \geq 2$) be i.i.d. with values in $[0, B]$, sample mean $\bar{X}$, and sample variance $V_n = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. Then with probability at least $1 - \delta$,*

$$\mu \leq \bar{X} + \sqrt{\frac{2 V_n \ln(2/\delta)}{n}} + \frac{7 B \ln(2/\delta)}{3(n-1)}. \quad (19)$$

Intuition

Read (19) as: *you pay the spread you actually observed* (the $\sqrt{V_n/n}$ term), plus a *fixed entrance fee* proportional to B/n for the privilege of estimating the variance from the same data. When the observations are tightly clustered ($\sqrt{V_n} \ll B$) and n is moderate, the first term is far below Hoeffding’s $B\sqrt{\ln(2/\delta)/2n}$ and the entrance fee is negligible. When n is very small, the entrance fee dominates everything.

Remark 2.2 (Why this matters at $E_r = 5$ –7 episodes). In RISP, the per-niche certificate averages the episode risks of the $E_r \in \{5, \dots, 7\}$ episodes stored in niche r ’s memory. The entire design of the niche structure—episodes are routed to a niche precisely because the niche head’s decision behavior on them is similar—aims to make the *within-niche* spread of episode risks small, i.e. $V_n \ll B^2$. Empirical Bernstein is the inequality that can, in principle, convert that design property into a tighter certificate, because it is the only one in this family whose radius adapts to the realized within-niche concentration. Hoeffding cannot see the clustering at all. The sobering quantitative reality at $n \in \{5, \dots, 7\}$ is worked out in the next example.

Example 2.1 (Estimation radii with 6 episodes). Take $n = 6$ episodes, regret range $B = 0.4$, confidence $\delta = 0.1$, so $\ln(2/\delta) = \ln 20 \approx 2.9957$.

Hoeffding. The two-sided radius (18) is

$$B\sqrt{\frac{\ln(2/\delta)}{2n}} = 0.4 \times \sqrt{\frac{2.9957}{12}} = 0.4 \times \sqrt{0.24964} = 0.4 \times 0.49964 \approx 0.1999 \approx 0.20.$$

The certificate is “mean regret within ± 0.20 ”—*half the entire range* $[0, 0.4]$. As a quantitative guarantee this is nearly vacuous.

Empirical Bernstein. Suppose the six stored episode regrets are tightly clustered with sample standard deviation $\sqrt{V_n} = 0.05$. The variance term of (19) is

$$\sqrt{\frac{2 \times 0.0025 \times 2.9957}{6}} = \sqrt{0.0024964} \approx 0.050,$$

a four-fold improvement—but the entrance fee is

$$\frac{7 \times 0.4 \times 2.9957}{3 \times 5} = \frac{8.388}{15} \approx 0.559,$$

which alone exceeds the whole range B . At $n = 6$ the empirical-Bernstein certificate is *worse* than Hoeffding’s; the variance adaptivity only pays once n is large enough that the $B/(n-1)$ term recedes (here, roughly $n \gtrsim 40$ for these constants).

Important

Example 2.1 is the honest small-sample accounting that frames every finite-sample statement in this document: at $E_r = 5\text{--}7$ episodes, *no* distribution-free concentration inequality certifies the per-niche mean regret to a useful tolerance. The role of these bounds in the RISP theory is therefore *structural*, not numerical: they identify *which* quantities control transfer (the within-niche mean and variance of episode risk, and only those), and they predict the correct *qualitative* ordering of mechanisms (lower within-niche variance $\Rightarrow$ tighter transfer). That prediction is exactly what the variance-penalized objective of Section 5 acts on, and it is tested empirically rather than certified theoretically.

3 Distances Between Distributions and the Pinsker Bridge

RISP inherits from the invariant predict-and-optimize line of work [6] guarantees of the form “within a regime, the divergence between the training mixture of episodes and a deployment episode is at most ϵ .” Such statements are about *distributions*; what we need are statements about *regret*. This section builds the standard bridge: total variation converts distributional closeness into closeness of bounded expectations, and Pinsker’s inequality converts KL divergence into total variation.

Throughout, P and Q are probability measures on a common measurable space $(\Omega, \mathcal{F})$, and we fix a dominating σ -finite measure μ (e.g. $\mu = P + Q$) with densities $p = dP/d\mu$, $q = dQ/d\mu$. All statements are independent of the choice of μ .

3.1 Total Variation Distance

Definition 3.1 (Total variation). The **total variation distance** between P and Q is

$$\text{TV}(P, Q) = \sup_{A \in \mathcal{F}} |P(A) - Q(A)|. \quad (20)$$

Proposition 3.1 (Variational and L_1 characterizations). *With p, q as above,*

$$\text{TV}(P, Q) = \frac{1}{2} \int_{\Omega} |p - q| d\mu = \sup_{g: \Omega \rightarrow [0,1]} |\mathbb{E}_P[g] - \mathbb{E}_Q[g]|, \quad (21)$$

where the supremum is over all measurable functions g with values in $[0, 1]$, and is attained at $g = \mathbf{1}_{A^*}$ with $A^* = \{p \geq q\}$.

Proof. Write $\Delta = p - q$ and note the key cancellation

$$\int_{\Omega} \Delta d\mu = \int p d\mu - \int q d\mu = 1 - 1 = 0 \implies \int_{A^*} \Delta d\mu = \int_{(A^*)^c} (-\Delta) d\mu, \quad (22)$$

where $A^* = \{\Delta \geq 0\}$; both sides of the implication equal $\frac{1}{2} \int |\Delta| d\mu$ since $\int |\Delta| = \int_{A^*} \Delta + \int_{(A^*)^c} (-\Delta)$.

Step 1: every $[0, 1]$ -valued g is dominated. For any such g ,

$$\mathbb{E}_P[g] - \mathbb{E}_Q[g] = \int g \Delta d\mu = \int_{A^*} g \Delta d\mu + \int_{(A^*)^c} g \Delta d\mu \leq \int_{A^*} 1 \cdot \Delta d\mu + \int_{(A^*)^c} 0 \cdot \Delta d\mu = \frac{1}{2} \int |\Delta| d\mu,$$

using $0 \leq g \leq 1$, $\Delta \geq 0$ on A^* , and $\Delta < 0$ on $(A^*)^c$. Applying the same bound to $1 - g$ (also $[0, 1]$ -valued) gives $\mathbb{E}_Q[g] - \mathbb{E}_P[g] \leq \frac{1}{2} \int |\Delta|$ as well (note $\mathbb{E}_Q[g] - \mathbb{E}_P[g] = \mathbb{E}_P[1 - g] - \mathbb{E}_Q[1 - g]$). Hence $\sup_g |\mathbb{E}_P[g] - \mathbb{E}_Q[g]| \leq \frac{1}{2} \int |\Delta| d\mu$.

Step 2: attainment. The choice $g = \mathbf{1}_{A^*}$ achieves $\mathbb{E}_P[g] - \mathbb{E}_Q[g] = \int_{A^*} \Delta d\mu = \frac{1}{2} \int |\Delta| d\mu$ by (22). So the supremum over $[0, 1]$ -valued g equals $\frac{1}{2} \int |\Delta|$.

Step 3: events. Indicators $\mathbf{1}_A$ are particular $[0, 1]$ -valued functions, so $\sup_A |P(A) - Q(A)| \leq \sup_g |\mathbb{E}_P g - \mathbb{E}_Q g|$; and the optimizer found in Step 2 *is* an indicator, so the two suprema coincide, and both equal $\frac{1}{2} \int |p - q| d\mu$. $\square$

Intuition

TV answers: *over all bounded tests normalized to $[0, 1]$, how differently can the two distributions score?* The optimal test is the likelihood-ratio region $\{p \geq q\}$. The $[0, 1]$ -function form—rather than the events-only form—is the one we will actually use, because a bounded loss rescaled into $[0, 1]$ is such a function but is not an indicator.

3.2 KL Divergence and Gibbs' Inequality

Definition 3.2 (Kullback–Leibler divergence). If $P \ll Q$ (i.e. $q = 0 \Rightarrow p = 0$, μ -a.e.),

$$\text{KL}(P \| Q) = \int_{\{p>0\}} p \log \frac{p}{q} d\mu = \mathbb{E}_P \left[\log \frac{p}{q} \right], \quad (23)$$

with the conventions $0 \log 0 = 0$; if $P \not\ll Q$ we set $\text{KL}(P \| Q) = +\infty$.

Proposition 3.2 (Gibbs' inequality / non-negativity). $\text{KL}(P \| Q) \geq 0$, with equality if and only if $P = Q$.

Proof. Assume $P \ll Q$ (otherwise $\text{KL} = \infty > 0$). Then

$$-\text{KL}(P \| Q) = \mathbb{E}_P \left[\log \frac{q}{p} \right] \leq \log \mathbb{E}_P \left[\frac{q}{p} \right] = \log \int_{\{p>0\}} q d\mu \leq \log 1 = 0,$$

where the first inequality is Jensen's inequality for the concave function $\log$, and the second uses $\int_{\{p>0\}} q d\mu \leq \int q d\mu = 1$. For the equality case: $\log$ is *strictly* concave, so Jensen is an equality iff q/p is P -a.s. constant; combined with $\int_{\{p>0\}} q = 1$ (forced by the second inequality being tight) the constant is 1, i.e. $p = q$ μ -a.e., i.e. $P = Q$. $\square$

3.3 Pinsker's Inequality

We give the classical proof [7] in two moves: a data-processing (two-point) reduction via the log-sum inequality, then the scalar inequality for Bernoulli pairs.

Lemma 3.1 (Log-sum inequality). For nonnegative reals $a_1, \dots, a_m$ and positive reals $b_1, \dots, b_m$,

$$\sum_{i=1}^m a_i \log \frac{a_i}{b_i} \geq \left(\sum_{i=1}^m a_i \right) \log \frac{\sum_{i=1}^m a_i}{\sum_{i=1}^m b_i}, \quad (24)$$

with the convention $0 \log 0 = 0$. The same holds with sums replaced by integrals of nonnegative functions.

Proof. Let $\phi(t) = t \log t$ for $t > 0$, $\phi(0) = 0$; ϕ is convex on $[0, \infty)$ since $\phi''(t) = 1/t > 0$. Set $b = \sum_i b_i$ and apply Jensen's inequality with weights b_i/b to the points $t_i = a_i/b_i$:

$$\sum_{i=1}^m \frac{b_i}{b} \phi\left(\frac{a_i}{b_i}\right) \geq \phi\left(\sum_{i=1}^m \frac{b_i}{b} \cdot \frac{a_i}{b_i}\right) = \phi\left(\frac{\sum_i a_i}{b}\right).$$

Multiplying both sides by b and expanding ϕ :

$$\sum_i b_i \cdot \frac{a_i}{b_i} \log \frac{a_i}{b_i} = \sum_i a_i \log \frac{a_i}{b_i} \geq b \cdot \frac{\sum_i a_i}{b} \log \frac{\sum_i a_i}{b} = \left(\sum_i a_i \right) \log \frac{\sum_i a_i}{\sum_i b_i}.$$

The integral version is Jensen for the same ϕ with the normalized measure $b(\omega) d\mu / \int b d\mu$. $\square$

Lemma 3.2 (Two-point (data-processing) reduction). *For any event $A \in \mathcal{F}$, writing $a = P(A)$, $b = Q(A)$ and $\text{kl}(a \| b) = a \log \frac{a}{b} + (1-a) \log \frac{1-a}{1-b}$ for the Bernoulli KL,*

$$\text{KL}(P \| Q) \geq \text{kl}(P(A) \| Q(A)). \quad (25)$$

Proof. Split the defining integral over the partition $\{A, A^c\}$ and apply the integral log-sum inequality (Lemma 3.1) on each cell:

$$\text{KL}(P \| Q) = \int_A p \log \frac{p}{q} d\mu + \int_{A^c} p \log \frac{p}{q} d\mu \geq \left(\int_A p \right) \log \frac{\int_A p}{\int_A q} + \left(\int_{A^c} p \right) \log \frac{\int_{A^c} p}{\int_{A^c} q} = \text{kl}(P(A) \| Q(A)). \quad \square$$

Lemma 3.3 (Scalar Pinsker). *For all $a, b \in [0, 1]$ (with $\text{kl}(a \| b) = \infty$ when undefined),*

$$\text{kl}(a \| b) \geq 2(a - b)^2. \quad (26)$$

Proof. If $b \in \{0, 1\}$ and $a \neq b$ the left side is $+\infty$ and the claim is trivial; if $a = b$ both sides vanish. So fix $b \in (0, 1)$ and define on $(0, 1)$ (extending continuously to the endpoints where finite)

$$f(a) = \text{kl}(a \| b) - 2(a - b)^2 = a \log \frac{a}{b} + (1 - a) \log \frac{1 - a}{1 - b} - 2(a - b)^2.$$

Then $f(b) = 0$, and differentiating in a :

$$f'(a) = \log \frac{a}{b} - \log \frac{1 - a}{1 - b} - 4(a - b), \quad f'(b) = 0,$$

$$f''(a) = \frac{1}{a} + \frac{1}{1 - a} - 4 = \frac{1}{a(1 - a)} - 4 \geq 0,$$

since $a(1 - a) \leq 1/4$ for $a \in (0, 1)$ (AM-GM: $a(1 - a) \leq \left(\frac{a + (1 - a)}{2}\right)^2 = \frac{1}{4}$). Thus f' is nondecreasing and vanishes at $a = b$: $f' \leq 0$ on $(0, b]$ and $f' \geq 0$ on $[b, 1)$, so f attains its minimum over $(0, 1)$ at $a = b$, where it is 0. Hence $f(a) \geq 0$ for all a , which is (26). $\square$

Theorem 3.1 (Pinsker's inequality). *For all probability measures P, Q ,*

$$\text{TV}(P, Q) \leq \sqrt{\frac{1}{2} \text{KL}(P \| Q)}. \quad (27)$$

Proof. If $\text{KL}(P \| Q) = \infty$ there is nothing to prove. Otherwise let $A^* = \{p \geq q\}$, so that by Proposition 3.1, $\text{TV}(P, Q) = P(A^*) - Q(A^*)$. Chaining Lemmas 3.2 and 3.3,

$$\text{KL}(P \| Q) \geq \text{kl}(P(A^*) \| Q(A^*)) \geq 2(P(A^*) - Q(A^*))^2 = 2 \text{TV}(P, Q)^2.$$

Rearranging gives (27). $\square$

3.4 The Pinsker Bridge: from Divergence to Regret

Theorem 3.2 (Bounded-loss transfer / Pinsker bridge). *Let $\ell : \Omega \rightarrow [0, B]$ be any measurable loss, and write $L_P = \mathbb{E}_P[\ell]$, $L_Q = \mathbb{E}_Q[\ell]$. Then*

$$|L_P - L_Q| \leq B \cdot \text{TV}(P, Q) \leq B \sqrt{\frac{1}{2} \text{KL}(P \| Q)}. \quad (28)$$

Proof. The function $g = \ell/B$ takes values in $[0, 1]$, so by the variational characterization (Proposition 3.1),

$$|L_P - L_Q| = B |\mathbb{E}_P[g] - \mathbb{E}_Q[g]| \leq B \sup_{g' : 0 \leq g' \leq 1} |\mathbb{E}_P[g'] - \mathbb{E}_Q[g']| = B \cdot \text{TV}(P, Q).$$

The second inequality in (28) is Pinsker (Theorem 3.1). $\square$

Key Point

Role in the RISP theory. Fix a regime r and apply Theorem 3.2 with: Ω the space of (feature, return) days; $\ell(\omega) = \rho(\hat{y}_\theta(x_\omega); y_\omega)$ the decision regret of niche head θ , bounded by $B = 2k w_{\max} y_{\max}$ (Proposition 1.1); P the mixture of the niche’s stored training episodes; Q a fresh deployment episode of the same regime. An Inv-PnCO-style assumption [6]—that episodes of one regime are divergence-close, $\text{KL}(Q\|P) \leq \epsilon_r$ (or $\text{TV} \leq \tau_r$)—then yields a *regret* transfer guarantee:

$$\text{fresh-episode regret} \leq \text{stored-episode regret} + B\sqrt{\epsilon_r/2},$$

which composes with the concentration bounds of Section 2 (controlling the stored-episode average) to give the episode-transfer lemma of Part II. The bridge is what lets a statement about *data distributions* (checkable by regime statistics) control a statement about *money* (regret of induced portfolios), for *any* head simultaneously—no union bound over heads is needed because (28) is deterministic given the divergence.

Remark 3.1 (Honest looseness accounting). Every appearance of $B = 2k w_{\max} y_{\max}$ in (28) imports the worst case of Proposition 1.1: the forecast shorting itself into the k worst assets while the oracle rides the k best, all at the extreme return $y_{\max}$. Empirically observed decision regrets in the companion experiments range roughly *fifty times smaller* than B . The bridge’s *form*—transfer error scaling as (regret scale) $\times \sqrt{\text{divergence}}$ —is the content we rely on; its *constant* should be read as a certificate of structure, not of magnitude. Tightening B to a high-probability regret envelope (e.g. a quantile of the realized regret distribution) is possible but converts the deterministic bridge into a probabilistic one; we keep the loose constant and flag it.

4 Online Learning: Hedge, Exponentiated Gradient, and Their Limits

The RISP gate maintains a probability vector over niches and updates it multiplicatively in observed decision quality. This update has three classical pedigrees—Hedge [8], exponentiated gradient [9], and multiplicative weights [10]—and one classical guarantee, the $O(\sqrt{T \log N})$ adversarial regret bound [11]. This section develops all of it, and then states precisely why the classical guarantee does *not* cover RISP’s use of the update.

4.1 The Experts Problem and the Hedge Update

Definition 4.1 (Prediction with expert advice). There are N **experts**. At each round $t = 1, \dots, T$: the learner chooses a distribution $p_t \in \Delta_N = \{p \in [0, 1]^N : \sum_i p_i = 1\}$; an adversary simultaneously chooses a loss vector $\ell_t \in [0, 1]^N$; the learner suffers $\langle p_t, \ell_t \rangle$ and observes all of ℓ_t . The learner’s **regret** is

$$R_T = \sum_{t=1}^T \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t=1}^T \ell_{t,i}, \quad (29)$$

the gap to the best *single* expert in hindsight.

Definition 4.2 (Hedge / multiplicative weights). Fix a learning rate $\eta > 0$. Initialize $w_{1,i} = 1$ for all i and $p_1 = (1/N, \dots, 1/N)$. After observing ℓ_t ,

$$w_{t+1,i} = w_{t,i} e^{-\eta \ell_{t,i}}, \quad p_{t+1,i} = \frac{w_{t+1,i}}{\sum_{j=1}^N w_{t+1,j}} = \frac{p_{t,i} e^{-\eta \ell_{t,i}}}{\sum_j p_{t,j} e^{-\eta \ell_{t,j}}}. \quad (30)$$

This lineage runs through universal portfolio selection [12] and online portfolio optimization [13]: there, the “experts” are constant-rebalanced portfolios or trading strategies and the weights track wealth shares. In RISP the experts are niche heads and the weights drive both selection and training-resource allocation—a difference we return to in Section 4.5.

4.2 Exponentiated Gradient as Mirror Descent with Entropy

The exponential form (30) is not an ad hoc choice: it is the *exact* solution of a regularized linear step where the regularizer is the KL divergence on the simplex [9].

Proposition 4.1 (EG update solves the KL-regularized linear step). *Fix p_t in the relative interior of Δ_N and $\eta > 0$. The unique solution of*

$$p_{t+1} = \arg \min_{p \in \Delta_N} \left\{ \eta \langle \ell_t, p \rangle + \text{KL}(p \| p_t) \right\}, \quad \text{KL}(p \| p_t) = \sum_{i=1}^N p_i \log \frac{p_i}{p_{t,i}}, \quad (31)$$

is exactly the Hedge update (30).

Proof. The objective $J(p) = \eta \langle \ell_t, p \rangle + \text{KL}(p \| p_t)$ is strictly convex on Δ_N ($\langle \ell_t, \cdot \rangle$ is linear and $\text{KL}(\cdot \| p_t)$ is strictly convex, being a sum of strictly convex scalar functions $p_i \mapsto p_i \log(p_i/p_{t,i})$), so the minimizer is unique. Moreover the entropy term’s derivative blows up at the boundary ($\partial_{p_i} p_i \log p_i = \log p_i + 1 \rightarrow -\infty$ as $p_i \rightarrow 0^+$), which forces the minimizer into the relative interior; hence the nonnegativity constraints are inactive and we may use a Lagrangian with only the equality constraint. Set

$$\mathcal{L}(p, \lambda) = \sum_{i=1}^N \left[\eta \ell_{t,i} p_i + p_i \log \frac{p_i}{p_{t,i}} \right] + \lambda \left(\sum_{i=1}^N p_i - 1 \right).$$

Stationarity in each coordinate:

$$\frac{\partial \mathcal{L}}{\partial p_i} = \eta \ell_{t,i} + \log \frac{p_i}{p_{t,i}} + 1 + \lambda = 0 \quad \Longleftrightarrow \quad p_i = p_{t,i} e^{-\eta \ell_{t,i}} e^{-(1+\lambda)}.$$

The factor $e^{-(1+\lambda)}$ is a positive constant independent of i ; the constraint $\sum_i p_i = 1$ pins it to $(\sum_j p_{t,j} e^{-\eta \ell_{t,j}})^{-1}$. This is precisely (30). $\square$

Intuition

The step (31) balances two pulls: “move mass toward low-loss coordinates” (the linear term) against “do not move far from where you are, measured *multiplicatively*” (the KL term). Measuring movement by KL rather than Euclidean distance is what makes the update respect the simplex geometry: probabilities can shrink toward zero but never jump across it, weights never go negative, and a coordinate’s *relative* change $p_{t+1,i}/p_{t,i}$ depends only on its own loss and the normalizer. This is the precise sense in which the RISP gate is a “well-behaved simplex ascent”: bounded relative updates, invariance to common loss shifts, and no projection step that could zero out a niche in one round.

4.3 The Classical Regret Guarantee

Theorem 4.1 (Hedge regret bound [8, 11]). *For losses $\ell_t \in [0, 1]^N$ and any $\eta > 0$, Hedge guarantees*

$$R_T \leq \frac{\ln N}{\eta} + \frac{\eta T}{8}, \quad (32)$$

and with the tuned rate $\eta = \sqrt{8 \ln N / T}$, $R_T \leq \sqrt{(T/2) \ln N} = O(\sqrt{T \log N})$.

Proof (potential-function argument). Let $W_t = \sum_{i=1}^N w_{t,i}$ with the weights of Definition 4.2, and consider the potential $\ln W_t$.

Per-round drop. Since $p_{t,i} = w_{t,i}/W_t$,

$$\ln \frac{W_{t+1}}{W_t} = \ln \sum_{i=1}^N \frac{w_{t,i}}{W_t} e^{-\eta \ell_{t,i}} = \ln \mathbb{E}_{i \sim p_t} [e^{-\eta \ell_{t,i}}] \leq -\eta \mathbb{E}_{i \sim p_t} [\ell_{t,i}] + \frac{\eta^2}{8} = -\eta \langle p_t, \ell_t \rangle + \frac{\eta^2}{8},$$

where the inequality is Hoeffding's lemma (Lemma 2.1) applied to the random variable $-\eta(\ell_{t,I} - \mathbb{E} \ell_{t,I})$, $I \sim p_t$, supported on an interval of width $\eta \cdot 1$. Summing over t and telescoping,

$$\ln W_{T+1} - \ln W_1 \leq -\eta \sum_{t=1}^T \langle p_t, \ell_t \rangle + \frac{\eta^2 T}{8}. \quad (33)$$

Comparator floor. For any fixed expert i^* , since all weights are positive,

$$\ln W_{T+1} \geq \ln w_{T+1,i^*} = \ln \left(1 \cdot \prod_{t=1}^T e^{-\eta \ell_{t,i^*}} \right) = -\eta \sum_{t=1}^T \ell_{t,i^*}, \quad (34)$$

while $\ln W_1 = \ln N$.

Combine. Chaining (34) $\leq \ln W_{T+1}$ with (33):

$$-\eta \sum_t \ell_{t,i^*} - \ln N \leq -\eta \sum_t \langle p_t, \ell_t \rangle + \frac{\eta^2 T}{8} \implies \sum_t \langle p_t, \ell_t \rangle - \sum_t \ell_{t,i^*} \leq \frac{\ln N}{\eta} + \frac{\eta T}{8}.$$

Taking i^* to be the best expert gives (32); substituting the tuned η gives $2\sqrt{(\ln N)(T)/8} = \sqrt{(T/2) \ln N}$. $\square$

Remarkably, the bound holds against an adversary who sees the learner's distribution before choosing losses—but, crucially, an adversary whose losses are *handed to fixed experts*, a point we scrutinize below.

4.4 The Replicator Limit

For small η , the discrete multiplicative update becomes the replicator dynamics of evolutionary game theory—the lens through which the winner-take-all niche competition of RISP (and of GAUSE before it [14]) is analyzed.

Proposition 4.2 (Small- η replicator limit). *For the Hedge update (30) with bounded losses,*

$$p_{t+1,i} - p_{t,i} = \eta p_{t,i} (\langle p_t, \ell_t \rangle - \ell_{t,i}) + O(\eta^2), \quad (35)$$

so as $\eta \rightarrow 0$ the rescaled-time trajectories solve the replicator ODE $\dot{p}_i = p_i(\bar{f} - f_i)$ with “fitness” $f_i = \ell_{t,i}$ (lower loss = higher fitness, $\bar{f} = \langle p, \ell \rangle$ the population mean).

Proof. Expand the exponentials to first order: $e^{-\eta \ell_{t,i}} = 1 - \eta \ell_{t,i} + O(\eta^2)$, uniformly in i by boundedness. The normalizer is

$$\sum_j p_{t,j} e^{-\eta \ell_{t,j}} = 1 - \eta \langle p_t, \ell_t \rangle + O(\eta^2), \quad \text{so} \quad \left(\sum_j p_{t,j} e^{-\eta \ell_{t,j}} \right)^{-1} = 1 + \eta \langle p_t, \ell_t \rangle + O(\eta^2).$$

Multiplying,

$$p_{t+1,i} = p_{t,i} (1 - \eta \ell_{t,i}) (1 + \eta \langle p_t, \ell_t \rangle) + O(\eta^2) = p_{t,i} [1 + \eta (\langle p_t, \ell_t \rangle - \ell_{t,i})] + O(\eta^2). \quad \square$$

Intuition

A niche grows in gate weight iff it beats the *current weighted average* of all niches; the growth rate is proportional to both its advantage and its current weight. This is exactly the dynamics under which competitive exclusion and niche differentiation are usually derived—and it is a *dynamical-systems* object (fixed points, basins, exclusion), not a regret object.

4.5 What the Classical Guarantee Does Not Cover

Important

The adversarial regret bound does not transfer to RISP. Theorem 4.1 compares the learner to the best *fixed* expert under loss sequences that are *exogenous*: expert i ’s loss at round t does not depend on how much weight the learner has been giving expert i . In RISP both premises fail, for the same reason they fail in GAUSE [14]:

1. *Losses are endogenous.* Niche i ’s decision quality at round t depends on its head parameters, which were trained with data and gradient budget allocated *in proportion to past gate weights*. The loss sequence $\ell_{t,i}$ is a functional of the learner’s own trajectory $p_{1:t-1}$ —outside the oblivious- and even the standard adaptive-adversary model, where “expert i ’s loss” must remain well-defined independently of the comparison.
2. *The comparator is meaningless.* “The best fixed niche in hindsight” is not a coherent benchmark when the niche’s identity (its trained parameters) is itself an artifact of the weight history; under a different weight history the “same” niche would be a different predictor. A regret bound against an incoherent comparator is vacuous even if formally true.
3. *Winner-take-all coupling.* The RISP assignment concentrates training on high-weight niches, which is the very mechanism that creates specialization; this positive feedback is what the replicator analysis (Proposition 4.2) studies and what a worst-case regret analysis is designed to neutralize.

Accordingly, RISP uses the EG/Hedge update for its *structural* virtues established in Proposition 4.1 and the intuition following it—a strictly interior, multiplicatively bounded, geometry-respecting ascent on the simplex—and *not* for the bound (32). Every stability claim about the coupled dynamics is proved separately in Part II; none is inherited from Theorem 4.1.

4.6 Sleeping Experts and Adaptive Regret: the Gap RISP Addresses

Non-stationarity has its own classical formalisms, and it is important to locate precisely where they stop short of the RISP setting.

Definition 4.3 (Sleeping experts [15]). In the sleeping-experts (specialists) problem, at each round t an adversary additionally reveals an **awake set** $A_t \subseteq [N]$; only awake experts make predictions, the learner must put its mass on A_t , and asleep experts are unevaluated. The standard guarantee is per-expert: for every expert i ,

$$\sum_{t: i \in A_t} \langle p_t, \ell_t \rangle - \sum_{t: i \in A_t} \ell_{t,i} \leq O(\sqrt{T_i \log N}), \quad T_i = |\{t : i \in A_t\}|, \quad (36)$$

i.e. the learner competes with each specialist *on the rounds where that specialist is awake*.

Definition 4.4 (Adaptive and strongly adaptive regret [16, 17]). The **adaptive regret** of an algorithm is its worst regret over any contiguous interval,

$$\text{AR}_T = \max_{1 \leq s \leq e \leq T} \left[\sum_{t=s}^e \langle p_t, \ell_t \rangle - \min_i \sum_{t=s}^e \ell_{t,i} \right]; \quad (37)$$

an algorithm is **strongly adaptive** [17] if for every interval length τ its regret on every length- τ interval is $O(\text{poly}(\log T) \cdot \sqrt{\tau})$. Such algorithms track the best expert *per regime segment* rather than globally.

Key Point

Why dormant *fixed* experts retain trivially. In both frameworks an expert is a fixed function (or a fixed external advisor). When expert i sleeps for a thousand rounds, nothing happens to it: its parameters do not exist or do not change, and when it wakes its predictions are exactly what they would have been. Retention of dormant competence is *free*—it is an assumption, not an achievement. The guarantees (36) and (37) are therefore statements purely about the *weighting* layer: how fast the learner re-concentrates on the right expert after a regime switch.

Remark 4.1 (The precise gap). RISP operates in a setting these formalisms do not model: **capacity-bounded *learning* experts under reward-driven training allocation**. Formally: each expert i carries parameters $\theta_{t,i}$ in a bounded-capacity class; at each round a shared training budget is divided according to (a function of) the gate weights p_t , and $\theta_{t+1,i}$ is produced by gradient steps on data weighted by expert i ’s allocation; the loss $\ell_{t,i} = \ell(\theta_{t,i}; \text{episode}_t)$ is thereby a functional of the entire weight history. In this setting dormancy is no longer safe by assumption: an expert that receives weight (hence budget, hence gradient updates on *out-of-niche* data, or no updates while the shared feature extractor drifts) can have its dormant competence *overwritten*—the continual-learning failure mode studied for mixture-of-experts learners in [18] and in the GAUSE retention analysis [14]. The sleeping-experts guarantee says nothing here because the premise “the specialist’s awake-round losses are fixed targets” is false; strongly adaptive algorithms recover the right *weights* quickly but have no mechanism to ensure there remains a competent expert to weight. RISP’s contribution lives exactly in this gap: the niche memory and the variance-penalized per-niche objective are mechanisms for making the “expert exists and is still competent when its regime returns” premise *true*, after which adaptive-weighting logic can do its classical job.

5 Invariant Risk and the Failure of ERM Across Environments

The last foundation explains *what is being learned inside each niche*. A niche head sees a handful of episodes of “its” regime; those episodes share the regime’s stable structure but differ in incidental, episode-specific correlations. Pooled empirical risk minimization (ERM) over the episodes is the default—and it fails in a precise, derivable way: it loads on the incidental correlations exactly in proportion to how unbalanced the small sample of episodes happens to be. This section sets up the standard two-feature construction, derives the failure quantitatively, shows that the *across-episode variance of per-episode loss* is a statistic that detects exactly the failure, and positions the RISP per-niche objective in the invariant-risk literature [19, 20, 21].

5.1 Environments, Episodes, and Pooled ERM

Definition 5.1 (Multi-environment learning). Data arrive grouped into **episodes** (environments) $e \in \mathcal{E}$; episode e carries a distribution P_e over observation–target pairs. A training

set consists of samples from E episodes $e_1, \dots, e_E$ drawn from an environment distribution $\mathcal{Q}$; at deployment a *fresh* episode $e' \sim \mathcal{Q}$ is encountered. For a predictor h and loss ℓ , write $L_e(h) = \mathbb{E}_{P_e}[\ell(h(x), y)]$ for the per-episode risk. **Pooled ERM** minimizes $\frac{1}{E} \sum_{j=1}^E L_{e_j}(h)$ (empirically); the **invariant** ideal is a predictor whose risk—and whose optimality—is stable across all e in the support of $\mathcal{Q}$.

In RISP, $\mathcal{E}$ is the set of episodes routed to one niche, $E = E_r \in \{5, \dots, 7\}$, and ℓ is the SPO+ surrogate of Section 1.3; for the analysis of this section we use the squared loss, where everything is exactly computable, and return to the SPO+ case at the end.

5.2 The Two-Feature Construction

Definition 5.2 (Invariant vs. spurious feature model). Within episode e , independent draws are generated as

$$x_{\text{inv}} \sim \mathcal{N}(0, 1), \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad y = \theta x_{\text{inv}} + \varepsilon, \quad x_{\text{sp}} = \gamma_e y + \nu, \quad \nu \sim \mathcal{N}(0, \sigma_\nu^2), \quad (38)$$

with $x_{\text{inv}}, \varepsilon, \nu$ mutually independent. The **invariant** coefficient θ is the same in every episode; the **spurious** (anti-causal) coefficient γ_e is drawn once per episode, i.i.d. across episodes with $\mathbb{E}[\gamma] = 0$ and $\text{Var}[\gamma] = \sigma_\gamma^2 > 0$. Write $s^2 := \text{Var}(y) = \theta^2 + \sigma_\varepsilon^2$. A linear predictor is $h_{a,b}(x) = a x_{\text{inv}} + b x_{\text{sp}}$, and the **invariant predictor** is $h_{\theta,0}$.

Intuition

x_{inv} is a genuine driver of y (a stable regime characteristic); x_{sp} is a symptom—it is generated *from* y , with an episode-specific gain γ_e (an incidental correlation: e.g. a factor that happened to co-move with returns during those particular weeks). Because x_{sp} is correlated with y within each episode, regression happily uses it; because the correlation's sign and size are episode-specific accidents, doing so buys in-sample fit with out-of-episode fragility. Averaging the γ_e 's over infinitely many episodes would cancel them ($\mathbb{E}\gamma = 0$); the problem is that a niche has 5–7 episodes, and the empirical mean $\bar{\gamma} = \frac{1}{E} \sum_j \gamma_{e_j}$ is a $\Theta(\sigma_\gamma E^{-1/2})$ -sized accident that does *not* cancel.

Proposition 5.1 (Pooled-OLS limit loads on the spurious feature). *Consider pooled least squares over E training episodes with equal weights and infinite within-episode samples (so all second moments are exact). Write $\bar{\gamma} = \frac{1}{E} \sum_j \gamma_{e_j}$, $\hat{m}_2 = \frac{1}{E} \sum_j \gamma_{e_j}^2$, $\hat{\sigma}_\gamma^2 = \hat{m}_2 - \bar{\gamma}^2$. The pooled OLS coefficients $(\hat{a}, \hat{b}) = \arg \min_{a,b} \frac{1}{E} \sum_j L_{e_j}(h_{a,b})$ are*

$$\hat{b} = \frac{\bar{\gamma} \sigma_\varepsilon^2}{D}, \quad \hat{a} = \theta \cdot \frac{\sigma_\nu^2 + s^2 \hat{\sigma}_\gamma^2}{D}, \quad D := \sigma_\nu^2 + s^2 \hat{\sigma}_\gamma^2 + \bar{\gamma}^2 \sigma_\varepsilon^2 > 0. \quad (39)$$

In particular $\hat{b} \neq 0$ whenever $\bar{\gamma} \neq 0$: the pooled solution loads on x_{sp} in proportion to the empirical mean of the episode coefficients, which for small E is nonzero with probability one (for continuous γ) and of typical size $\sigma_\gamma/\sqrt{E}$.

Proof. Within episode e , using (38) and independence:

$$\mathbb{E}[x_{\text{inv}}^2] = 1, \quad \mathbb{E}[x_{\text{inv}} x_{\text{sp}}] = \gamma_e \mathbb{E}[x_{\text{inv}} y] = \gamma_e \theta, \quad \mathbb{E}[x_{\text{sp}}^2] = \gamma_e^2 s^2 + \sigma_\nu^2, \quad \mathbb{E}[x_{\text{inv}} y] = \theta, \quad \mathbb{E}[x_{\text{sp}} y] = \gamma_e s^2.$$

Pooling with equal weights averages these moments over the E training episodes, so the pooled normal equations $\bar{\Sigma}(\hat{a}, \hat{b})^\top = \bar{v}$ have

$$\bar{\Sigma} = \begin{pmatrix} 1 & \bar{\gamma} \theta \\ \bar{\gamma} \theta & \hat{m}_2 s^2 + \sigma_\nu^2 \end{pmatrix}, \quad \bar{v} = \begin{pmatrix} \theta \\ \bar{\gamma} s^2 \end{pmatrix}.$$

The determinant is

$$\det \bar{\Sigma} = \hat{m}_2 s^2 + \sigma_\nu^2 - \bar{\gamma}^2 \theta^2 = \sigma_\nu^2 + s^2(\hat{m}_2 - \bar{\gamma}^2) + \bar{\gamma}^2(s^2 - \theta^2) = \sigma_\nu^2 + s^2 \hat{\sigma}_\gamma^2 + \bar{\gamma}^2 \sigma_\varepsilon^2 = D > 0.$$

By Cramer's rule,

$$\hat{b} = \frac{1 \cdot \bar{\gamma} s^2 - \bar{\gamma} \theta \cdot \theta}{D} = \frac{\bar{\gamma}(s^2 - \theta^2)}{D} = \frac{\bar{\gamma} \sigma_\varepsilon^2}{D}, \quad \hat{a} = \frac{(\hat{m}_2 s^2 + \sigma_\nu^2) \theta - \bar{\gamma} \theta \cdot \bar{\gamma} s^2}{D} = \frac{\theta(\sigma_\nu^2 + s^2 \hat{\sigma}_\gamma^2)}{D}. \quad \square$$

Note the mechanism: $\hat{b} \propto \bar{\gamma}$, an average of mean-zero accidents that a law of large numbers would kill—if E were large. ERM is not wrong asymptotically; it is wrong at $E = 5-7$, which is the only regime RISP lives in. Pooled ERM also genuinely *prefers* this loading in-sample:

Lemma 5.1 (ERM strictly descends into the spurious direction). *The per-episode risk of $h_{a,b}$ is*

$$L_e(h_{a,b}) = (\theta - a - b\gamma_e \theta)^2 + (1 - b\gamma_e)^2 \sigma_\varepsilon^2 + b^2 \sigma_\nu^2, \quad (40)$$

and at the invariant predictor $(a, b) = (\theta, 0)$ the pooled risk has $\frac{\partial}{\partial b} \frac{1}{E} \sum_j L_{e_j} = -2\bar{\gamma} \sigma_\varepsilon^2$. Hence whenever $\bar{\gamma} \neq 0$, moving b slightly in the direction of $\text{sign}(\bar{\gamma})$ strictly reduces pooled training risk below the invariant predictor's.

Proof. The residual is $y - ax_{\text{inv}} - bx_{\text{sp}} = (\theta - a - b\gamma_e \theta)x_{\text{inv}} + (1 - b\gamma_e)\varepsilon - b\nu$ (substitute $x_{\text{sp}} = \gamma_e(\theta x_{\text{inv}} + \varepsilon) + \nu$); squaring and using independence and unit/known variances gives (40). Differentiate in b :

$$\frac{\partial L_e}{\partial b} = -2\gamma_e \theta (\theta - a - b\gamma_e \theta) - 2\gamma_e \sigma_\varepsilon^2 (1 - b\gamma_e) + 2b\sigma_\nu^2 \xrightarrow{(a,b)=(\theta,0)} 0 - 2\gamma_e \sigma_\varepsilon^2 + 0.$$

Averaging over the training episodes gives $-2\bar{\gamma} \sigma_\varepsilon^2$, nonzero iff $\bar{\gamma} \neq 0$. $\square$

Proposition 5.2 (Expected excess risk on a fresh episode). *For any fixed predictor $h_{a,b}$ and a fresh episode $e' \sim \mathcal{Q}$ (i.e. $\gamma_{e'}$ a fresh draw with mean 0 and variance σ_γ^2 , independent of the training episodes),*

$$\mathbb{E}_{\gamma_{e'}}[L_{e'}(h_{a,b})] - L_{e'}(h_{\theta,0}) = (\theta - a)^2 + b^2(\sigma_\nu^2 + s^2 \sigma_\gamma^2) \geq 0, \quad (41)$$

with equality iff $(a, b) = (\theta, 0)$; note $L_{e'}(h_{\theta,0}) = \sigma_\varepsilon^2$ deterministically. For the pooled-OLS solution (39), the leading term for small $|\bar{\gamma}|$ is

$$\hat{b}^2(\sigma_\nu^2 + s^2 \sigma_\gamma^2) = \frac{\bar{\gamma}^2 \sigma_\varepsilon^4 (\sigma_\nu^2 + s^2 \sigma_\gamma^2)}{D^2}, \quad \mathbb{E}[\bar{\gamma}^2] = \frac{\sigma_\gamma^2}{E}, \quad (42)$$

while $(\theta - \hat{a})^2 = O(\bar{\gamma}^4)$. The expected damage of pooling thus decays only as $1/E$ —negligible for hundreds of episodes, material for $E = 5-7$.

Proof. Take the expectation of (40) over $\gamma_{e'}$ with $\mathbb{E}\gamma_{e'} = 0$, $\mathbb{E}\gamma_{e'}^2 = \sigma_\gamma^2$:

$$\mathbb{E}[(\theta - a - b\gamma_{e'} \theta)^2] = (\theta - a)^2 + b^2 \theta^2 \sigma_\gamma^2, \quad \mathbb{E}[(1 - b\gamma_{e'})^2] \sigma_\varepsilon^2 = \sigma_\varepsilon^2 + b^2 \sigma_\varepsilon^2 \sigma_\gamma^2.$$

Summing with $b^2 \sigma_\nu^2$ and subtracting $L_{e'}(h_{\theta,0}) = \sigma_\varepsilon^2$ (immediate from (40) at $(\theta, 0)$) gives (41), using $\theta^2 + \sigma_\varepsilon^2 = s^2$. For the rate: $\hat{b} = \bar{\gamma} \sigma_\varepsilon^2 / D$ from Proposition 5.1, and $\theta - \hat{a} = \theta \bar{\gamma}^2 \sigma_\varepsilon^2 / D$ (subtract (39) from θ), whose square is $O(\bar{\gamma}^4)$. Finally $\mathbb{E}[\bar{\gamma}^2] = \text{Var}(\bar{\gamma}) = \sigma_\gamma^2 / E$ for i.i.d. mean-zero γ_{e_j} . $\square$

Example 5.1 (Numbers at niche scale). Take $\theta = 1$, $\sigma_\varepsilon^2 = 0.5$ (so $s^2 = 1.5$), $\sigma_\nu^2 = 1$, and $E = 6$ training episodes with $\gamma = (0.4, -0.2, 0.3, -0.1, 0.2, 0.0)$ —a sample from a mean-zero law that happens, as small samples do, to lean positive: $\bar{\gamma} = 0.1$, $\hat{m}_2 = 0.34/6 \approx 0.0567$, $\hat{\sigma}_\gamma^2 \approx 0.0467$. Then $D = 1 + 1.5 \times 0.0467 + 0.01 \times 0.5 \approx 1.075$ and

$$\hat{b} = \frac{0.1 \times 0.5}{1.075} \approx 0.047, \quad \hat{a} \approx 1 \times \frac{1.070}{1.075} \approx 0.995.$$

With true $\sigma_\gamma^2 = 0.05$, the expected fresh-episode excess (41) is approximately $0.047^2(1 + 1.5 \times 0.05) + 0.005^2 \approx 0.0023$ —about 0.5% of the irreducible risk $\sigma_\varepsilon^2 = 0.5$ per fresh episode, incurred systematically, with the sign of the error flipping adversarially on episodes where $\gamma_{e'}$ opposes $\bar{\gamma}$.

5.3 The Variance Statistic Detects Exactly the Loading

The decisive structural fact about the construction is visible in (40): the per-episode risk depends on the episode *only through the products* $b\gamma_e$. A predictor that ignores the spurious feature has the *same* risk in every episode; a predictor that uses it has episode-dependent risk. Across-episode variance is therefore a *detector* of spurious loading.

Proposition 5.3 (Variance identifies invariance). *Let $\text{Var}_e[\cdot]$ denote the empirical variance across the training episodes $e_1, \dots, e_E$, and suppose at least two of the γ_{e_j} are distinct. Then for the model (38):*

1. *If $b = 0$, then $L_{e_j}(h_{a,0}) = (\theta - a)^2 + \sigma_\varepsilon^2$ for every j , so $\text{Var}_e[L_e(h_{a,0})] = 0$.*
2. *If $b \neq 0$, then $\text{Var}_e[L_e(h_{a,b})] > 0$, except for at most finitely many exceptional values of (a, b) (those making the quadratic $\gamma \mapsto L(\gamma)$ in (40) take equal values at all the distinct observed γ_{e_j} , possible only when fewer than three distinct values are observed and (a, b) is tuned to the pair). With three or more distinct γ_{e_j} , $b \neq 0$ implies $\text{Var}_e[L_e] > 0$ without exception.*

Consequently the population of zero-variance predictors is exactly the $b = 0$ axis (generically), on which the pooled mean risk is minimized at the invariant predictor $a = \theta$: minimizing $\text{Var}_e[L_e] + \beta \mathbb{E}_e[L_e]$ with $\beta > 0$ and a sufficiently large variance weight recovers $h_{\theta,0}$, whereas pooled ERM (β -only) strictly prefers $\hat{b} \neq 0$ by Lemma 5.1.

Proof. (1) Set $b = 0$ in (40): every γ_e -dependent term carries a factor b , leaving $(\theta - a)^2 + \sigma_\varepsilon^2$, constant across episodes; the empirical variance of a constant list is 0.

(2) Regard (40) as a polynomial in γ :

$$L(\gamma) = \underbrace{b^2\theta^2\gamma^2 - 2b\theta(\theta - a)\gamma + (\theta - a)^2}_{\text{first term}} + \underbrace{\sigma_\varepsilon^2 b^2\gamma^2 - 2\sigma_\varepsilon^2 b\gamma + \sigma_\varepsilon^2 + b^2\sigma_\nu^2}_{\text{second term}},$$

with leading coefficient $b^2(\theta^2 + \sigma_\varepsilon^2) = b^2s^2 > 0$ when $b \neq 0$: a nondegenerate quadratic. The empirical variance vanishes iff $L(\gamma_{e_j})$ is the same number for all j . A nondegenerate quadratic takes any given value at most twice, so if three or more distinct γ_{e_j} are observed, equality of all values is impossible and $\text{Var}_e[L_e] > 0$. With exactly two distinct values $\gamma \neq \gamma'$, equality forces the quadratic's axis of symmetry to bisect them, i.e. pins one algebraic relation between a and b —the finitely-many-exceptions caveat. The final claims restate Lemma 5.1 and part (1) together with the strict positivity in part (2): for large variance weight, any $b \neq 0$ is dominated by its $b = 0$ projection, and among $b = 0$ predictors the mean term selects $a = \theta$. $\square$

Key Point

The variance statistic is not a generic regularizer; in this construction it is a *consistent test* for the precise pathology of Proposition 5.1. Pooled ERM cannot see the difference between fitting structure and fitting accidents, because both reduce mean training risk (Lemma 5.1); the across-episode variance sees *only* the accidents, because structure—by definition of being shared across the niche’s episodes—contributes nothing to it (Proposition 5.3). This is also exactly the quantity the small-sample transfer story of Section 2 wants minimized: a smaller across-episode spread is what would tighten an empirical-Bernstein-style certificate (Remark 2.2) and what shrinks the variance in Cantelli’s one-sided bound (Theorem 2.2).

5.4 Relation to IRM, REx, and Causal Invariance; the RISP Objective

Three strands of the invariance literature meet here. **Invariant causal prediction** [21] formalizes the target: under interventions that vary across environments, the conditional distribution of the response given its *causal parents* is invariant, and (under conditions) the causal feature set is identifiable as the one producing invariant residuals. In Definition 5.2, x_{inv} is a parent and x_{sp} a child of y ; the invariant predictor is the causal one. **IRM** [19] pursues this target through a bilevel program—learn a representation such that one classifier on top of it is simultaneously per-environment optimal—in practice relaxed to a gradient penalty $\sum_e \|\nabla_w|_{w=1} L_e(w \cdot h)\|^2$. **REx** [20] observes that a simpler scalarization often suffices and is better behaved: penalize the spread of per-environment risks directly, minimizing $\text{Var}_e[L_e(h)] + \beta \mathbb{E}_e[L_e(h)]$ -type objectives (V-REx); robustness to the spread of risks extrapolates to environments whose “accidents” are larger than any seen in training.

The RISP per-niche objective is the V-REx scalarization applied to a *decision-focused* episode risk. For niche r with stored episodes $\mathcal{E}_r$ and head parameters θ_r , define the SPO+ episode risk $R_e(\theta_r) = \frac{1}{|e|} \sum_{(x,y) \in e} \ell_{\text{SPO+}}(\hat{y}_{\theta_r}(x); y)$ and train by

$$J_r(\theta_r) = \text{Var}_{e \in \mathcal{E}_r}[R_e(\theta_r)] + \beta \mathbb{E}_{e \in \mathcal{E}_r}[R_e(\theta_r)], \quad \beta > 0. \quad (43)$$

Two substitutions distinguish (43) from textbook REx. First, the risk inside the variance is the SPO+ *decision* risk, so “invariance” is demanded of the quantity that matters (Section 1): a head may have episode-varying *forecast* error and still be exactly invariant in decision quality, and the objective correctly declines to punish it. Second, the environments are not given exogenously but are the episodes routed to the niche—the regime-indexed environment structure of the invariant predict-and-optimize formulation [6]—so the variance is computed within a (putative) regime, where invariance is actually achievable, rather than across all of history, where it is not.

Important

Variance alone ($\beta = 0$) is degenerate. The zero-variance set contains far more than invariant-and-good predictors: *any* head whose episode risks are constant achieves $J_r = 0$ at $\beta = 0$, including uniformly terrible ones. Concretely, a head whose forecasts induce the same fixed bad portfolio in all states has $R_e \equiv$ (the same poor value) on every episode—zero variance, maximal complacency; in the squared-loss model of Proposition 5.3, the entire plane $b = 0$ is zero-variance regardless of how wrong a is. The mean term is therefore not a tuning nicety but a load-bearing component: it is the only part of (43) that distinguishes invariant-good from invariant-bad. This degeneracy predicts, qualitatively and in advance, the $\beta = 0$ collapse observed in the companion experiments: post-reactivation decision regret 0.0356 for the variance-only objective versus 0.0321 for plain retained ERM—the ablation does not merely lose the invariance benefit, it underperforms doing nothing clever at all, exactly as a degenerate objective should.

Remark 5.1 (What carries over from squared loss to SPO+). The exact algebra of Propositions 5.1–5.3 is special to squared loss, where risks are quadratics in γ_e . What transfers to the SPO+ risk is the structural trichotomy, none of whose ingredients used quadraticity: (i) pooled ERM’s first-order preference for episode-specific signal scales with the *empirical mean* of the accident process and is therefore $\Theta(E^{-1/2})$ -sized at niche scale; (ii) any loading on episode-specific signal makes the per-episode risk *episode-dependent*, hence visible to the across-episode variance; (iii) zero variance plus low mean jointly—and only jointly—certify “invariantly good.” These three facts, with SPO+ risks in place of squared risks and with the boundedness R_e -analogues supplied by Proposition 1.1, are the form in which Part II uses this section.

This completes the toolkit. Part II assembles it: the decision layer and SPO+ machinery of Section 1 define what each niche head optimizes (43); the EG gate of Section 4 allocates episodes and training resources to niches, deliberately forgoing the adversarial guarantee (Section 4.5) in favor of analyzable replicator dynamics; and the retention and transfer theorems combine the concentration inequalities of Section 2 with the Pinsker bridge of Section 3 to state precisely when a dormant niche’s certified competence survives until its regime returns—the capability gap, identified in Remark 4.1, that the sleeping-experts literature leaves open and RISP is built to close.

Part II

The Environment: Regimes, Episodes, and Dormancy

The mechanism documented in Part III is an answer to a question posed entirely by the environment: regimes that arrive in persistent blocks (Section 6), episodes as the true statistical unit (Section 7), and the combinatorics of dormancy, during which a capacity-limited learner is quietly destroyed (Section 8). Every design choice in RISP is a response to a quantitative fact established in this part.

6 Regime-Switching Markets

6.1 The Regime Process

Definition 6.1 (Block-structured regime process). Fix a finite regime set $\mathcal{R} = \{1, \dots, R\}$. A **block-structured (semi-Markov) regime process** is a sequence of pairs

$$(\rho_1, \ell_1), (\rho_2, \ell_2), (\rho_3, \ell_3), \dots \quad \rho_b \in \mathcal{R}, \ell_b \in \{1, 2, \dots\},$$

where ρ_b is the regime of the b -th *block* and ℓ_b its duration in days, with $\rho_{b+1} \neq \rho_b$ (blocks are maximal). The day-level regime label r_t is the piecewise-constant expansion: writing $T_b = \sum_{b' < b} \ell_{b'}$ for the start of block b ,

$$r_t = \rho_b \quad \text{for } T_b < t \leq T_b + \ell_b. \quad (44)$$

The process is semi-Markov when ρ_{b+1} depends on the past only through ρ_b (a transition kernel on $\mathcal{R}$) while ℓ_b is drawn from a regime-specific duration distribution G_{ρ_b} on $\{1, 2, \dots\}$.

Hamilton’s Markov-switching model [22]—the canonical formalization of regimes in econometrics—is the special case in which r_t is a first-order Markov chain: block durations are then forced to be geometric, $\Pr(\ell = k) = P_{rr}^{k-1}(1 - P_{rr})$ with mean $1/(1 - P_{rr})$. The semi-Markov form frees the duration law G_r , which matters because the experimental stream uses bounded, non-geometric blocks (25–60 days common, 15–30 days crisis) and because real regime durations are empirically far from geometric [23, 24]. Nothing in Part III depends on the duration law beyond persistence itself.

The concrete instantiation used throughout: $R = 4$ regimes $\mathcal{R} = \{\text{trend, range, highvol, crisis}\}$; common regimes arrive in blocks of 25–60 days; crisis arrives in short blocks of 15–30 days and only on every *fourth* cycle through the common regimes, so consecutive crisis episodes are separated by roughly 250–450 days of dormancy; horizon $T \approx 2370$ days (≈ 9.5 trading years); each day carries features $X_t \in \mathbb{R}^{n \times d}$ ($n = 20$ assets, $d = 20$ features per asset) and realized returns $y_t \in \mathbb{R}^n$.

6.2 Why Block Persistence (Not I.I.D. Regime Draws) Is the Whole Game

It is tempting to model regime-switching as $r_t \stackrel{iid}{\sim} \pi$ for some distribution π on $\mathcal{R}$. That model destroys both of the phenomena this project is about.

Proposition 6.1 (Persistence creates local stationarity and dormancy). *Let r_t be a regime process over $\mathcal{R}$.*

1. **(I.i.d. draws have no usable present.)** *If $r_t \stackrel{iid}{\sim} \pi$, the conditional law of r_{t+1} given the entire past equals π : knowing today’s regime says nothing about tomorrow’s, every day is a fresh mixture draw, and the optimal single-day strategy is mixture-optimal, not any regime-specialist.*
2. **(I.i.d. draws have no dormancy.)** *Under i.i.d. draws, $\Pr(\text{regime } r \text{ dormant} > D \text{ days}) = (1 - \pi(r))^D$, exponentially small in D . Under the block process, dormancy is the sum of intervening block durations and is typically hundreds of days when r sits on a long cycle—as crisis does here (250–450 days by construction).*

Proof. (1) is immediate from independence. (2): with $\pi(\text{crisis}) \approx 1/13$ (one short crisis block per ≈ 12 common blocks), an i.i.d. stream would produce a 250-day crisis dormancy with probability $(1 - 1/13)^{250} \approx e^{-19.2} \approx 5 \times 10^{-9}$ —essentially never—whereas the block schedule produces one every cycle. The block construction makes long dormancy the rule rather than a large-deviation event. $\square$

Persistence is what makes specialization purchasable

Persistence does two jobs at once. *Within* a block, the world is locally stationary for 25–60 days, long enough for a specialist to be trained, evaluated, and trusted—this is what makes a regime a *niche* worth owning. *Between* blocks of the same regime, the world is silent about that regime for weeks to years—this is what makes memory a problem. Under i.i.d. draws neither holds: there is nothing to specialize in (job one fails) and nothing is ever dormant (job two never arises). Every result in this document lives in the gap between those two timescales.

6.3 The Causal Labelers

All eleven arms of the experiment receive the regime label $\hat{r}_t$ from a shared **causal labeler**: a deterministic function of information available *strictly before* the decision at day t is made. Two labelers are used, deliberately spanning a simple/structured spectrum, so that no conclusion depends on one labeling artifact.

Labeler L1: rolling-percentile volatility band $\times$ trend sign. At day t , using returns $y_1, \dots, y_{t-1}$ only:

1. **Trailing volatility.** Realized volatility $v_t = \text{sd}(\bar{y}_{t-w_v}, \dots, \bar{y}_{t-1})$ over a trailing window of w_v days, $\bar{y}_s$ the cross-sectional mean return on day s .
2. **Rolling percentile thresholds.** The empirical q^{high} - and q^{crisis} -quantiles $\tau_t^{\text{high}}, \tau_t^{\text{crisis}}$ of the trailing history $\{v_s : s \leq t-1\}$ —quantiles of *past* volatilities only, re-estimated daily from the filtration at $t-1$, never from the full sample.
3. **Trend statistic.** A trailing trend signal m_t (sign and magnitude of the mean of $\bar{y}_{t-w_m}, \dots, \bar{y}_{t-1}$, or a fast-minus-slow moving-average gap), again from days $\leq t-1$.
4. **Label.**

$$\hat{r}_t = \begin{cases} \text{crisis} & v_t \geq \tau_t^{\text{crisis}}, \\ \text{highvol} & \tau_t^{\text{high}} \leq v_t < \tau_t^{\text{crisis}}, \\ \text{trend} & v_t < \tau_t^{\text{high}} \text{ and } |m_t| \geq \tau^m, \\ \text{range} & v_t < \tau_t^{\text{high}} \text{ and } |m_t| < \tau^m. \end{cases}$$

Labeler L2: forward-only filtered two-state volatility model $\times$ trend sign. L2 replaces the percentile band with a probabilistic state model:

1. Maintain a two-state Markov-switching volatility model (low/high-vol states with state-dependent return variance), parameters estimated on an expanding window through day $t-1$ and refit periodically from past data only.
2. Run the **Hamilton filter** [22] *forward only* for the one-step-ahead predictive probability $p_t = \Pr(s_t = \text{high-vol} \mid \bar{y}_1, \dots, \bar{y}_{t-1})$. The Kim *smoothed* probability, which conditions on the future, is never computed.
3. Label crisis when p_t exceeds a high threshold and trailing volatility is extreme, highvol when p_t exceeds a moderate threshold, otherwise trend/range by the same trend statistic m_t as in L1.

Leakage analysis. Write $\mathcal{F}_{t-1} = \sigma(X_1, y_1, \dots, X_{t-1}, y_{t-1}, X_t)$ for the information legitimately available when the day- t decision is placed (today’s features are observable; today’s returns are not). Both labelers satisfy the **causality contract**

$$\hat{r}_t \in \mathcal{F}_{t-1} \quad (\text{in fact } \hat{r}_t \in \sigma(y_1, \dots, y_{t-1})), \quad (45)$$

because every ingredient—trailing volatility, rolling quantiles, trend statistics, filter recursions, parameter refits—is a function of returns through $t - 1$. Three subtler channels are also closed: (i) *contemporaneous leak*: the label for day t never uses y_t itself (a one-day embargo between the last return used and the day labeled); (ii) *refit leak*: parameters and thresholds are re-estimated only on data through $t - 1$, so no full-sample estimate back-propagates into early labels; (iii) *smoothing leak*: L2 uses filtered, never smoothed, probabilities—the smoother is two-sided, and a single smoothed label can encode information from arbitrarily far in the future.

Label leakage: the classic regime-paper sin

A large fraction of the regime-switching literature evaluates strategies against regime labels that *could not have been known at the time*. The recurring offenders:

- **Smoothed state probabilities:** using Kim-smoothed $\Pr(s_t \mid y_{1:T})$ as “the regime at t ”—the smoother conditions on the future and labels turning points in advance.
- **Full-sample thresholds:** defining *highvol* as the top volatility quintile *of the whole sample*—the 1990 boundary then depends on 2008.
- **Peak/trough dating:** NBER-style bull/bear datings require future data to *confirm* a peak; using the dated label at the peak itself is hindsight.
- **Full-sample parameters with filtered probabilities:** the leak hides in $\hat{\theta}$, not in the recursion.

Each inflates the apparent value of regime-conditioning for the same reason published return predictors decay out of sample [25]: the evaluation quietly conditions on information the strategy never had. Everything downstream—specialist training, competition windows, pinning—consumes only the causal label $\hat{r}_t \in \mathcal{F}_{t-1}$ of (45); “reactivation” means the *causal labeler* says so, with whatever lag and misclassification that entails.

Remark 6.1 (The label is an input, not a contribution). RISP takes $\hat{r}_t$ as given and identical across all arms: the experiment isolates *what to do with* a regime signal (allocation, training, memory), not how to detect regimes—a separate, well-studied problem [26, 27]. A shared causal label removes regime-detection skill as a confound.

7 Episodes as Environments

7.1 Episodes

Definition 7.1 (Episode). For regime $r \in \mathcal{R}$, the e -th episode of r , written (r, e) , is the e -th maximal block of consecutive days on which $\hat{r}_t = r$, i.e. the e -th element of the subsequence of blocks $\{b : \rho_b = r\}$ in Definition 6.1. We write $\mathcal{T}_{r,e} \subset \{1, \dots, T\}$ for its day set and $|\mathcal{T}_{r,e}| = \ell_{r,e}$ for its length.

The central modeling move—inherited from invariant-prediction thinking and from the decision-focused learning literature [1, 6]—is to treat each episode as a distinct *environment*: the regime fixes what is stable, the episode index fixes what is incidental.

Definition 7.2 (Hyper-distribution assumption). Each regime r carries a **hyper-distribution** Q_r over environment parameters. Episode (r, e) draws its parameter $\gamma_{r,e} \sim Q_r$ independently across e , and the days within the episode are then i.i.d. given $(r, \gamma_{r,e})$:

$$\gamma_{r,e} \stackrel{iid}{\sim} Q_r, \quad (X_t, y_t) \mid (r, \gamma_{r,e}) \stackrel{iid}{\sim} P_{r,\gamma_{r,e}} \quad \text{for } t \in \mathcal{T}_{r,e}.$$

The **stable component** of regime r is whatever is shared by $P_{r,\gamma}$ across all γ in the support of Q_r ; the **episode-specific component** is the rest.

Remark 7.1 (Two honest idealizations). Definition 7.2 idealizes reality in two ways. (i) *Macro memory*. Real episodes of the same regime are not independent draws: the 2008 and 2011 crises share causal machinery and participants who remember the previous one. Positive dependence across e means the effective number of independent episodes is *smaller* than the count—scarcity (Section 7.2) is, if anything, understated. (ii) *Secular drift*. Q_r is treated as fixed over the horizon, but markets drift on decadal scales: predictors decay after publication [25], and the “stable” component of a regime is stable only relative to the horizon. The synthetic generator below satisfies both idealizations exactly, so failures of an arm on synthetic data cannot be blamed on them; transfer to long real panels inherits both caveats.

7.2 Episode Scarcity and Its Two Mitigations

The statistical currency of Definition 7.2 is the number of *episodes* E_r , not the number of days: any quantity that varies at the episode level (an episode-mean loss, an episode-specific coefficient) is estimated with variance $\propto 1/E_r$, no matter how many days each episode contains. For common regimes E_r is comfortable (dozens of 25–60-day blocks over $T \approx 2370$ days). For crisis it is brutal: in roughly 35 years of equity data one finds on the order of **5–7** crisis episodes, and fewer in shorter panels; in the synthetic stream, the every-fourth-cycle schedule produces a comparable handful over $T \approx 2370$ days. Two standard mitigations trade scarcity for dependence, and it is important to state exactly what dependence each one buys.

Definition 7.3 (Sub-episode blocks). Fix a block length B . The **sub-episode blocks** of episode (r, e) are the contiguous segments obtained by partitioning $\mathcal{T}_{r,e}$ into pieces of length B (discarding or merging a short remainder). Each block is treated as a pseudo-episode in episode-level statistics.

Sub-episode blocks multiply the apparent environment count by $\approx \ell_{r,e}/B$, but all blocks of one episode *share the same draw* $\gamma_{r,e}$, so block mean losses are positively correlated through γ , and cross-block variance *underestimates* cross-episode variance: the blocks inform about within-episode noise, not about Q_r . They are a variance mitigation for the stable component only; they cannot manufacture information about episode heterogeneity the panel does not contain.

Definition 7.4 (Cross-asset episodes). When the same regime episode is observed simultaneously on A assets or markets (here $n = 20$ assets share each episode), the per-asset day records $\{(x_{t,i}, y_{t,i}) : t \in \mathcal{T}_{r,e}\}$ for $i = 1, \dots, A$ may be treated as A parallel realizations of the episode environment.

Cross-asset episodes multiply the day count per episode by A , but introduce *cross-sectional dependence*: assets within a day share common shocks (market factors, and the episode draw $\gamma_{r,e}$ itself, common across assets in the generator below), so the effective sample size per day lies between 1 and A . Again: more data about the within-episode law, no new draws from Q_r .

Key Point

Both mitigations enlarge the denominator of *within-episode* estimates and leave the number of independent draws from Q_r untouched. Whatever must be learned *about episode heterogeneity*—in particular the variance-across-episodes term the INV objective penalizes—must come from E_r genuine episodes, and for crisis E_r is single-digit. This one fact sizes the episode library (≤ 6 retained episodes suffices, because there are never many more) and explains the crisis-pinning wrinkle of Section 10.

7.3 The Synthetic Generator: Heterogeneity Made Explicit

The synthetic stream instantiates Definition 7.2 exactly. For day t in episode (r, e) and asset i with feature vector $x_{t,i} \in \mathbb{R}^{20}$:

$$y_{t,i} = \underbrace{\theta_r \cdot (x_{t,i,1}, x_{t,i,2})}_{\text{stable, regime-level}} + \underbrace{\gamma_{r,e} x_{t,i,\text{sp}}}_{\text{episode-specific}} + b_r + \varepsilon_{t,i}, \quad (46)$$

with the following calibration:

- **Stable signal.** $\theta_r \in \mathbb{R}^2$, $\|\theta_r\| = 0.012$, directions well-separated across regimes—so a head trained on one regime is wrong (not merely stale) on another.
- **Episode-specific signal.** $\gamma_{r,e} \sim \mathcal{N}(0, (0.012 \cdot \text{het})^2)$, *redrawn independently each episode*, loading on a designated spurious feature x_{sp} ; $\text{het} \geq 0$ is the heterogeneity knob.
- **Distractors.** 16 of the 20 features carry no signal in any regime: overfitting bait at $n-d$ parity ($n = d = 20$).
- **Noise.** $\varepsilon_{t,i} \sim \mathcal{N}(0, \sigma_r^2)$, $\sigma_r = 0.02 \cdot (1, 0.75, 2, 3)$ for (trend, range, highvol, crisis): the rare regime is also the noisiest, by a factor of three.

Example 7.1 (What $\text{het} = 0$ versus $\text{het} = 2$ means). The standard deviation of the episode-specific coefficient is $\text{sd}(\gamma_{r,e}) = 0.012 \cdot \text{het}$, against a stable coefficient magnitude of $\|\theta_r\| = 0.012$. Concretely:

- **het = 0:** $\gamma_{r,e} \equiv 0$. Every episode of regime r has the *same* law; the spurious feature is a 17th distractor; Q_r is a point mass; plain ERM on pooled regime data is right, and a variance-across-episodes penalty has nothing to penalize.
- **het = 1:** $\text{sd}(\gamma) = 0.012 = \|\theta_r\|$. The spurious coefficient is at least as large as the stable one whenever $|\gamma_{r,e}| \geq 0.012$, probability $\Pr(|Z| \geq 1) \approx 0.317$ for $Z \sim \mathcal{N}(0, 1)$: in roughly one episode in three, the most profitable in-episode feature is the one that betrays you next episode.
- **het = 2:** $\text{sd}(\gamma) = 0.024 = 2\|\theta_r\|$. Now $|\gamma_{r,e}| \geq \|\theta_r\|$ with probability $\Pr(|Z| \geq 0.5) \approx 0.617$: in the *majority* of episodes the episode-specific signal dominates inside the episode. An in-episode ERM fit loads heavily on x_{sp} —correctly, for that episode—then carries a coefficient of typical size 0.024 with a freshly randomized sign into the next episode.

The probe window (the first 15 days after a reactivation, Definition 8.2) is precisely where this bites: the learner meets a new draw of $\gamma_{r,e}$ before any in-episode data has accumulated, so whatever γ -chasing is baked into its head is realized as pure decision regret.

The generator is a forgetting–overfitting double trap

Equation (46) arms two independent failure modes. The *regime* structure (separated θ_r , long dormancy) punishes arms that forget; the *episode* structure ($\gamma_{r,e}$ redrawn) punishes arms that remember too literally—memorize the last crisis episode including its γ and you import a coin-flip coefficient into the next one. In the eleven arms of Section 11, the allocation axis addresses the first trap, the objective axis (ERM vs. INV) the second.

8 Dormancy Combinatorics and Memory Models

8.1 Dormancy and the Probe Window

Definition 8.1 (Dormancy). The **dormancy** of regime r before episode $e + 1$ is the gap

$$D_r(e) = \min \mathcal{T}_{r,e+1} - \max \mathcal{T}_{r,e} - 1,$$

the number of days between the end of episode (r, e) and the start of $(r, e + 1)$ during which other regimes are active. In the synthetic stream, common regimes have dormancies on the order of one cycle (tens of days), while *crisis* has $D \approx 250$ –450 days.

Definition 8.2 (Probe window and forgetting deficit). A **reactivation** of r is the first day of an episode $(r, e + 1)$ with dormancy $D_r(e) \geq 90$ days. The **probe window** is the first 15 days of that episode. For an arm A , let $\bar{\rho}_{\text{probe}}(A)$ denote its mean per-day decision regret (Definition 9.2) over probe-window days. The **forgetting deficit** of arm A at dormancy D is

$$\Phi(D; A) = \mathbb{E}[\bar{\rho}_{\text{probe}}(A) \mid \text{dormancy } D] - \mathbb{E}[\bar{\rho}_{\text{probe}}(A^\circ)], \quad (47)$$

where A° is the *retention counterfactual*: the same arm serving the probe window with the head (and episode library) it held at the close of the previous episode of r , untouched by any intervening foreign training. $\Phi \approx 0$ means dormancy cost the arm nothing it had; Φ large means the dormancy machinery below did its damage.

The probe window is 15 days: long enough to average regime noise ($\sigma_{\text{crisis}} = 0.06$ per day) into a stable regret estimate, short enough that within-episode relearning cannot mask a cold start—it measures what you walked in with. All headline numbers in this document (0.0303–0.0320) are means of $\bar{\rho}_{\text{probe}}$ across reactivations and 20 seeds.

8.2 The Foreign-Update Process

During r ’s dormancy, the stream keeps producing days from the W_a *other* active regimes, and the holder of r ’s head may be made to *train on a foreign regime*. We abstract this as the **foreign-update process**: each dormancy day is a foreign-update day with probability p , where p encodes the arm: $p = 1$ for a monolith that trains daily on whatever is active; $p \approx$ (win rate) for an unpinned RISP specialist; $p = 0$ for a pinned owner whose regime is dormant. The damage per foreign update is what the two memory models specify.

Definition 8.3 (Hard and soft memory models). Each specialist has capacity $K < R$: at most K regime heads (linear SPO+ heads, Section 9), each with an episode buffer of up to 6 episodes $\times$ 40 days.

- **HARD (LRU eviction)**. Heads occupy discrete slots ordered by recency of training. Training on a held regime refreshes its recency. Training on an *unheld* regime at full capacity instantiates a new head and **evicts** the least-recently-used head *together with its episode buffer*—competence and memory are destroyed jointly and completely.

- **SOFT (interference decay).** No head is ever evicted. Instead, every foreign update multiplies *all other held heads* by $(1 - \rho_{\text{int}})$ with $\rho_{\text{int}} = 0.06$: after N foreign updates a dormant head retains strength $(1 - \rho_{\text{int}})^N$ of its trained weights.

The hard model is the cache-eviction picture; the soft model is the gradient-interference picture of catastrophic forgetting in neural networks [28, 29, 30], where training on task B perturbs the very weights encoding task A . Progressive networks [31] avoid interference by freezing columns—a structural cousin of what pinning will achieve in Section 10.

8.3 Hard Model: The Coupon-Collector Clock

Under HARD, a dormant head dies when enough *distinct* foreign regimes have been learned to push it out of the LRU order. The clean question: a learner holds the dormant head and has K slots; how long until it has trained on K distinct foreign regimes (each claiming a slot, the K -th overflowing capacity and evicting the dormant head, which—having received no training all dormancy—is always the least recently used)?

Proposition 8.1 (Expected time to eviction). *Suppose each dormancy day is independently a foreign-update day with probability p , and on an update day the active regime is uniform on the W_a foreign regimes, independently across update days. Let T_K be the number of days until K distinct foreign regimes have been trained on ($K \leq W_a$). Then*

$$\mathbb{E}[T_K] = \frac{1}{p} \sum_{j=0}^{K-1} \frac{W_a}{W_a - j}. \quad (48)$$

Proof. Stage j begins when exactly j distinct foreign regimes have been collected, $0 \leq j \leq K - 1$. On each day of stage j , the event “collect a new regime today” requires an update day (probability p) on which the active regime is one of the $W_a - j$ uncollected ones (probability $(W_a - j)/W_a$, by uniformity and independence). The stage- j waiting time is therefore geometric with success probability $p(W_a - j)/W_a$ and expectation $W_a/(p(W_a - j))$. The stages are traversed in order, so by linearity of expectation $\mathbb{E}[T_K] = \sum_{j=0}^{K-1} W_a/(p(W_a - j))$, which is (48). $\square$

Example 8.1 (Worked numbers: $W_a = 3$, $K = 2$, $p = \frac{1}{2}$). A learner holds the dormant crisis head with capacity $K = 2$; the $W_a = 3$ common regimes are active; it trains on the active regime every other day on average. By (48),

$$\mathbb{E}[T_2] = \frac{1}{0.5} \left(\frac{3}{3} + \frac{3}{2} \right) = 2 \times 2.5 = 5 \text{ days} :$$

$1/p = 2$ expected days for any foreign update to claim the spare slot, plus $3/(0.5 \times 2) = 3$ for an update from one of the remaining two regimes. Five days—against a crisis dormancy of 250–450 days: the dormant head’s expected lifetime is roughly 1% of the dormancy it must survive.

How certain is eviction by day D ? For the worked example we can compute the survival probability exactly.

Proposition 8.2 (Survival probability, exact and bounded). *In the setting of Proposition 8.1 with $W_a = 3$, $K = 2$:*

$$\Pr(\text{head survives } D \text{ days}) = 3 \left(1 - \frac{2p}{3} \right)^D - 2(1 - p)^D. \quad (49)$$

In general ($K \leq W_a$), a union bound over which $K - 1$ regimes were the only ones updated gives the binomial-tail bound

$$\Pr(\text{survive } D) \leq \binom{W_a}{K-1} \left(1 - p \frac{W_a - K + 1}{W_a} \right)^D.$$

Proof. Survival through day D means at most $K - 1 = 1$ distinct regime was trained on. Condition on $N_D \sim \text{Bin}(D, p)$ update days: given $N_D = m \geq 1$, all m uniform draws land on a single regime with probability $3 \cdot (1/3)^m$; given $m = 0$, survival is certain. Hence

$$\Pr(\text{survive}) = (1-p)^D + \sum_{m \geq 1} \binom{D}{m} p^m (1-p)^{D-m} \cdot 3 \cdot 3^{-m} = (1-p)^D + 3 \left[(1-p+\frac{p}{3})^D - (1-p)^D \right],$$

which is (49). General bound: survival requires the updated regimes to lie in some $(K - 1)$ -subset S ; for fixed S , each day avoids “update outside S ” with probability $1 - p(W_a - K + 1)/W_a$; union over the $\binom{W_a}{K-1}$ subsets (this matches the leading term of (49) at $W_a = 3$, $K = 2$). $\square$

Example 8.2 (Eviction probabilities by dormancy length). With $p = 0.5$, evaluating (49) (note $1 - 2p/3 = 2/3$, $1 - p = 1/2$):

D (days)	$3(2/3)^D$	$2(1/2)^D$	$\Pr(\text{evicted by } D)$
5	0.39506	0.06250	$1 - 0.33256 = 0.667$
10	0.05202	0.00195	$1 - 0.05007 = 0.950$
20	0.00090	1.9×10^{-6}	$1 - 0.00090 = 0.9991$
40	2.7×10^{-7}	1.8×10^{-12}	$1 - 2.7 \times 10^{-7}$

By $D = 20$ days—half of one common-regime block—eviction is a 99.9% event; at a crisis dormancy of $D = 250$ survival is $3(2/3)^{250} \approx 3 \times 10^{-44}$. Under the hard model an unprotected crisis head *does not survive dormancy*, full stop. The i.i.d.-day idealization is conservative here: the real stream arrives in persistent blocks, each delivering many updates from one regime, so distinct-regime collection is at least as fast at the block scale.

Example 8.3 (LRU mechanics, step by step). A capacity- $K=2$ learner ends a crisis episode holding heads for $\{\text{crisis}, \text{trend}\}$, with **crisis** most recent. The dormancy stream then activates **trend**, **range**, **highvol**, ... and the learner trains on whatever is active. Slot contents (ordered LRU $\rightarrow$ MRU; each head carries its episode buffer):

Event	Active regime	Action	Slots after (LRU $\rightarrow$ MRU)
1	trend	hit: refresh recency	[crisis, trend]
2	range	miss at full capacity: evict crisis + buffer	[trend, range]
3	crisis reactivates	miss: cold start , evict trend	[range, crisis ^{new}]

At event 2—the *first* foreign block from an unheld regime—the crisis head and its entire episode library are gone. At event 3 the probe window is served by a freshly initialized head: full cold-start regret, then a 15–30-day scramble to relearn what was once known, and the cycle repeats.

8.4 Soft Model: Geometric Erosion and the Half-Life

Under SOFT nothing is evicted; instead each foreign update multiplies every other held head by $(1 - \rho_{\text{int}})$ with $\rho_{\text{int}} = 0.06$. After N_D foreign updates during a dormancy, the dormant head retains

$$\text{retained strength} = (1 - \rho_{\text{int}})^{N_D} = 0.94^{N_D}, \quad (50)$$

of its trained weights (the head decays toward initialization; it is never refreshed because its regime emits no training days). The natural summary is the **half-life in foreign updates**: the N solving $(1 - \rho)^N = \frac{1}{2}$,

$$N_{1/2} = \frac{\log(1/2)}{\log(1 - \rho_{\text{int}})} = \frac{-0.693147}{\log(0.94)} = \frac{0.693147}{0.061875} \approx 11.2 \text{ foreign updates}, \quad (51)$$

using $\log(0.94) = -0.061875$. Eleven foreign updates—about two competition windows’ worth of foreign training, or a week and a half of daily monolith training—halve the head. The full erosion schedule:

Foreign updates N	11	12	25	50	100	250
Retained 0.94^N	0.506	0.476	0.213	0.045	0.0021	1.9×10^{-7}

For a monolith that trains every day ($N_D = D$), a crisis dormancy of 250–450 days leaves $0.94^{250} \approx 2 \times 10^{-7}$ of the head: a cold start without a discrete eviction event. A learner suffering one foreign update per 20-day window over the same dormancy takes $N_D \approx 12$ –22 updates, retention 0.48–0.25: degraded, not destroyed. The soft model grades the damage by the *training cadence* of the allocation rule—exactly the quantity the eleven arms vary.

Two models, one phenomenon, bounded from both ends

HARD and SOFT bracket the same constraint—finite capacity under foreign training—from opposite extremes. HARD is the worst case per event: one overflow update destroys a head completely, at a stopping time whose law Propositions 8.1–8.2 pin down. SOFT is the gentlest plausible case: no single update matters, yet the geometric decay (50) reaches the same endpoint at the dormancies in play ($0.94^{250} \approx 10^{-7}$). A conclusion that holds under *both* is about bounded capacity itself, not a bookkeeping rule; and a mechanism that protects a head must both prevent overflow updates (HARD) and drive the foreign-update count to zero (SOFT). Pinning (Section 10) does both at once: after a covering one-per-regime assignment, each specialist’s foreign-update probability is structurally $p = 0$.

Remark 8.1 (Why Φ is defined against the retention counterfactual). The forgetting deficit (47) subtracts the regret of the arm’s *own frozen past self*, not of an oracle, isolating dormancy damage from competence: a mediocre head retained perfectly has $\Phi = 0$ (a skill problem, not a memory problem); an excellent head lost to eviction has large Φ even at moderate probe regret. The eleven-arm comparison of Section 11 needs both coordinates—probe regret for the bottom line, Φ for the diagnosis.

Part III

The RISP Mechanism

Part II established the problem: regimes recur after dormancies that no capacity-limited, reward-chasing learner survives, while episode heterogeneity punishes whoever survives by memorizing too literally. This part builds the mechanism that answers it: specialists (Section 9), windowed competition with multiplicative affinity updates and an absorbing pin (Section 10), and the placement of RISP among ten alternatives in a two-axis design space (Section 11).

9 The Specialist

9.1 State: Identity, Competence, Memory

Definition 9.1 (Specialist state). The population contains $N = 4$ specialists. Specialist i at day t carries three coupled state components:

1. **Identity: niche affinity** $\alpha_i \in \Delta^R$, initialized uniform ($\alpha_{i,r} = 1/R = 0.25$): the specialist’s *claim* on each regime, updated only by winning competition windows (Section 10); it controls the incumbency bonus and, ultimately, pinning.

2. **Competence: regime heads** $\{w_{i,r} : r \in H_i(t)\}$, one linear SPO+ head per *held* regime, $|H_i(t)| \leq K < R$, mapping features to predicted returns $\hat{y}_{t,i'} = w_{i,r} \cdot x_{t,i'}$ and thence to a decision via the optimizer (Definition 9.2).
3. **Memory: episode library** $\{\mathcal{B}_{i,r} : r \in H_i(t)\}$, at most 6 episodes $\times$ 40 days per held regime of (X_t, y_t) records keyed by episode—created with the head and **evicted with the head** (hard model); the raw material for both training variants below.

The three components are deliberately welded together: affinity determines where the specialist wins; winning triggers training, which builds the head and writes the buffer; head and buffer determine future window performance, which feeds back into affinity. Identity, competence, and memory rise together and (under eviction) fall together.

Definition 9.2 (Decision regret). Fix a compact feasible decision set $\mathcal{A} \subset \mathbb{R}^n$ (a polytope of admissible portfolios over the $n = 20$ assets), and let $a^*(v) \in \arg \max_{a \in \mathcal{A}} v^\top a$ with optimal value $z^*(v) = \max_{a \in \mathcal{A}} v^\top a$. The **per-day decision regret** of a prediction $\hat{y}_t$ against realized returns y_t is

$$\rho(\hat{y}_t, y_t) = z^*(y_t) - y_t^\top a^*(\hat{y}_t) \geq 0 : \quad (52)$$

the return foregone by acting on the prediction instead of on the truth. Regret depends on $\hat{y}_t$ only through the *decision* it induces—two predictions with different numbers but the same induced portfolio have identical regret. This is the decision-focused viewpoint of Elmachetoub and Grigas [1]: the loss is downstream decision quality, not prediction error.

9.2 One Day in the Life

Each day t , the causal labeler emits $\hat{r}_t$ (visible to everyone), and the system follows one of two paths.

Pinned case ($\hat{r}_t$ pinned to owner $o(\hat{r}_t)$, Definition 10.2):

1. The owner alone serves the live decision $a_t = a^*(\hat{y}_t^{(o)})$ using $w_{o,\hat{r}_t}$.
2. The day’s record (X_t, y_t) is appended to the owner’s buffer $\mathcal{B}_{o,\hat{r}_t}$ (current episode, capped at 40 days; oldest of the ≤ 6 episodes rotated out when a new one opens).
3. On its training cadence, the owner takes a training step (Section 9.3) on $\mathcal{B}_{o,\hat{r}_t}$. *No other specialist decides, trains, or updates anything.*

Unpinned case:

1. **Shadow decisions.** Every specialist i forms $\hat{y}_t^{(i)}$ (with $w_{i,\hat{r}_t}$ if held; with the zero-initialized default head, hence a default decision, if not) and its shadow decision $a_t^{(i)} = a^*(\hat{y}_t^{(i)})$. The live decision is served by the current best claimant (highest $\alpha_{i,\hat{r}_t}$).
2. **Regret accounting.** Each specialist’s realized regret $\rho_t^{(i)} = \rho(\hat{y}_t^{(i)}, y_t)$ is appended to the open competition window for regime $\hat{r}_t$. Windows are regime-gated: only days with $\hat{r}_t = r$ advance r ’s window, so a window spans regime interruptions without contamination.
3. **Window close.** When r ’s window reaches $W_c = 20$ in-regime days, a winner is declared (Section 10), the winner’s affinity takes one exponentiated-gradient step, and *the winner alone* trains on the window’s data (appended to its buffer for r ; if it holds no head for r , one is instantiated now—the moment HARD eviction or SOFT interference strikes its *other* heads).
4. **Pin check.** If $\alpha_{i^*,r} \geq 0.95$, regime r is pinned to i^* permanently.

Why losers do not train

The most consequential line in the loop is the quiet one: only the window winner trains. Training is simultaneously the reward (the winner improves at r) and the cost-bearer (training on r is a foreign update against every other head the trainer holds). A specialist that keeps losing on r never pays r 's interference cost—its own niche's head sits at $p = 0$ in the foreign-update process of Section 8. Competition is not just selecting who improves; it is rationing who gets *damaged*.

9.3 The SPO+ Training Step

Training must improve *decisions*, so the loss is a decision-aware surrogate: regret (52) itself is piecewise constant in $\hat{y}$ (it changes only when the induced $\arg \max$ jumps), hence useless for gradients. The SPO+ surrogate of Elmachoub and Grigas [1], written here in the return-maximization convention, is

$$\ell_{\text{SPO}+}(\hat{y}, y) = \max_{a \in \mathcal{A}} \{(2\hat{y} - y)^\top a\} - 2\hat{y}^\top a^*(y) + z^*(y). \quad (53)$$

Lemma 9.1 (SPO+ is a convex upper bound on regret). *$\ell_{\text{SPO}+}(\cdot, y)$ is convex in $\hat{y}$, satisfies $\ell_{\text{SPO}+}(y, y) = 0$, and $\ell_{\text{SPO}+}(\hat{y}, y) \geq \rho(\hat{y}, y)$ for all $\hat{y}$. A subgradient in $\hat{y}$ is*

$$g(\hat{y}, y) = 2(a^*(2\hat{y} - y) - a^*(y)). \quad (54)$$

Proof. Convexity: the first term of (53) is a maximum of affine functions of $\hat{y}$, the rest linear or constant. At $\hat{y} = y$ the three terms are $z^*(y) - 2z^*(y) + z^*(y) = 0$. Upper bound: evaluating the max at the feasible point $a^*(\hat{y})$,

$$\ell_{\text{SPO}+}(\hat{y}, y) \geq 2\hat{y}^\top (a^*(\hat{y}) - a^*(y)) + (z^*(y) - y^\top a^*(\hat{y})),$$

where the first bracket is ≥ 0 because $a^*(\hat{y})$ maximizes $\hat{y}^\top a$, and the second is exactly $\rho(\hat{y}, y)$. Subgradient: Danskin's theorem on the max term gives $2a^*(2\hat{y} - y)$; the linear term contributes $-2a^*(y)$. $\square$

With the linear head $\hat{y}_{t,i} = w^\top x_{t,i}$, the chain rule turns (54) into a gradient in w : $\nabla_w \ell_{\text{SPO}+}(\hat{y}_t, y_t) = \sum_{i=1}^n g_i(\hat{y}_t, y_t) x_{t,i}$.

Example 9.1 (SPO+ arithmetic on two assets). Let $\mathcal{A} = \{e_1, e_2\}$, truth $y = (0.01, -0.01)$, prediction $\hat{y} = (-0.002, 0.004)$ (ranking inverted). Then $a^*(y) = e_1$, $z^*(y) = 0.01$, $a^*(\hat{y}) = e_2$, so $\rho = 0.01 - (-0.01) = 0.02$. For SPO+: $2\hat{y} - y = (-0.014, 0.018)$, maximized by e_2 at 0.018; the linear term is $-2\hat{y}^\top e_1 = 0.004$; hence $\ell_{\text{SPO}+} = 0.018 + 0.004 + 0.01 = 0.032 \geq \rho = 0.02$. The subgradient (54) is $g = 2(e_2 - e_1) = (-2, 2)$: descent pushes $\hat{y}_1$ up and $\hat{y}_2$ down—directly toward repairing the ranking that caused the bad decision, not toward shrinking prediction error per se.

9.4 ERM versus INV over the Episode Library

A training step consumes the episode buffer $\mathcal{B}_{i,r} = \{B_1, \dots, B_E\}$, $E \leq 6$ episodes. Write

$$\bar{\ell}_e(w) = \frac{1}{|B_e|} \sum_{t \in B_e} \ell_{\text{SPO}+}(\hat{y}_t(w), y_t), \quad \bar{\ell}(w) = \frac{1}{E} \sum_{e=1}^E \bar{\ell}_e(w)$$

for the per-episode and pooled mean surrogate losses. The two variants:

ERM variant. Sample one episode $e \sim \text{Unif}\{1..E\}$ and take a subgradient step on $\bar{\ell}_e(w)$. In expectation this descends the pooled objective $\bar{\ell}(w)$: all episode-days are equal, and the head converges toward the buffer-pooled SPO+ minimizer—including whatever episode-specific γ structure dominates the buffer.

INV variant. Descend the variance-penalized objective

$$J(w) = \text{Var}_e[\bar{\ell}_e(w)] + \beta \mathbb{E}_e[\bar{\ell}_e(w)] = \frac{1}{E} \sum_{e=1}^E (\bar{\ell}_e(w) - \bar{\ell}(w))^2 + \beta \bar{\ell}(w), \quad (55)$$

which demands not just low mean loss but *evenly* low loss across episodes—the invariance principle transplanted into decision-focused learning [6].

Proposition 9.1 (INV gradient). *The gradient of (55) is*

$$\nabla_w J(w) = \frac{1}{E} \sum_{e=1}^E \left[2(\bar{\ell}_e(w) - \bar{\ell}(w)) + \beta \right] \nabla_w \bar{\ell}_e(w). \quad (56)$$

Proof. Differentiate the variance term of (55):

$$\nabla_w \text{Var}_e = \frac{1}{E} \sum_e 2(\bar{\ell}_e - \bar{\ell})(\nabla \bar{\ell}_e - \nabla \bar{\ell}) = \frac{2}{E} \sum_e (\bar{\ell}_e - \bar{\ell}) \nabla \bar{\ell}_e - \underbrace{\frac{2}{E} \nabla \bar{\ell} \sum_e (\bar{\ell}_e - \bar{\ell})}_{=0}.$$

The cross term vanishes identically because the centered residuals $\bar{\ell}_e - \bar{\ell}$ sum to zero by definition of $\bar{\ell}$ —this is the one step worth seeing explicitly, as it is why the INV gradient is a clean *reweighting* of per-episode gradients rather than a new object. Adding $\beta \nabla \bar{\ell} = (\beta/E) \sum_e \nabla \bar{\ell}_e$ and collecting terms gives (56). $\square$

Reading the INV weights

Equation (56) gives episode e the gradient weight $2(\bar{\ell}_e - \bar{\ell}) + \beta$: an episode at the buffer average gets weight β (plain ERM); a worse-than-average episode gets weight $> \beta$; and a better-than-average episode with $\bar{\ell}_e < \bar{\ell} - \beta/2$ gets a *negative* weight—gradient *ascent* on episodes served suspiciously well. That last case is the anti-overfitting tooth. In the generator (46), loading on the stable direction θ_r lowers all $\bar{\ell}_e$ roughly equally (no variance contribution), while loading on x_{sp} helps the episodes whose $\gamma_{r,e}$ sign matches and hurts the rest—pumping the variance term. INV strips γ -chasing weight while keeping θ -weight: the objective-level answer to the het trap of Example 7.1, complementary to the allocation-level answer (pinning). At het = 0 the penalty has nothing real to penalize and INV pays a small price—which is why the experiment carries both A5 (ERM) and A6 (INV) rather than declaring a winner a priori.

10 Competition Dynamics

10.1 Windows, Winners, and the Exponentiated-Gradient Update

Definition 10.1 (Competition window and winner). For an unpinned regime r , the competition window collects, for each specialist i , its shadow decision regrets over $W_c = 20$ in-regime days. At window close the **winner** is

$$i^* = \arg \min_{i \in \{1..N\}} \left[\bar{\rho}_{\text{win}}^{(i)} - \lambda(\alpha_{i,r} - \frac{1}{R}) \right], \quad \lambda = 0.3, \quad (57)$$

where $\bar{\rho}_{\text{win}}^{(i)}$ is i 's mean window regret: lowest regret wins, with an **incumbency bonus** $\lambda(\alpha_{i,r} - 1/R)$ subtracted for affinity above uniform. The winner then applies one **exponentiated-gradient (EG) step** to its own affinity vector,

$$\alpha_{i^*,r} \leftarrow \alpha_{i^*,r} e^\eta, \quad \text{then renormalize } \alpha_{i^*} \text{ over } \Delta^R, \quad \eta = 0.6, \quad (58)$$

and trains on the window's data. Losers' affinity vectors are untouched.

Two structural remarks. First, the bonus is centered at $1/R$, so an uncommitted specialist gets zero bonus, and since $\sum_r (\alpha_{i,r} - 1/R) = 0$, incumbency advantages across regimes are *zero-sum*: claiming r harder structurally weakens the claim everywhere else. Second, (58) belongs to the multiplicative-update family of exponentiated gradient [9], Hedge [8], and multiplicative weights [10]—inverted: each *specialist* runs the update over *regimes* (what shall I be?), rather than a master algorithm weighting experts (whom shall I trust?).

Because only the r -coordinate is multiplied and the rest are renormalized proportionally, the update has a two-point reduction: writing $\alpha = \alpha_{i^*,r}$ before the win,

$$\alpha' = \frac{\alpha e^\eta}{\alpha e^\eta + (1 - \alpha)}, \quad \text{equivalently} \quad \underbrace{\log \frac{\alpha'}{1 - \alpha'}}_{\text{log-odds}} = \log \frac{\alpha}{1 - \alpha} + \eta. \quad (59)$$

Each win adds exactly η to the log-odds of regime r within the winner's identity. Affinity is a win counter, read through a sigmoid.

Example 10.1 (Worked EG trace: uniform to pinned in seven wins). Start at $\alpha = 1/R = 0.25$ and let the same specialist win consecutive windows for r (winning nothing else in between). With $\eta = 0.6$, $e^{0.6} = 1.82212$. First win, by (59):

$$\alpha_1 = \frac{0.25 \times 1.82212}{0.25 \times 1.82212 + 0.75} = \frac{0.45553}{1.20553} = 0.37787.$$

Second win:

$$\alpha_2 = \frac{0.37787 \times 1.82212}{0.37787 \times 1.82212 + 0.62213} = \frac{0.68852}{1.31065} = 0.52533.$$

Iterating (all values rounded from full-precision iteration):

Wins k	0	1	2	3	4	5	6	7	8
α_k	0.250	0.378	0.525	0.668	0.786	0.870	0.924	0.957	0.976

The pin threshold 0.95 is crossed at win 7. The closed form confirms it: $\alpha_k \geq 0.95$ iff the odds satisfy $\frac{\alpha_0}{1 - \alpha_0} e^{\eta k} = \frac{1}{3} e^{0.6k} \geq 19$, i.e. $e^{0.6k} \geq 57$, i.e. $k \geq \ln 57 / 0.6 = 4.043 / 0.6 = 6.74$, so $k = 7$. Seven clean wins— $7 \times 20 = 140$ in-regime days, roughly three to four common-regime episodes—separate a cold uniform specialist from permanent ownership.

10.2 Rich-Get-Richer: The Symmetry-Breaking Engine

The EG update alone would make ownership a random walk among near-identical specialists. What breaks the symmetry is the coupling to training: *winners train* $\Rightarrow$ *training improves the head and deepens the buffer* $\Rightarrow$ *a better head wins the next window more often* $\Rightarrow$ *winners win again*.

Formally, let q_m be the probability that the current leader on regime r wins window m . Early on, heads are near-identical and $q_m \approx$ the tie-breaking rate, but the first (possibly lucky) win hands the winner a window's worth of training on r ; under the separated- θ_r generator this lowers its expected window regret, so $q_{m+1} > q_m$; and the affinity bonus in (57) adds a

deterministic tilt $\lambda(\alpha_{i,r} - \alpha_{j,r})$ on top. The win process is a reinforcing urn: the leader’s log-odds climb at rate η per win, while losers—who do not train on r —suffer no foreign updates from it. Once one specialist leads on each regime, the system is in the basin of the covering one-per-regime assignment.

This is competitive exclusion in the sense of Li [14]: one species per niche at equilibrium, with reward competition as the exclusion force—and the empirical record matches the mechanism’s prediction exactly: in **all 20 seeds**, the population converges to a covering one-specialist-per-regime assignment, with each common regime pinned at $\alpha \geq 0.95$.

10.3 The Batching Question: A Pre-Registered Prediction and Its Refutation

How large must W_c be? The pre-registered reasoning was a single-window concentration argument. Suppose two specialists’ true mean per-day regrets on r differ by $\Delta > 0$, with per-day regrets bounded in $[0, B]$. The window comparison errs when the worse specialist posts the lower sample mean; by Hoeffding’s inequality on the per-day regret differences (range $2B$, mean Δ),

$$\Pr(\text{wrong winner}) \leq \exp\left(-\frac{W_c \Delta^2}{2B^2}\right), \quad \text{so} \quad W_c \geq \frac{2B^2}{\Delta^2} \ln \frac{1}{\delta} \quad (60)$$

guarantees error $\leq \delta$. Early in competition Δ is small by construction (symmetric initialization), so (60) blows up, and the natural prediction follows: *small windows produce noisy winners, noisy winners scatter the EG updates, and specialization should require large W_c* —batching looked necessary, with $W_c = 1$ expected to fail outright.

The empirical refutation, stated plainly

The pre-registered prediction was wrong. Sweeping $W_c \in \{1, 5, 20, 60\}$ over 20 seeds, post-reactivation regret is **flat**: 0.0303–0.0309 across the sweep, with $W_c = 1$ *marginally best*. Batching within the window is not necessary for convergence, and the analysis that said otherwise mislocated where the averaging happens.

The corrected mechanism story. Inequality (60) bounds the error of a *single* comparison, but nothing in the architecture consumes single comparisons: the EG recursion (59) sums $\eta \cdot \mathbf{1}\{\text{win}\}$ in log-odds space across windows, and reaching the pin requires $\approx \ln 57/\eta \approx 6.7$ *net* wins, not one correct vote. If the better specialist wins each window with probability q above its rivals’ rates, its lead in win counts—hence in log-odds—grows linearly in the number of windows with noise $O(\sqrt{\text{windows}})$: the law of large numbers operates *across* windows, in α -space, regardless of W_c . Shrinking W_c from 20 to 1 makes each vote noisier but delivers $20\times$ more votes per regime-day, and the two effects approximately cancel; small windows also start the rich-get-richer training loop earlier (the first win arrives after 1 day, not 20), the plausible source of $W_c = 1$ ’s marginal edge. The affinity is already an exponentially-weighted tally of the win history—the EG accumulation *is* the cross-window averaging—so a second averaging layer inside the window was redundant.

Do not quietly upgrade the corrected story to a prediction

The flat W_c sweep is an empirical result on this stream (synthetic generator, $R = 4$, $N = 4$, $\eta = 0.6$), not a theorem. The corrected account is a post-hoc narrative consistent with the data and with standard multiplicative-weights analyses [8, 10], but the experiment that would test *it* (varying η jointly with W_c) was not pre-registered. One box above records a refuted prediction; this box keeps the replacement story from enjoying a credibility it has not yet earned.

10.4 Pinning

Definition 10.2 (Pin). When a window winner’s updated affinity satisfies $\alpha_{i^*,r} \geq 0.95$, regime r is **pinned** to owner $o(r) = i^*$, permanently: thereafter r ’s live decisions are served by $o(r)$ alone, training on r -days goes to $o(r)$ alone, and the competition machinery for r (shadow decisions, windows, EG updates) shuts down.

What changes at the pin is the *modality* of protection. Before it, the owner-apparent is protected statistically: it keeps winning, so it keeps being the one who trains, but a bad run could in principle let a rival in. After it, protection is structural: ownership no longer consults the reward stream at all—the assignment becomes reward-independent, not as an assumption but as a phase the dynamics enter and cannot leave. Once all regimes are pinned one-per-specialist (the observed endpoint in every seed), each specialist trains only on its own regime: $p = 0$, so under HARD it never overflows and under SOFT it suffers zero interference. The coupon-collector clock of Proposition 8.1 never starts; the half-life (51) never ticks. Dormancy becomes silence, with the head and the six-episode library untouched—a distributed persistent memory.

Two ablations calibrate how much of the benefit is the pin itself versus the competition that produces it:

- **Never-pin:** disabling Definition 10.2 costs 0.0320 versus 0.0307 pinned post-reactivation regret—the residual price of merely statistical protection: perpetual shadow competition occasionally hands windows, hence foreign training, to non-owners, and serving can flicker at reactivations.
- **No incumbency bonus:** $\lambda = 0$ yields 0.0314 versus 0.0307. The bonus *accelerates* convergence to ownership but does not cause it—the rich-get-richer loop breaks symmetry on its own. Causality lives in the win–train coupling; λ is a catalyst.

The crisis wrinkle. The rare regime exposes a quantization effect in the pin rule. A crisis episode lasts 15–30 days and yields at most one complete $W_c = 20$ window—often zero—and crisis episodes are themselves few (Section 7.2). Across the 20 seeds, the crisis regime’s top affinity reaches only $\alpha \in [0.67, 0.87]$: by the trace of Example 10.1, exactly 3–5 EG wins from uniform ($\alpha_3 = 0.668$, $\alpha_5 = 0.870$)—the affinity arithmetic independently confirms that the stream does not contain the ≈ 7 complete crisis windows formal pinning requires. The substantive outcome is nonetheless clean: in every seed the crisis regime has a *stable unique owner*, so the covering assignment is achieved; only the formal threshold is starved. The deployment lesson: pin on *ownership stability* (the same specialist winning a regime’s windows across k consecutive episodes), not on the raw $\alpha \geq 0.95$ threshold, whose win-count requirement assumes window-rich regimes. For rare regimes, windows—not days, not episodes—are the binding currency.

11 The Eleven Arms as Points in Design Space

The companion experiment runs eleven arms. They are not eleven competing architectures; they are a factorial probe of a two-axis design space, and each arm exists to occupy a cell or test an edge.

Axis 1: allocation reward-coupling. How does the regime $\rightarrow$ learner assignment respond to the *current* reward stream? Three positions: **coupled** (chases recent performance forever), **transient-then-structural** (RISP: coupled during competition, independent after the pin), and **independent** (lottery or oracle). Part II showed why this axis governs forgetting: coupled allocation keeps every learner’s foreign-update probability $p > 0$ during dormancy, and both memory models convert $p > 0$ over 250–450 days into a destroyed head.

Axis 2: training objective. Pooled **ERM** or the variance-penalized **INV** of (55) on the buffered episodes. This axis governs the het trap of Example 7.1: ERM imports episode-specific γ -weight into the probe window; INV strips it.

Arm	Allocation rule	Coupling	Objective
A1	monolith: train on whatever is active	coupled (degenerate)	ERM
A2	MoE learned router	coupled	ERM
A3	recent-performance capital shares	coupled (smooth)	ERM
A4	random fixed niches	independent	ERM
A5	RISP competition $\rightarrow$ pin	transient $\rightarrow$ structural	ERM
A6	RISP competition $\rightarrow$ pin	transient $\rightarrow$ structural	INV
A7	monolith	coupled (degenerate)	INV
A8a	Hedge over fixed strategies	coupled (weights only)	none (no training)
A8b	Hedge over learning experts	coupled	ERM
A9	oracle assignment from day 1	independent	ERM
A10	oracle assignment from day 1	independent	INV

A1 (monolith-ERM). One capacity- K learner; the allocation rule is the degenerate *always me*: it serves and trains on whatever $\hat{r}_t$ is active. Its foreign-update process during any dormancy has $p = 1$: under HARD it is the learner of Example 8.3, evicted within one foreign block; under SOFT it retains $0.94^{250} \approx 2 \times 10^{-7}$ over a crisis dormancy. *Prediction*: catastrophic forgetting under both memory models; worst-tier probe regret.

A2 (MoE learned router). N experts plus a gate matrix $G \in \mathbb{R}^{R \times N}$; on each day the gate serves expert $j_t = \arg \max_j G[\hat{r}_t, j]$ with probability 0.9 and a uniformly random expert with probability $\epsilon = 0.1$ (ϵ -greedy); the served expert trains on the day; the gate updates

$$G[\hat{r}_t, j_t] \ += \ \text{lr} \cdot (\text{baseline} - \rho_t),$$

with the baseline a running mean regret—below-baseline performance raises the routing weight. This is reward-coupled allocation *forever*, and exploration alone guarantees damage: during a crisis dormancy of D days the crisis expert is dragged into foreign training at rate $\geq \epsilon/N$ per day, so $\mathbb{E}[\text{foreign updates}] \geq \epsilon D/N = 0.1 \times (250\text{--}450)/4 = 6.25\text{--}11.25$ —about one full half-life (51) before counting routing churn; under HARD a single explored foreign block can evict. MoE routing in continual settings is studied in Li et al. [18]; the failure here is not expressiveness but the absence of any reward-independent protection phase. *Prediction*: forgets despite having one expert per regime available.

A3 (recent-performance allocator). All learners persist; capital shares are a softmax of recent performance, and each learner’s *training intensity equals its share*. The instructive feature is the **softmax floor**: a softmax is strictly positive, so every learner receives a minimum share $s_{\min} > 0$ of every day’s training, foreign days included. The floor *is* the erosion rate: a dormant regime’s natural owner accrues foreign updates at rate $\geq s_{\min}$ per day, retention $\approx 0.94^{s_{\min}D}$ under SOFT—graded and unavoidable. *Prediction*: graded (not cliff-like) forgetting; better than A1/A2, worse than any covering reward-independent arm.

A4 (random fixed niches). Each specialist is assigned a regime by lottery at $t = 0$, fixed forever: perfectly reward-independent, zero foreign updates, $\Phi \approx 0$ on every *covered* regime. The defect is coverage: with $N = R = 4$ and independent uniform draws, all four regimes are covered with probability $4!/4^4 = 24/256 \approx 9.4\%$ —in over 90% of lotteries some regime has no owner and is served at default-decision quality forever. Reward-independence is necessary but not sufficient: the assignment must also cover. *Prediction*: bimodal—oracle-grade retention on covered regimes, worst-case service on uncovered ones.

A5 / A6 (RISP-ERM / RISP-INV). The mechanism of Sections 9–10, differing only in the buffer objective (ERM step vs. INV gradient (56)). RISP sits at the distinguished point of axis 1: reward-coupled *exactly long enough* to discover a covering assignment (solving A4’s coverage failure), then structurally independent (solving A1–A3’s protection failure). *Predictions*: both converge to covering pinned assignments and retain through dormancy (confirmed: 20/20 seeds, common regimes pinned ≥ 0.95 , pinned regret 0.0307); A6 outperforms A5 on probe windows as het grows, since the probe meets a fresh $\gamma_{r,e}$ (Example 7.1).

A7 (monolith-INV). A1’s allocation with A6’s objective, blocking a confound: if RISP-INV beats the monolith, is it the mechanism or just the better loss? INV reshapes *what* is learned from the buffer, but the buffer itself is evicted (HARD) or the head eroded (SOFT) by the allocation; no objective can preserve parameters it does not control. *Prediction*: $A7 \approx A1$ on retention—the objective axis cannot rescue a failure on the allocation axis.

A8a / A8b (Hedge over fixed / learning experts). A8a runs Hedge [8] over a pool of *fixed* strategies: weights $w_i \propto \exp(-\eta_H \sum_s \rho_s^{(i)})$, no training anywhere. The allocation is reward-coupled, but with nothing trainable there is nothing to forget: A8a is structurally immune to dormancy, and bounded by the best fixed strategy rather than the best learned head. A8b keeps the Hedge allocation but lets the weighted experts train on the days they are trusted with: trainability returns, and with it the foreign-update process. *A8a vs. A8b isolates whether dormancy damage comes from reward-coupled allocation per se or from training under it* (the latter). *Predictions*: A8a immune to forgetting but capped within episodes; A8b recovers in-episode adaptation and re-imports forgetting.

A9 / A10 (oracle assignment, ERM / INV). The regime $\rightarrow$ specialist map is given correctly at $t = 0$ —pinning before day one—and each owner trains only on its regime, with ERM (A9) or INV (A10). These are the ceilings: reward-independent, covering, zero discovery cost. The A5/A6 vs. A9/A10 gap prices the *discovery phase* (contested serving, occasional wrong winners, foreign training before ownership settles); the A9 vs. A10 gap reads off the pure objective effect with allocation held perfect, mirroring A5 vs. A6.

The arms are an experiment about classes, not architectures

Each arm represents a *cell* in (coupling $\times$ objective) space, and every headline conclusion is a contrast between cells with one axis held fixed: A5 vs. A2/A3 tests allocation class at fixed objective; A6 vs. A5 and A10 vs. A9 test objective class at fixed allocation; A7 shows the objective axis cannot compensate for the allocation axis; A8a vs. A8b shows the damage mechanism is training-under-coupling, not coupling itself; A4 shows reward-independence without coverage is not enough; A9/A10 price discovery. Substituting a different router, softmax temperature, or fixed pool would not move any cell. The claim under test is never “RISP beats architecture X”; it is that *retention through dormancy requires an assignment that is both reward-independent and covering, and competition-then-pinning is a cheap emergent source of exactly that conjunction*—with the objective axis an orthogonal dial governing probe-window robustness to episode heterogeneity, not retention.

Remark 11.1 (Where the empirical record currently stands). The companion experiments confirm the RISP cell directly: 20/20 seeds reach a covering one-per-regime assignment; common regimes pin at $\alpha \geq 0.95$; crisis acquires a stable unique owner at $\alpha \in [0.67, 0.87]$ without a formal pin; never-pinning costs 0.0320 vs. 0.0307; $\lambda = 0$ costs 0.0314 vs. 0.0307; and the W_c sweep is flat at 0.0303–0.0309 with $W_c = 1$ marginally best—the latter refuting the pre-registered batching prediction of Section 10.3. The refutation is reported with the same prominence as the confirmations; a design-space map is only as trustworthy as its falsified cells.

Part IV

The Theory in Full

This part assembles the foundations of Part I into the four results that organize the RISP account: the episode-transfer lemma controlling the invariance gap, the eviction-rate lemma controlling the forgetting deficit, the decomposition proposition that yields the 2×2 prediction (including its super-additive signature), and the break-even proposition that prices retention against carrying costs. We close with the Gaussian construction that exhibits ERM’s across-episode failure in closed form, and with the honest remarks on what the coupled competition dynamics are *not* proven to do. Throughout, the register is that of the GAUSE companion theory [14]: idealized-model results that explain mechanisms and generate falsifiable orderings, with the quantitative weight carried by the controlled experiments of Part V.

12 Assumptions, Stated and Audited

Definition 12.1 (Episode risk). For regime r , episode environment ξ with law P_ξ on $\mathcal{X} \times \mathcal{Y}$, and head w , the episode risk is $L_\xi(w) = \mathbb{E}_{(x,y) \sim P_\xi} [\rho(Xw; y)]$, with ρ the decision regret of Part I, $\rho \in [0, B]$, $B = 2k w_{\max} y_{\max}$.

Definition 12.2 (Assumption A1: episode exchangeability). For each regime r there is a hyper-distribution Q_r over episode environments such that the historical episodes $\xi_{r,1}, \dots, \xi_{r,E_r}$ and the deployed episode $\xi_{r,e'}$ are i.i.d. draws from Q_r . Write $\mu_r(w) = \mathbb{E}_{\xi \sim Q_r} L_\xi(w)$ and $\sigma_r^2(w) = \text{Var}_{\xi \sim Q_r} L_\xi(w)$.

Important

A1 idealizes twice. *Independence* fails when macro memory links episodes (the 2011 crisis is causally downstream of 2008); *identical distribution* fails under secular drift (market microstructure in 1998 differs from 2020 in ways no regime label captures). We use A1 exactly as Inv-PnCO uses its environment assumption [6]: as the formal rendering of “the next crisis is a fresh draw from the crisis distribution,” with the experiments designed so that A1 holds by construction (the synthetic generator) and the limitations section of the companion paper carrying the real-world caveat. Nothing in this part silently strengthens A1.

Definition 12.3 (Assumption A2: dormancy and foreign updates). While regime r is dormant for D steps, an allocation mechanism A routes learning. A is *reward-driven* if the specialist holding r is made to learn a currently active regime with per-step probability at least $p > 0$; A is *reward-independent* if the assignment of r is measurable with respect to information independent of the realized reward stream and, at convergence, routes to r ’s owner only regimes in its fixed niche set.

The $p > 0$ clause is not vacuous bookkeeping: it is verified *by construction* for each reward-driven arm in our inventory (the monolith learns the active regime with probability 1; the router’s ϵ -greedy exploration plus reward-chasing gate gives $p \geq \epsilon/N$; the recent-performance allocator’s softmax floor gives $p \geq v_{\min} > 0$), and it is exactly the quantity a fund can audit in its own allocator: *what fraction of a losing desk’s capacity is reassigned per period?*

13 The Invariance Gap

Lemma 13.1 (Episode transfer). *Under A1, for any fixed head w and any $\delta \in (0, 1]$, with probability at least $1 - \delta$ over $\xi_{e'} \sim Q_r$,*

$$L_{\xi_{e'}}(w) \leq \mu_r(w) + \sigma_r(w) \sqrt{\frac{1-\delta}{\delta}} \leq \mu_r(w) + \frac{\sigma_r(w)}{\sqrt{\delta}}.$$

Proof. Cantelli’s inequality (Part I) applied to the real random variable $Z = L_{\xi}(w)$, $\xi \sim Q_r$, which has mean $\mu_r(w)$ and variance $\sigma_r^2(w)$: for $\tau > 0$, $\Pr[Z - \mu_r \geq \tau] \leq \sigma_r^2/(\sigma_r^2 + \tau^2)$. Setting the right side to δ and solving, $\tau = \sigma_r \sqrt{(1-\delta)/\delta}$, which is at most $\sigma_r/\sqrt{\delta}$. No boundedness or distributional shape is used; only the existence of the second moment, guaranteed by $Z \in [0, B]$. $\square$

Key Point

The bound is monotone in *both* $\mu_r(w)$ and $\sigma_r(w)$, and in nothing else. Any training objective that drives the pair (μ_r, σ_r) toward the lower-left of its Pareto frontier optimizes the guarantee; the RISP objective $\text{Var}_e + \beta \mathbb{E}_e$ is the Lagrangian scalarization of exactly that frontier, with β the exchange rate. For each tail level δ there is a $\beta(\delta)$ whose minimizer over any convex model class attains the frontier point optimizing the Lemma’s bound (standard scalarization; for non-convex classes the scalarization reaches the convex hull of the frontier, a caveat we inherit silently from every penalty method in the invariance literature [20]).

Lemma 13.2 (Estimation price of few episodes). *Fix w . With E_r training episodes drawn i.i.d. from Q_r and $\hat{\mu}_r(w)$, $\hat{\sigma}_r(w)$ the empirical mean and standard deviation of $\{L_{\xi_e}(w)\}_{e=1}^{E_r}$: for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$|\hat{\mu}_r(w) - \mu_r(w)| \leq B \sqrt{\frac{\log(2/\delta)}{2E_r}}, \quad |\hat{\sigma}_r(w) - \sigma_r(w)| \leq B \sqrt{\frac{2 \log(2/\delta)}{E_r - 1}}.$$

Proof. The mean bound is Hoeffding’s inequality (Part I) for i.i.d. $[0, B]$ -valued variables. For the deviation bound, note that the empirical standard deviation of $[0, B]$ -bounded variables is a function of the sample with bounded differences: replacing one L_{ξ_e} changes $\hat{\sigma}_r$ by at most $B/\sqrt{E_r - 1}$ (each summand $(L_e - \hat{\mu})^2$ moves by at most B^2 terms damped by the $1/(E_r - 1)$ normalization; a one-line calculation, or see [5] for the sharper empirical-Bernstein form). McDiarmid’s inequality then gives the stated tail, and $\mathbb{E}[\hat{\sigma}_r] \leq \sigma_r$ plus the same concentration controls the bias direction. $\square$

Important: the plug-in caveat

Lemmas 13.1–13.2 hold for *fixed* w . The trained head $\hat{w}_r$ is a function of the training episodes, so applying them at $w = \hat{w}_r$ is a plug-in heuristic, not a uniform-convergence guarantee; making it uniform costs a complexity term over the head class (linear heads in $d = 20$ dimensions would admit a standard covering-number argument) that we do not need for the mechanism claims and therefore do not carry. This is the same register at which Inv-PnCO deploys its Theorem 1, and we flag rather than hide it. The practical content survives untouched: with $E_r = 6$ episodes and $B = 0.4$, the mean’s estimation radius at $\delta = 0.1$ is $B\sqrt{\log 20/12} \approx 0.20$ —half the regret range—which is the formal statement of *why* crisis-grade invariance cannot be certified from six episodes alone and why the sub-episode and cross-asset enlargements of Part II are design defaults rather than afterthoughts.

Corollary 13.1 (Bridge to divergence-style guarantees). *If a representation Φ renders episode laws ε -invariant in KL — $KL(P_\xi(\cdot \mid \Phi) \parallel \bar{P}_r(\cdot \mid \Phi)) \leq \varepsilon$ for Q_r -almost-every ξ —then every head w factoring through Φ satisfies $\sup_\xi |L_\xi(w) - \bar{L}_r(w)| \leq B\sqrt{\varepsilon/2}$, hence $\sigma_r(w) \leq B\sqrt{\varepsilon/2}$, and Lemma 13.1 applies with that variance.*

Proof. Immediate from the divergence-to-regret bridge of Part I ($|L_P - L_{P'}| \leq B \text{TV} \leq B\sqrt{KL/2}$, Pinsker) applied between each P_ξ and the mixture $\bar{P}_r$; a uniformly c -close-to-mean random variable has standard deviation at most c . $\square$

This corollary is the formal connective tissue between the Inv-PnCO guarantee language (divergences between solution distributions [6]) and the regret currency of our decomposition: one Pinsker step, at the explicit and admittedly loose price B .

14 The Forgetting Deficit

Lemma 14.1 (Eviction rate under reward-driven allocation). *Hard memory, capacity K ; during r ’s dormancy let $W_a \geq K$ distinct regimes be active, let each dormant step independently deliver a foreign learning assignment to r ’s owner with probability $\geq p$, and conditional on an assignment let the regime be drawn from the active set with each regime’s probability $\geq q_{\min} > 0$. Let U_D be the number of distinct previously-unheld regimes learned in D steps. Then r ’s head (and its episode buffer) is evicted on $\{U_D \geq K\}$, and*

$$\Pr[U_D < K] \leq \Pr[\text{Binom}(D, pq_{\min}(W_a - K + 1)) < K] \leq \exp\left(-2D(pq_{\min}(W_a - K + 1) - \frac{K}{D})_+^2\right),$$

so $\Pr[\text{eviction}] \rightarrow 1$ exponentially in D . Moreover the expected eviction time is bounded by the coupon-collector sum

$$\mathbb{E}[\text{time to } U = K] \leq \frac{1}{p} \sum_{j=0}^{K-1} \frac{1}{q_{\min}(W_a - j)}.$$

Under the soft model with per-foreign-update decay ρ_{int} , the retained strength after N_D foreign updates is $(1 - \rho_{\text{int}})^{N_D}$, and $N_D \geq \text{Binom}(D, p)$ stochastically, so $\mathbb{E}[\text{strength}] \leq (1 - p\rho_{\text{int}})^D \rightarrow 0$.

Proof. While fewer than K distinct foreign regimes have been acquired, at least $W_a - K + 1$ of the active regimes are still unheld, so each step delivers a *new* distinct foreign regime with probability at least $p q_{\min}(W_a - K + 1)$. The count of such successes in D steps thus stochastically dominates the stated binomial while $U < K$, giving the first inequality; the second is Hoeffding’s bound for the binomial lower tail. The waiting-time bound is the sum of the K geometric means with success probabilities $p q_{\min}(W_a - j)$, $j = 0, \dots, K-1$ (the coupon-collector computation of Part II). LRU evicts r precisely when K distinct foreign heads have displaced it, since the dormant r is by definition the least recently used. For the soft model, each foreign update multiplies r ’s strength by $(1 - \rho_{\text{int}})$; taking expectations over the binomial count gives the product form via $\mathbb{E}[(1 - \rho)^{\text{Binom}(D,p)}] = (1 - p\rho)^D$. $\square$

Example 14.1 (Numbers at our calibration). $K = 2$, $W_a = 3$ active common regimes, $p \approx 1$ (the monolith always learns the active regime), $q_{\min} \approx 1/3$ at block scale: the expected eviction time is on the order of two block lengths (≈ 50 – 100 days), and at crisis dormancy $D \approx 280$ days the eviction probability is numerically indistinguishable from 1. This is why the E3 dormancy sweep finds the retention gap already saturated at $D = 63$ – 126 : the eviction event is binary, happens early in the dormancy, and cannot happen twice.

Proposition 14.1 (Dichotomy: reward-independence zeroes the deficit). *Define the forgetting deficit $\Phi(D; A) = \Pr[r \text{’s head lost at reactivation under } A]$ (hard) or $1 - \mathbb{E}[\text{retained strength}]$ (soft). Then: (i) for any reward-driven A satisfying A2, $\Phi(D; A) \rightarrow 1$ as $D \rightarrow \infty$ at the rate of Lemma 14.1; (ii) for any reward-independent covering A , $\Phi(D; A) = 0$ for all D in the hard model, and in the soft model the owner’s strength in r decays only through the regime’s own drift, with zero contribution from foreign updates.*

Proof. (i) is the lemma. (ii): coverage gives an owner holding r within capacity; reward-independence means the assignment map does not respond to the realized reward stream, so during r ’s dormancy the owner is assigned only regimes in its fixed niche set. If its niche set is $\{r\}$ (the converged one-per-regime case) it receives no updates at all; if its set contains other regimes, those are by construction within capacity, so no eviction is triggered ($U_D = 0$ counts only *unheld* acquisitions) and—in the implemented soft model, where decay attaches to foreign-*regime* updates—no interference is triggered either. The head and the episode buffer at reactivation are bit-identical to their state at dormancy onset. $\square$

Key Point

The dichotomy is qualitative, not quantitative: the two classes are different *functions* of D (one $\rightarrow 1$, one $\equiv 0$), which is what licenses the experimental design—arm classes, not parameter sweeps, separate them, and the E1 table confirms the class boundary at $p \sim 10^{-6}$ while E3 confirms the saturation shape. Coverage is the second conjunct and not decorative: random fixed niches are reward-independent yet retain only what they own, and their doubled seed variance in E1 is the measured price of leaving coverage to chance.

15 The Decomposition and the 2×2

Definition 15.1 (Relearning cost). C_{relearn} is the expected excess of a from-prior head’s mean probe-window regret over a converged retained head’s, for the same regime and probe length N_p . It is an empirical constant of the learning problem (model class, optimizer, signal-to-noise), not a theorem; the E6 audit shows it collapsing as signal strength rises, which is the honest reason the entire mechanism has a signal-to-noise boundary.

Proposition 15.1 (Post-reactivation regret decomposition). *Under A1–A2, when regime r reactivates after dormancy D into a fresh episode, served under allocation A by a head trained on E_r episodes, then at tail level δ , with the convention that the invariance terms are evaluated at the served retained head $\hat{w}_r$,*

$$\mathbb{E}[\rho_{\text{react}}(r)] \leq \underbrace{\hat{\mu}_r(\hat{w}_r) + \frac{\hat{\sigma}_r(\hat{w}_r)}{\sqrt{\delta}} + O\left(B\sqrt{\frac{\log(1/\delta)}{E_r}}\right)}_{G_{\text{inv}}} + \underbrace{\Phi(D; A) C_{\text{relearn}}}_{G_{\text{forget}}},$$

where the $O(\cdot)$ collects the estimation radii of Lemma 13.2.

Proof. Condition on the retention event R ($\Pr[R] = 1 - \Phi$). On R the served head is $\hat{w}_r$; Lemma 13.1 bounds its fresh-episode risk by $\mu_r + \sigma_r/\sqrt{\delta}$, and Lemma 13.2 converts population to empirical quantities at the stated radii. On R^c ($\Pr = \Phi$) the served head is fresh; by Definition 15.1 its probe regret exceeds the retained benchmark by C_{relearn} in expectation, and the benchmark itself is bounded by the same invariance terms. Total expectation: $\mathbb{E}[\rho_{\text{react}}] \leq (1 - \Phi) G_{\text{inv}} + \Phi (G_{\text{inv}} + C_{\text{relearn}}) = G_{\text{inv}} + \Phi C_{\text{relearn}}$. $\square$

Proposition 15.2 (Independent controls; super-additivity where it matters). *G_{forget} is a function of $(A, D, K, \text{memory model})$ alone and is zeroed by any reward-independent covering A (Proposition 14.1) regardless of the training objective; G_{inv} is a function of $(\text{objective}, Q_r, E_r)$ alone and is optimized by the variance-penalized objective (Lemma 13.1 discussion) regardless of A . The bound is therefore additive in the two design axes. The realized interaction is predicted super-additive in one specific cell: under reward-driven A with $\Phi \approx 1$, the served head at reactivation is fresh whatever objective trained the evicted one, so improving G_{inv} moves nothing observable—the invariance axis is inert in the reward-driven row.*

Remark 15.1 (The measured interaction). Define $I = (A1 - A5) + (A1 - A7) - (A1 - A6)$ on post-reactivation regret: $I = 0$ is additivity, $I < 0$ is synergy. The companion experiments measure $I = -0.0015 \pm 0.0005$ per-seed (95% CI excluding zero), realized almost entirely through $A1 - A7 \approx 0$ ($p = 0.88$)—the inert cell, exactly where Proposition 15.2 puts it. We are careful about logical direction: the additive *bound* motivates but does not entail the super-additive *realization*; the prediction that A7 specifically would sit at A1’s level was pre-registered from the mechanism (an evicted head serves nobody) and is the account’s sharpest falsifiable signature—a “both tricks help” story without the two-clock structure would have A7 strictly between A1 and A6.

16 Break-Even Dormancy: Pricing Retention

Proposition 16.1 (The retention window). *Charge carrying cost c per idle specialist-day and let V convert one unit of mean probe regret to value over the N_p -day probe. Over one dormancy–reactivation cycle of length D , retaining r ’s specialist pays iff*

$$c D \leq \Phi(D; A_{\text{rd}}) C_{\text{relearn}} N_p V.$$

Since Φ is increasing and saturates at 1 while cD grows linearly, the solution set is an interval $[D_{\min}, D_{\max}]$ (possibly empty): $D_{\min}$ where Φ ramps (eviction becomes likely enough to fear), $D_{\max} = C_{\text{relearn}} N_p V / c$ where cumulative carrying cost overtakes the one-shot saving. The window is empty iff $c > \max_D \Phi(D) C_{\text{relearn}} N_p V / D$, i.e., for carrying costs above a threshold set entirely by measurable quantities.

Example 16.1 (At the E3 calibration). $C_{\text{relearn}} \approx 0.004$ regret per probe day, $N_p = 15$: the cycle saving is $\approx 0.06 V$. At a carrying cost of 1% of the daily probe saving ($c = 4 \times 10^{-5} V$),

$D_{\max} \approx 1,500$ trading days and $D_{\min} \approx 40\text{--}60$ (where E3 shows Φ ramping); at 10%, $D_{\max} \approx 150$ days—tighter than one crisis cycle, so a fund at that cost level should *not* retain a dedicated crisis specialist and should plan to relearn; the indifference threshold sits near 15%. The proposition’s value is not a universal verdict but the conversion of an organizational argument (“keep the crisis desk?”) into three measurable numbers.

17 What the Competition Dynamics Are Not Proven to Do

Two honest closings. First, *we prove no convergence theorem for the coupled winner-take-all dynamics*: specialists’ window wins depend on heads, heads on training, training on wins—an endogenous feedback for which adversarial-regret guarantees of the EG update do not apply (Part I) and for which we know of no clean fixed-point argument covering the full mechanism. What we have is the GAUSE precedent (the same dynamics converge to covering one-per-regime assignments across all its domains and protocols [14]), our own twenty-for-twenty convergence record with the crisis-regime affinity wrinkle reported, and the single-window concentration remark—whose pre-registered implication (batching necessary) the W_c ablation *refuted*, teaching the corrected lesson that cross-window EG accumulation is the operative averager. Second, *the decomposition’s two terms are bounds, not identities*: a mechanism could in principle beat the bound’s additivity (and ours measurably does, in the predicted direction and cell). The theory’s role is to say which knob owns which term and where each term vanishes—claims the eleven-arm design tests directly—not to predict regret to the fourth decimal. Where the theory and the experiments disagreed, this document reports the experiment and revises the story, twice (batching; the high-SNR prototype that became E6).

Part V

Experimental Evidence in Full

The theory of the preceding parts makes a small number of sharp, falsifiable claims: that a reward-independent niche assignment retains dormant-regime competence where every reward-coupled allocator relearns; that an invariance objective inside a niche buys robustness that an invariance objective inside a monolith cannot; that the two mechanisms interact super-additively; and that all of this is conditional on the environment actually containing persistent, re-activating conditional structure. This part subjects each claim to its experiment. We present the numbers exhaustively—every arm, every metric, every pre-registered comparison with its raw and corrected p -value—and we present the null results with the same care as the positive ones, because the protocol’s credibility rests on its ability to return “no” (Sections 20 and 22.2).

18 Methodology and Statistical Apparatus

18.1 The arms

Eleven arms share an identical decision layer (top- K portfolio construction over $n = 20$ assets from predicted returns) and identical data per seed; they differ only in how predictive capacity is organized and trained.

The synthetic market has $n = 20$ assets, $d = 20$ features, $R = 4$ latent regimes with dormancy/reactivation dynamics, per-specialist capacity $K = 2$, and 20 seeds per configuration. Evaluation is restricted to the second half of each stream so that all arms are past their transient.

Table 1: The eleven arms. “Assignment” is how specialists (or capacity slots) are bound to regimes; “Objective” is the within-specialist training loss. Oracle arms receive the true regime→specialist map and serve as skylines, not competitors.

Arm	Name	Assignment	Objective
A1	monolith-ERM	none (single model)	ERM
A2	MoE router	learned gate, reward-driven	ERM
A3	recent-perf allocator	capital $\propto$ recent performance	ERM
A4	random fixed niches	random, frozen at init	ERM
A5	RISP-ERM	learned, dormancy-pinned	ERM
A6	RISP-INV (full system)	learned, dormancy-pinned	invariance
A7	monolith-INV	none (single model)	invariance
A8a	Hedge over fixed experts	exponential weights, frozen experts	none
A8b	Hedge over learners	exponential weights, learning experts	ERM
A9	oracle-pinned-ERM	oracle map, pinned	ERM
A10	oracle-pinned-INV	oracle map, pinned	invariance

18.2 The seeded paired design

Every seed $s \in \{1, \dots, 20\}$ fixes the entire data-generating draw: the regime schedule, the per-regime parameters, the feature noise, and the realized returns. All eleven arms are then run on the *same* stream for seed s . Consequently the between-seed component of variance—which regime schedule was drawn, how harsh the reactivations happened to be—is common to all arms, and arm differences within a seed are estimates of the treatment effect alone. This is the experimental-design analogue of common random numbers: it does not change the means but sharply reduces the variance of *contrasts*.

We nonetheless report unpaired Welch tests on the per-seed means (Section 18.4) rather than paired t -tests. This is deliberately conservative: the paired test would have strictly more power (the seed-level draws are positively correlated across arms), so any comparison that survives the unpaired test would also survive the paired one, while the converse need not hold. Where a paired analysis materially changes the reading (the λ ablation in Section 22.1), we say so explicitly.

18.3 The three metrics

All metrics are means of the per-day *decision regret*

$$r_t = V_t^\star - V_t,$$

where V_t is the realized value of the arm’s top- K portfolio on day t and $V_t^\star$ the value of the best top- K portfolio in hindsight for that day. Regret is measured in the decision space, not the prediction space, following the smart-predict-then-optimize critique [1]: prediction error and decision quality can dissociate, and our research question lives entirely on the decision side.

Let $\mathcal{T}$ be the evaluation window (second half of the stream). A *reactivation event* is a day on which a regime that has been dormant for at least 90 days re-enters the schedule. The *probe window* of an event is its first 15 evaluation days. Then:

$$\text{post_react} = \text{mean of } r_t \text{ over the union of probe windows in } \mathcal{T}, \quad (61)$$

$$\text{steady} = \text{mean of } r_t \text{ over } \mathcal{T} \setminus \{\text{probe windows}\}, \quad (62)$$

$$\text{overall} = \text{mean of } r_t \text{ over all of } \mathcal{T}. \quad (63)$$

post_react is the primary endpoint: it isolates the days on which retention can matter at all. **steady** measures the price paid for the mechanism when nothing is reactivating, and **overall** aggregates both. Each seed contributes one number per metric per arm; all inference is on these $n = 20$ per-seed means, never on the (serially dependent) daily values.

18.4 Welch’s t -test

For arms a and b with per-seed means $\bar{x}_a, \bar{x}_b$, sample variances s_a^2, s_b^2 , and $n_a = n_b = 20$, the Welch statistic is

$$t = \frac{\bar{x}_a - \bar{x}_b}{\sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}}, \quad (64)$$

referred to a t distribution with the Welch–Satterthwaite degrees of freedom

$$\nu = \frac{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}\right)^2}{\frac{(s_a^2/n_a)^2}{n_a - 1} + \frac{(s_b^2/n_b)^2}{n_b - 1}}. \quad (65)$$

We use the unequal-variance form rather than the pooled Student t because variance heterogeneity across arms is not an edge case here—it is a *finding*. A4’s defining signature is inflated between-seed variance (its post-reactivation 95% CI half-width is 0.00126 against A5’s 0.00070, a factor of 1.8; Section 19.2), and A6’s high-SNR collapse comes with CI half-widths four to six times A1’s (Section 22.2). Pooling variances across such pairs would miscalibrate exactly the comparisons we care most about. With $n_a = n_b = 20$ and roughly comparable variances, ν ranges over roughly 30–38 in our comparisons; the test is essentially a z -test with a mild small-sample correction.

18.5 Holm–Bonferroni and a proof that it controls FWER

The E1 analysis involves a pre-registered family of $m = 11$ pairwise comparisons (Table 3). With 11 tests at level 0.05 each, the chance of at least one false rejection under the global null is bounded only by 0.55 by Bonferroni’s inequality—useless. We therefore control the family-wise error rate (FWER) with Holm’s step-down procedure, which dominates plain Bonferroni at no assumption cost.

Definition 18.1 (Holm step-down procedure). Order the p -values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$ with corresponding hypotheses $H_{(1)}, \dots, H_{(m)}$. Let

$$k^* = \min \left\{ k : p_{(k)} > \frac{\alpha}{m - k + 1} \right\}$$

(with $k^* = m + 1$ if no such k exists). Reject $H_{(1)}, \dots, H_{(k^*-1)}$ and retain the rest. Equivalently, report adjusted p -values $\tilde{p}_{(k)} = \max_{j \leq k} \min\{(m - j + 1)p_{(j)}, 1\}$ and reject where $\tilde{p} \leq \alpha$. The `holm_p` entries in our result files are these adjusted values.

Proposition 18.1 (Holm controls FWER at level α). *Under arbitrary dependence among the p -values, the probability that Holm’s procedure rejects at least one true null hypothesis is at most α .*

Proof. Let I_0 be the index set of true nulls and $m_0 = |I_0|$; assume $m_0 \geq 1$ (otherwise there is nothing to prove). Suppose the procedure rejects at least one true null, and let k be the *first* step (in the sorted order) at which a true null is rejected. All hypotheses rejected at steps $1, \dots, k - 1$ are then false nulls, of which there are $m - m_0$ in total; hence $k - 1 \leq m - m_0$, i.e. $k \leq m - m_0 + 1$, and therefore

$$\frac{\alpha}{m - k + 1} \leq \frac{\alpha}{m_0}.$$

Because step k rejected, $p_{(k)} \leq \alpha/(m - k + 1) \leq \alpha/m_0$, and since the true null rejected at step k has p -value $p_{(k)}$,

$$\min_{i \in I_0} p_i \leq p_{(k)} \leq \frac{\alpha}{m_0}.$$

So the event “at least one true null is rejected” implies the event $\{\min_{i \in I_0} p_i \leq \alpha/m_0\}$. For each true null, $\Pr(p_i \leq \alpha/m_0) \leq \alpha/m_0$ (validity of the individual test), so by the union bound

$$\Pr\left(\min_{i \in I_0} p_i \leq \frac{\alpha}{m_0}\right) \leq m_0 \cdot \frac{\alpha}{m_0} = \alpha. \quad \square$$

Multiple-testing discipline is not optional in this literature: the data mining of trading-strategy families is precisely the failure mode documented by the reality-check and SPA literature [32, 33], by the deflated Sharpe ratio [34], and by the backtesting critiques of Harvey and Liu [35] and López de Prado [36]. Our family is small (11), fixed in advance, and corrected; that is the minimum standard those papers demand.

18.6 Confidence intervals and the choice of 20 seeds

All reported intervals are per-arm 95% confidence intervals on the mean over seeds,

$$\bar{x} \pm t_{19, 0.975} \frac{s}{\sqrt{20}}, \quad t_{19, 0.975} = 2.093, \quad (66)$$

where s is the between-seed standard deviation. The `ci95` fields in the result files are the half-widths of these intervals.

Why 20 seeds? Back out the per-seed SD from Eq. (66): on the primary endpoint the major arms have $s \approx 0.0014$ – 0.0027 (A1: 0.0020; A5: 0.0015; A6: 0.0014; A4: 0.0027). The tightest pre-registered contrast is the invariance axis A6 vs. A5, with observed difference 0.00142. The Welch standard error for that pair is $\sqrt{(0.0015^2 + 0.0014^2)/20} \approx 0.00046$, giving $t \approx 3.1$ and the observed raw $p = 0.0061$ —comfortably detectable, with power ≈ 0.85 at $\alpha = 0.05$ for an effect of that size. Halving the seed count would have left the invariance axis at the edge of detectability after Holm correction; doubling it would have bought little, since the binding constraint on the smallest effects we care about (e.g. A6 vs. A10) is that they are *predicted to be null*. Twenty seeds is the point where every effect we predicted to exist is powered and every effect we predicted not to exist is bounded by a usefully narrow CI.

18.7 The noise-floor convention

Raw regret levels conflate two things: irreducible per-day noise (even the oracle-assigned, invariance-trained A10 cannot drive regret to zero in a stochastic market) and the mechanism-dependent excess above that floor. We adopt the convention

$$\text{floor} = \text{steady}(\text{A10}) = 0.0250, \quad \text{excess}(\cdot) = \text{metric}(\cdot) - 0.0250. \quad (67)$$

A10’s steady state is the best any arm in this family achieves with oracle assignment, oracle-free training, and no reactivation stress; treating it as the irreducible floor makes relative improvements interpretable. A 10.3% improvement in raw regret (A6 vs. A1, post-reactivation) is a 38.1% reduction in excess-over-floor—the latter is the honest measure of how much of the *addressable* regret the mechanism removes (Section 19.3).

18.8 Pre-registration discipline

Before the final 20-seed E1 run, we fixed in writing: (i) the eleven-arm roster and all hyperparameters; (ii) the three metrics and the primary endpoint (`post_react`); (iii) the family of eleven pairwise comparisons and the Holm correction over exactly that family; (iv) a directional prediction for every comparison, including four predicted *nulls* (A2 $\approx$ A1, A7 $\approx$ A1, A4 $\approx$ A5, A6 $\approx$ A10); and (v) the E0 structure gate for real data, with the commitment to run E1s and report it regardless of the gate’s outcome. The predicted nulls matter as much as the predicted effects: a theory that only predicts differences can survive almost any data, whereas predicting

that invariance alone does *nothing* post-reactivation (A7≈A1) is a claim the data could easily have destroyed.

Intuition

The entire E1 design is a 2×2 factorial hiding inside an eleven-arm lineup: {monolith, niche} $\times$ {ERM, INV} is the square (A1, A5, A7, A6), and everything else is a control that pins down *why* the square comes out the way it does. A2 and A3 show that reward-coupled allocation does not reproduce the niche effect; A4 shows assignment quality matters beyond mere partitioning; A8a/A8b bracket the system between “memory without adaptation” and “adaptation without memory”; A9/A10 show how close learned pinning gets to the oracle ceiling.

Important

What these statistics do not establish. Every p -value in this part is a statement about *mean decision regret on a specified generator or historical sample, under a specified decision layer, relative to other arms of the same family*. None of the following is claimed, and the reader should resist inferring any of it: (i) no claim of positive risk-adjusted return (“alpha”) on any market—regret relative to a daily hindsight oracle is not profit; (ii) no claim about live trading, transaction costs, slippage, capacity, or market impact; (iii) no claim that the synthetic generator is a calibrated model of any asset class; (iv) no claim that the real-data nulls of Section 20 would persist under other universes, frequencies, or feature sets—they are scoped to the data actually tested. The expected post-publication fate of any genuine market anomaly is decay [25]; our claims are deliberately about *mechanism*, which is the only thing a synthetic-plus-null-real evidence base can support.

19 E1: The Headline Dissociation, Number by Number

19.1 The full table

Table 2 reports all eleven arms on all three metrics, with 95% CIs, from the archived 20-seed run ($K = 2$, heterogeneity 1.0, hard memory, probe 15, minimum dormancy 90 days).

Table 2: E1 headline results. Mean daily decision regret ($\times 10^{-2}$, lower is better) $\pm$ 95% CI half-width over 20 seeds. Excess = post-react mean minus the noise floor 2.50×10^{-2} (Eq. (67)). Bold: best non-oracle arm per column.

Arm	post_react	overall	steady	Excess	Post/steady
A1 monolith-ERM	3.424 $\pm$ 0.095	2.690 $\pm$ 0.044	2.459 $\pm$ 0.058	0.925	1.39
A2 router	3.408 $\pm$ 0.110	2.695 $\pm$ 0.052	2.481 $\pm$ 0.070	0.909	1.37
A3 recent-perf	3.776 $\pm$ 0.096	2.940 $\pm$ 0.055	2.685 $\pm$ 0.072	1.277	1.41
A4 random-fixed	3.295 $\pm$ 0.126	2.827 $\pm$ 0.114	2.717 $\pm$ 0.124	0.795	1.21
A5 RISP-ERM	3.214 $\pm$ 0.070	2.750 $\pm$ 0.064	2.644 $\pm$ 0.084	0.715	1.22
A6 RISP-INV	3.072 $\pm$ 0.066	2.602 $\pm$ 0.059	2.507 $\pm$ 0.074	0.572	1.23
A7 monolith-INV	3.435 $\pm$ 0.100	2.663 $\pm$ 0.044	2.428 $\pm$ 0.062	0.935	1.41
A8a Hedge-fixed	4.390 $\pm$ 0.090	3.815 $\pm$ 0.069	3.704 $\pm$ 0.096	1.890	1.19
A8b Hedge-learners	3.407 $\pm$ 0.100	2.672 $\pm$ 0.046	2.443 $\pm$ 0.057	0.907	1.39
A9 oracle-pinned-ERM	3.210 $\pm$ 0.077	2.736 $\pm$ 0.066	2.624 $\pm$ 0.088	0.710	1.22
A10 oracle-pinned-INV	3.053 $\pm$ 0.078	2.590 $\pm$ 0.059	2.499 $\pm$ 0.075	0.553	1.22

Three structural facts are visible before any test is run. First, the `post_react` column separates the arms far more than the `steady` column: retention is a property of reactivation

days. Second, the post/steady ratio splits the arms into two clean families: everything with a reward-coupled or absent assignment (A1, A2, A3, A7, A8b) sits at 1.37–1.41, and everything with a fixed or pinned assignment (A4, A5, A6, A8a, A9, A10) sits at 1.19–1.23. The ratio is a fingerprint of the mechanism, independent of the level. Third, the **steady** column shows the honest cost: pinning is not free. A5’s steady regret (2.644) is *worse* than A1’s (2.459), because partitioned specialists see less data per niche; A7 actually posts the best steady value in the table (2.428), better even than A10. RISP wins **overall** because reactivation days are frequent and expensive enough that retention pays for the partitioning cost—not because it dominates everywhere.

19.2 The pre-registered comparisons

Table 3: The eleven pre-registered comparisons on **post_react**. Δ = first arm minus second arm ($\times 10^{-2}$; negative favors the first arm). Verdicts at FWER $\alpha = 0.05$ (Holm). “Null (pred.)” denotes comparisons pre-registered as approximate equalities.

#	Comparison	Δ	Welch p	Holm p	Verdict
1	A6 vs A1	−0.353	9.2×10^{-7}	1.0×10^{-5}	rejected: A6 < A1
2	A6 vs A7	−0.363	1.2×10^{-6}	1.2×10^{-5}	rejected: A6 < A7
3	A6 vs A8b	−0.335	4.5×10^{-6}	4.1×10^{-5}	rejected: A6 < A8b
4	A5 vs A1	−0.210	1.3×10^{-3}	1.0×10^{-2}	rejected: A5 < A1
5	A6 vs A5	−0.142	6.1×10^{-3}	4.3×10^{-2}	rejected: A6 < A5
6	A5 vs A2	−0.194	6.3×10^{-3}	4.3×10^{-2}	rejected: A5 < A2
7	A6 vs A9	−0.138	1.1×10^{-2}	5.7×10^{-2}	<i>marginal</i> (fails Holm)
8	A5 vs A4	−0.081	0.281	1.0	null (pred.)
9	A6 vs A10	+0.019	0.719	1.0	null (pred.)
10	A2 vs A1	−0.016	0.827	1.0	null (pred.)
11	A7 vs A1	+0.011	0.882	1.0	null (pred.)

We now walk through each pre-registered prediction.

P1: A6 < A5 < A1 (both axes contribute). Confirmed at every link. A5 beats A1 by 0.210×10^{-2} (Holm $p = 0.010$): pinned niches alone, with plain ERM inside them, retain dormant-regime competence. A6 beats A5 by a further 0.142×10^{-2} (Holm $p = 0.043$): the invariance objective adds value *on top of* retention. The chain A6 < A1 is the largest effect in the family (Holm $p = 1.0 \times 10^{-5}$).

P2: A7 $\approx$ A1 post-reactivation (the dissociation). Confirmed as a null: $\Delta = +0.011 \times 10^{-2}$, raw $p = 0.882$. The invariance objective, trained in a monolith, does *nothing* for post-reactivation regret—indeed A7’s point estimate is fractionally worse than A1’s. This is the single most theory-laden prediction in the family: invariance can only protect parameters that still exist, and a monolith overwrites its dormant-regime parameters regardless of how they were trained. Note the contrast with **steady**, where A7 is the best arm in the table: invariance helps the monolith *between* reactivations and is helpless *at* them.

P3: A2 $\approx$ A1 (a reward-driven router is a monolith in disguise). Confirmed as a null: $\Delta = -0.016 \times 10^{-2}$, raw $p = 0.827$. The learned gate receives no gradient from a dormant regime and reallocates its experts’ capacity to live regimes; on reactivation it relearns like the monolith. The companion test A5 < A2 (Holm $p = 0.043$) confirms the positive side: dormancy-pinning beats the router by about as much as it beats the monolith.

P4: A3 is the worst learner. Confirmed. A3’s post-reactivation regret (3.776×10^{-2}) is the worst of any learning arm—only the frozen A8a is worse—and uniquely among the learners its damage is *not* confined to probe windows: its **overall** (2.940) and **steady** (2.685) are also the worst in the learner family. The failure mode is continuous erosion: the recent-performance allocator drives capital toward whatever just worked, and the capital floor (the minimum allocation that keeps a specialist training at all) is exactly where a temporarily-unlucky specialist’s data diet collapses. Every regime transition triggers a starve-and-relearn cycle, so A3 pays the relearning tax perpetually, not just after long dormancies.

P5: A8a retains but cannot adapt; A8b adapts but cannot retain. Confirmed on both jaws of the bracket. A8a (Hedge over frozen experts) has the *lowest* post/steady ratio in the table (1.19): its relative degradation at reactivation is minimal, because nothing it could forget was ever learned online—this is retention in its purest form. But its absolute level is catastrophic (**steady** = 3.704, 51% above A1), because frozen experts cannot track within-regime drift at all. A8b (Hedge over learning experts) is the mirror image: its level tracks the monolith almost exactly (**post_react** 3.407 vs 3.424; **steady** 2.443 vs 2.459; ratio 1.39 vs 1.39), because exponential weights over experts that all share the reward stream do not protect any expert’s parameters from being retrained. The sleeping-experts story, in other words, is not about the *weighting* scheme; it is about whether the experts’ parameters are insulated from reward-coupled reuse. A6 beats A8b by 0.335×10^{-2} (Holm $p = 4.1 \times 10^{-5}$).

P6: A4 $\approx$ A5 in mean, with higher variance. Confirmed. Random fixed niches land at 3.295×10^{-2} against A5’s 3.214 (raw $p = 0.281$, Holm 1.0): a random reward-independent assignment captures most of the retention effect, exactly as the theory says it should—retention comes from reward-independence, not from assignment quality. But A4’s between-seed SD is 0.0027 against A5’s 0.0015 (1.8 $\times$), and the signature is even starker on **steady** (CI 0.124 vs 0.084) and **overall** (CI 0.114 vs 0.064). Inspection of A4’s per-seed values shows why: most seeds are fine, but the seeds in which the random assignment leaves a regime with no well-matched specialist (coverage gaps) blow up the mean (per-seed **overall** values up to 0.0341 against a median near 0.0270). Learned pinning buys insurance against the coverage tail, not a better average draw.

P7: A6 $\approx$ A10 (the skyline is reached). Confirmed as a null: $\Delta = +0.019 \times 10^{-2}$, raw $p = 0.719$. The full system is statistically indistinguishable from the same system with an oracle-given assignment. Learned dormancy-pinning recovers essentially all of the oracle assignment’s value; the corresponding ERM-side check is even cleaner (A5 3.214 vs A9 3.210, not in the test family but visibly identical).

P8 (exploratory edge): A6 vs A9. A6’s point estimate beats the *oracle-assigned ERM* arm by 0.138×10^{-2} —learned assignment plus invariance edging out perfect assignment without it. Raw $p = 0.011$, but the Holm-adjusted value is 0.057, which fails the 0.05 FWER threshold. We report this exactly as it is: suggestive, consistent with the rest of the pattern, and not established at the pre-registered standard. Nothing downstream depends on it.

Key Point

Seven of seven predicted differences went the predicted direction; six of seven survived Holm at $\alpha = 0.05$ (the seventh, $A6 < A9$, is marginal at 0.057). All four predicted nulls came back null with raw $p \geq 0.28$. No comparison in the pre-registered family contradicted a prediction.

19.3 The axes arithmetic

Decomposing the headline effect along the factorial square:

$$\text{retention axis (A1} \rightarrow \text{A5): } \frac{3.424 - 3.214}{3.424} = 6.1\%, \quad (68)$$

$$\text{invariance axis (A5} \rightarrow \text{A6): } \frac{3.214 - 3.072}{3.214} = 4.4\%, \quad (69)$$

$$\text{joint (A1} \rightarrow \text{A6): } \frac{3.424 - 3.072}{3.424} = 10.3\%. \quad (70)$$

In excess-over-floor terms (Eq. (67)) the joint effect is

$$1 - \frac{3.072 - 2.500}{3.424 - 2.500} = 1 - \frac{0.572}{0.925} = 38.1\% : \quad (71)$$

the full system removes a bit over a third of the addressable post-reactivation regret, and lands within 3% of the oracle skyline’s excess (0.553 vs 0.572).

19.4 The interaction index: super-additivity

Define, per seed, the factorial interaction

$$I = (A1 - A5) + (A1 - A7) - (A1 - A6) = A1 - A5 - A7 + A6. \quad (72)$$

If the two mechanisms contributed additively, I would be zero: the joint gain would equal the sum of the marginal gains. Computed per seed on the paired design (here pairing is legitimate and free, since Eq. (72) is a within-seed contrast),

$$I = -0.0015 \pm 0.0005 \quad (\text{mean} \pm 95\% \text{ CI, } n = 20),$$

which excludes zero decisively. The sign is the interesting part: $I < 0$ means the joint gain $(A1 - A6) = 0.0035$ *exceeds* the sum of the marginals $(A1 - A5) + (A1 - A7) = 0.0021 + (-0.0001) = 0.0020$. The interaction is super-additive, and the arithmetic shows why in one line: the invariance marginal in the monolith is *zero* ($A1 - A7 = -0.0001$), yet inside niches invariance contributes 0.0014 ($A5 - A6$). Invariance is not an independent additive bonus; it is a multiplier on retained memories. A specialist that still holds its dormant regime’s parameters can hold *invariant* parameters, which transfer across the within-regime drift accumulated during dormancy; a monolith has nothing for the invariance penalty to protect at reactivation time. This is the empirical signature of the retention $\times$ robustness interaction derived in Part III, and it echoes the invariance-for-decision-making rationale of Krueger et al. [20] and Yan et al. [6]: invariance pays where the deployment distribution differs from the training distribution—which, for a retained dormant niche, is exactly the reactivation moment.

20 E0 and E1s: The Real-Data Nulls

A mechanism paper earns the right to a synthetic headline only by trying, honestly, to find the mechanism’s preconditions in real data, and reporting what it finds. We found that the precondition—persistent, re-activating, *decision-relevant* conditional structure—is absent in the real universes we tested. This section documents that null in full.

20.1 E0: the structure diagnostic

Protocol. E0 asks the gating question without running any learning arm at all: *if an oracle could condition the decision policy on regime labels, would that beat not conditioning?* The components:

1. **Universes.** Five crypto pairs (daily bars) and three commodities (daily). Evaluation spans 562 days (crypto) and 1,270 days (commodities).
2. **Causal labelers.** Two regime labelers, L1 (finer) and L2 (coarser), computed from *trailing* data only, so labels are known before the day’s decision and cannot leak the target. The realized regime distributions are imbalanced, especially under L2 on crypto (counts 517/517/31/59), which we note as a power limitation for the rare regimes.
3. **Decision metric.** The same top- K portfolio regret as E1 (Section 18.3)—a decision-space test, not a prediction-space test.
4. **Oracle conditioning.** Fit the best per-regime decision policy with hindsight model selection per regime (an upper bound on what any regime-aware learner could extract), and compare its mean regret to (a) the same machinery run on *block-shuffled* labels (preserving label marginals and block-length statistics, destroying the label–day alignment), and (b) a pooled, unconditioned policy.
5. **Statistic.** Gap % = (shuffled – real)/shuffled $\times$ 100 (positive iff real labels help), and $z = (\text{shuffled} - \text{real})/\text{sd}(\text{shuffled})$ over the shuffle distribution.

Table 4: E0 structure diagnostic on real data. Mean daily decision regret ($\times 10^{-2}$) for oracle regime-conditioning on real labels, on block-shuffled labels (mean $\pm$ SD of the shuffle distribution), and for the pooled unconditioned policy. Positive gap = real labels help. No cell shows exploitable structure.

Domain	Labeler	Real	Shuffled	Pooled	Gap (%)	z
Crypto ($n=562$ d)	L1	1.220	1.202 ± 0.027	1.195	−1.47	−0.66
Crypto ($n=562$ d)	L2	1.212	1.204 ± 0.047	1.195	−0.66	−0.17
Commodities ($n=1270$ d)	L1	0.834	0.832 ± 0.015	0.816	−0.25	−0.14
Commodities ($n=1270$ d)	L2	0.836	0.838 ± 0.009	0.816	+0.16	+0.16

Reading. Across both domains and both labelers, the real-label oracle never beats the shuffled-label oracle by more than 0.16%, all $|z| < 0.7$, and three of four point estimates are *negative*—real labels do marginally worse than shuffled ones. Most damning, the *pooled* policy is the best cell in every row: even hindsight-optimal regime conditioning loses to not conditioning at all, because splitting the sample by regime costs more in estimation variance than the (apparently nonexistent) regime structure returns. This is a null with teeth: it bounds not just RISP but *any* regime-conditioned method in this family, on this data, at this frequency, with these features.

Honest scoping. The null is scoped: five crypto pairs and three commodities, daily bars, one feature set, two specific (if reasonable) causal labelers, and a top- K decision layer. Intraday frequencies, larger cross-sections, alternative labelers (e.g. macro-indexed), or other decision layers could in principle revive the gate, and the protocol is designed to be rerun cheaply on any of them.

Relation to the GAUSE null. The predecessor project found a closely related null in *prediction* space: regime-aware prediction showed no real-vs-shuffled gap on financial domains [14]. That did not settle our question, and pretending otherwise would have been sloppy: the predict-then-optimize literature documents real cases where prediction-space and decision-space rankings dissociate [1, 6], so a decision-metric retest was a genuine open question, not a formality. E0 answers it: the dissociation does not rescue financial regime structure here. The prediction-space null and the decision-space null now agree.

20.2 E1s: the stitched-crypto experiment

E0 was the pre-registered gate for real-data claims, and it failed. Pre-registration committed us to running the full arm lineup on real data anyway and predicting *null separation*—a falsifiable prediction in its own right (if arms had separated despite the failed gate, the gate would be broken). E1s stitches real crypto regime blocks (block counts 15/13/7/9 across the four labeled regimes) into schedules with genuine dormancy (minimum 80 days, probe 15 days), 20 seeds of stitching randomization, ten arms (A10 requires generator access and is undefined on real data).

Table 5: E1s stitched real crypto. Mean daily decision regret ($\times 10^{-2}$) $\pm$ 95% CI, 20 seeds. All pairwise comparisons in the pre-registered family: raw Welch $p > 0.21$, all Holm $p = 1.0$.

Arm	post_react	overall
A1 monolith-ERM	1.396 ± 0.051	1.285 ± 0.017
A2 router	1.403 ± 0.050	1.296 ± 0.020
A3 recent-perf	1.398 ± 0.075	1.299 ± 0.025
A4 random-fixed	1.373 ± 0.057	1.273 ± 0.027
A5 RISP-ERM	1.381 ± 0.056	1.276 ± 0.025
A6 RISP-INV	1.344 ± 0.064	1.263 ± 0.025
A7 monolith-INV	1.410 ± 0.057	1.290 ± 0.018
A8a Hedge-fixed	1.325 ± 0.053	1.239 ± 0.023
A8b Hedge-learners	1.400 ± 0.059	1.285 ± 0.017
A9 oracle-pinned-ERM	1.342 ± 0.052	1.268 ± 0.027

All ten arms land in a band of $1.32\text{--}1.41 \times 10^{-2}$ post-reactivation; the six pre-registered comparisons return raw $p \in \{0.22, 0.23, 0.40, 0.57, 0.71, 0.95\}$ and Holm $p = 1.0$ throughout. Two details confirm that this is structure-absence rather than mere noise. First, the numerically best post-reactivation arm is *A8a*—the frozen-expert arm that cannot adapt to anything—which is exactly what one expects when there is nothing regime-specific to retain or relearn: not adapting is as good as adapting. Second, A9 (oracle labels!) is statistically indistinguishable from A6 with learned labels ($p = 0.95$): even perfect knowledge of the regime labels confers nothing, because the labels do not predict decision-relevant structure (which is what E0 measured directly).

Key Point

The protocol is falsifiable, and falsifies. The identical machinery—same arms, same metrics, same test family, same Holm correction—produces Holm $p \approx 10^{-5}$ separations where the generator contains dormant-but-persistent structure (E1), and a uniform $p > 0.2$ flatline where a pre-registered diagnostic says the structure is absent ($E0 \rightarrow E1s$). A pipeline that can only return positives is untrustworthy on synthetic data too; the E1s flatline is what makes the E1 separation credible.

21 E2–E4: Boundary Sweeps

The headline lives at one point in configuration space ($K = 2$, dormancy ≥ 90 , heterogeneity 1.0). E2–E4 sweep the three axes the theory says should govern the effect: capacity pressure, dormancy length, and cross-regime heterogeneity.

21.1 E2: capacity

The retention argument requires capacity pressure: a monolith forgets a dormant regime only if serving the live regimes forces reuse of its capacity. As $K \rightarrow R$ the pressure vanishes and the

retention gap should close. E2 sweeps $K \in \{1, 2, 3, 4\}$ ($R = 4$) under two memory models: *hard* (slot eviction) and *soft* (graded interference decay).

Table 6: E2 capacity sweep, `post_react` ($\times 10^{-2}$) $\pm$ 95% CI, hard memory. The pinned arms (A5, A6, A9) are exactly constant across K : each specialist serves only its own niches, so the capacity constraint never binds for them. Rightmost column: retention gap A1 – A5 ($\times 10^{-2}$).

K	A1	A2	A4	A5	A6	A9	A1–A5
1	3.427 ± 0.102	3.432 ± 0.100	3.667 ± 0.163	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.213
2	3.424 ± 0.095	3.408 ± 0.110	3.295 ± 0.126	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.210
3	3.303 ± 0.088	3.276 ± 0.064	3.202 ± 0.072	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.089
4	3.221 ± 0.082	3.228 ± 0.068	3.206 ± 0.078	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.007

Table 7: E2 capacity sweep, `post_react` ($\times 10^{-2}$) $\pm$ 95% CI, soft memory (no eviction; graded interference with overflow-scaled decay). Rightmost column: retention gap A1 – A5.

K	A1	A2	A4	A5	A6	A9	A1–A5
1	3.276 ± 0.079	3.285 ± 0.067	3.667 ± 0.163	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.062
2	3.277 ± 0.079	3.254 ± 0.077	3.318 ± 0.117	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.063
3	3.279 ± 0.079	3.265 ± 0.056	3.239 ± 0.071	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.065
4	3.277 ± 0.081	3.255 ± 0.064	3.273 ± 0.079	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.063

Four observations.

1. **Reward-independent arms are exactly flat.** A5, A6, and A9 post identical values at every K in both memory models. This is not a rounding artifact but a structural fact: a pinned specialist serves only its assigned niches, so its memory never overflows and the capacity model is inert for it. Flatness is the cleanest possible demonstration that the pinned arms’ advantage is *not* extra effective capacity—they use the same budget and simply never churn it.
2. **Gap closure at $K = R$ under hard memory.** The retention gap A1 – A5 falls from 0.213 at $K = 1$ to 0.007 at $K = 4$ —zero to within a tenth of a CI. With enough slots, the monolith never evicts, never forgets, and pinning buys nothing. The invariance edge survives, however: A6 (3.072) still beats everything at $K = 4$, confirming the axes are distinct (capacity governs retention, not robustness).
3. **Non-closure under soft memory—at any capacity.** With no eviction, capacity slots stop binding and the reward-driven arms go *flat* in K at ≈ 3.28 : gentler than hard eviction at tight capacity (3.43 at $K \leq 2$) but never closing the gap to the pinned arms (≈ 0.063 above A5, 0.21 above A6, at every K), because graded interference from foreign updates erodes dormant heads regardless of how many slots exist. The hard model predicts closure at $K = R$ and closes; the soft model predicts a capacity-independent residue and shows exactly that.
4. **A4’s coverage problem is worst at $K = 1$.** Random fixed niches post 3.667 ± 0.163 at $K = 1$ —the worst learner cell in the sweep—because with one slot per specialist a single unlucky assignment leaves a regime entirely uncovered. The cell’s CI (0.163) is the widest in the sweep, the variance signature of Section 19.2 amplified.

21.2 E3: dormancy length

Retention should matter more the longer a regime sleeps: the monolith’s relearning debt accumulates with dormancy, while a pinned specialist’s retained parameters do not decay (hard

memory). E3 sweeps the minimum dormancy threshold $D \in \{21, 63, 126, 252, 504\}$ days.

Table 8: E3 dormancy sweep ($\times 10^{-2}$, $\pm 95\%$ CI). Each D is a *different schedule generator* (shorter dormancies mean more frequent transitions and different steady-state mixes), so levels are comparable *within* a row block only, never across D . Rightmost columns: within- D retention+invariance gap A1 – A6, absolute and relative.

D	Arm	post_react	steady	A1 – A6 (post)	
21	A1	5.226 ± 0.148	3.225 ± 0.225	0.152	2.9%
	A2	5.241 ± 0.151	3.305 ± 0.213		
	A3	5.475 ± 0.169	3.504 ± 0.263		
	A5	5.198 ± 0.167	3.416 ± 0.220		
	A6	5.074 ± 0.142	3.293 ± 0.222		
63	A1	5.024 ± 0.202	2.719 ± 0.214	0.375	7.5%
	A2	5.028 ± 0.208	2.785 ± 0.226		
	A3	5.362 ± 0.173	2.973 ± 0.218		
	A5	4.818 ± 0.198	2.876 ± 0.208		
	A6	4.648 ± 0.193	2.776 ± 0.209		
126	A1	4.975 ± 0.160	2.550 ± 0.157	0.401	8.1%
	A2	4.960 ± 0.150	2.598 ± 0.156		
	A3	5.228 ± 0.147	2.747 ± 0.159		
	A5	4.787 ± 0.185	2.762 ± 0.141		
	A6	4.574 ± 0.181	2.649 ± 0.130		
252	A1	4.336 ± 0.098	2.633 ± 0.100	0.385	8.9%
	A2	4.344 ± 0.103	2.657 ± 0.096		
	A3	4.682 ± 0.103	2.811 ± 0.103		
	A5	4.095 ± 0.144	2.777 ± 0.099		
	A6	3.951 ± 0.103	2.652 ± 0.092		
504	A1	3.668 ± 0.059	2.530 ± 0.080	0.408	11.1%
	A2	3.646 ± 0.059	2.565 ± 0.080		
	A3	3.968 ± 0.049	2.713 ± 0.079		
	A5	3.454 ± 0.076	2.668 ± 0.077		
	A6	3.260 ± 0.061	2.533 ± 0.077		

Important

Do not compare levels across D blocks. A $D = 21$ generator produces a chaotic schedule in which *every* arm is perpetually post-transition (note the elevated steady values, 3.2–3.5, against 2.5–2.7 at $D = 504$); a $D = 504$ generator produces long calm stretches. The only licensed reading is the *within- D gap between arms*, which holds the schedule fixed.

The within- D gap A1 – A6 grows steeply from 0.152 at $D = 21$ to 0.375 at $D = 63$, then saturates in absolute terms ($0.375 \rightarrow 0.401 \rightarrow 0.385 \rightarrow 0.408$), while the relative gap continues to drift up ($7.5\% \rightarrow 11.1\%$) because the baseline level falls as schedules calm down. The shape is exactly the theory’s: at $D = 21$ the monolith’s parameters have barely drifted from the dormant regime, so there is little to retain; by $D \approx 63$ –126 (one to two quarters of dormancy) the monolith’s relearning debt is fully formed and further dormancy adds little, because the debt is bounded by “relearn from scratch.” The ordering $A6 < A5 < A1 \approx A2$, and A3 worst, holds at every D without exception.

21.3 E4: heterogeneity

The invariance axis should engage only insofar as regimes differ: at heterogeneity 0 all regimes share one mechanism and there is nothing to be invariant *across*. E4 sweeps the cross-regime heterogeneity scale $h \in \{0, 0.5, 1.0, 1.5, 2.0\}$ for the pinned arms.

Table 9: E4 heterogeneity sweep, `post_react` ($\times 10^{-2}$) $\pm$ 95% CI. Rightmost column: ERM–INV gap (A5 – A6).

h	A5 RISP-ERM	A6 RISP-INV	A9 oracle-pinned-ERM	A5–A6
0.0	2.998 ± 0.080	2.900 ± 0.079	2.997 ± 0.081	0.097
0.5	3.073 ± 0.081	2.948 ± 0.080	3.067 ± 0.073	0.125
1.0	3.214 ± 0.070	3.072 ± 0.066	3.210 ± 0.077	0.142
1.5	3.459 ± 0.087	3.249 ± 0.095	3.418 ± 0.094	0.210
2.0	3.749 ± 0.138	3.511 ± 0.124	3.679 ± 0.165	0.238

Three readings.

1. **Both arms degrade with heterogeneity, ERM faster.** Fitting endpoint slopes, A5 degrades at 0.376×10^{-2} per unit h against A6’s 0.306—the ERM arm is roughly 1.2 $\times$ more sensitive to heterogeneity than the INV arm, and the oracle-pinned ERM arm A9 sits in between (0.341). The cleaner statement is in the gap itself: the ERM–INV gap grows *monotonically* and roughly 2.4-fold across the sweep, from 0.0010 at $h = 0$ to 0.0024 at $h = 2$ (in raw units, $0.00097 \rightarrow 0.00238$). The invariance objective buys more the more the regimes differ—which is what an invariance objective is for.
2. **The $h = 0$ residual gap.** At zero heterogeneity the theory’s first-order prediction is gap = 0, yet A6 retains a small, CI-overlapping but consistently signed edge (0.097×10^{-2}). The likely mechanism is mundane and worth naming: with $d = 20$ features of which only a subset is causal, the cross-environment variance penalty also suppresses reliance on *spuriously correlated distractor features within* a niche—it acts as a targeted regularizer even when there is no cross-regime heterogeneity to be invariant to. We flag this as a confound on the pure invariance interpretation at low h : part of the A5–A6 gap at every h is plain regularization, and only the *growth* of the gap with h (the $0.0010 \rightarrow 0.0024$ monotone trend) is unambiguously attributable to invariance proper.
3. **A5 tracks A9 everywhere.** Learned pinning matches oracle pinning under ERM at every heterogeneity ($\leq 0.041 \times 10^{-2}$ apart, well inside CIs), extending the P7 skyline result along the whole sweep.

22 E5–E6: Ablations and the Inversion Audit

22.1 E5: ablations of the full system

E5 ablates A6’s four design choices one at a time, holding everything else at the headline configuration; the response variable is `post_react` ($\times 10^{-2}$, 20 seeds).

The invariance weight β . The response is a clean U-shape with minimum at $\beta = 1$ and a broad plateau over $\beta \in [0.25, 4]$ (all within $\sim 3\%$ of the minimum, CIs overlapping). Both extremes collapse, and the collapses are informative:

- At $\beta = 0$ the within-niche objective degenerates to the cross-environment variance term alone, and performance (3.555) falls not merely to the retained-ERM level but *below* it (A5 = 3.214; the variance-only arm is 10.6% worse). This is precisely the degeneracy the

Table 10: E5 β sweep (A6 variant, `post_react`, $\times 10^{-2}$). Reference points: A5 retained-ERM = 3.214 ± 0.070 ; headline A6 ($\beta = 1$) = 3.072 ± 0.066 .

β	0	0.25	1	4	16
<code>post_react</code>	3.555 ± 0.107	3.090 ± 0.079	3.072 ± 0.066	3.170 ± 0.076	3.522 ± 0.065

theory predicts (and that the V-REx analysis warns of [20]): a variance-only criterion is minimized by any predictor that *equalizes* risk across environments, including uniformly bad constant-like predictors. Equal risks are not low risks. The empirical collapse below the ERM reference is the theory’s degenerate solution showing up on schedule.

- At $\beta = 16$ the risk term swamps the variance term and performance (3.522) gives back the entire invariance gain and more, landing 14.6% above the $\beta = 1$ optimum and visibly worse than plain retained ERM. We read the overshoot beyond A5 cautiously (the heavily re-weighted objective also rescales effective step sizes), but the qualitative point stands: the benefit requires *both* terms in balance, and a factor-of-16 window around the default (0.25–4) is all flat—the choice $\beta = 1$ is not a tuned knife-edge.

Table 11: E5 W_c sweep (`post_react`, $\times 10^{-2}$).

W_c	1	5	20	60
<code>post_react</code>	3.033 ± 0.076	3.057 ± 0.069	3.072 ± 0.066	3.090 ± 0.070

The environment window W_c . We had entertained a deflationary explanation of the invariance gain: that chunking each niche’s stream into W_c -day pseudo-environments effectively *batches* the updates, and the gain is a batching/smoothing artifact rather than an invariance effect. The sweep refutes this cleanly, and we report it as the refutation it is: across a $60\times$ range of W_c the response is flat to within 0.057×10^{-2} —less than one CI half-width—and the numerically best cell is $W_c = 1$, i.e. *no* batching at all. If batching were the active ingredient, $W_c = 1$ would be the worst cell, not the best. The pseudo-environment construction matters for defining the variance penalty, not for any smoothing side effect.

The assignment-smoothing parameter λ . With λ off (0): 3.141 ± 0.082 ; on (0.3): 3.072 ± 0.066 . The unpaired CIs overlap, so by the conservative standard of Section 18.2 the effect is not individually established. The paired within-seed contrast, which is legitimate here, gives $+0.069 \pm 0.060$ ($\times 10^{-2}$, mean $\pm$ 95% CI)—a 2.2% improvement whose interval excludes zero, barely. Honest summary: λ helps a little, with weak evidence; nothing in the headline depends on it.

Pinning schedule: pinned vs. never. Removing dormancy-pinning entirely (assignments stay perpetually plastic) degrades `post_react` from 3.072 ± 0.066 to 3.197 ± 0.092 ; the paired contrast is $+0.126 \pm 0.059$ ($\times 10^{-2}$), a 3.9% loss that is clearly resolved. Never-pinning gives back most of the invariance-axis gain (the never-pin arm sits near the A5/A9 retained-ERM level of 3.21), which is consistent with the interaction story of Section 19.4: a perpetually plastic assignment lets reactivation-time reassignment churn destroy exactly the retained-invariant parameters the mechanism exists to protect.

22.2 E6: the SNR audit and the inversion

Development history, stated plainly. This experiment exists because our first prototype failed. The initial generator was, we later understood, calibrated to an effectively high signal-to-noise ratio, and the arm lineup showed no positive separation—RISP did not beat the monolith. The tempting path was to tune the generator until the predicted effect appeared and report that point. Instead we made SNR an explicit axis, swept it, and report the whole curve, including the region where our method *loses*. The audit below is the honest version of what could otherwise have been a silent hyperparameter search.

Table 12: E6 SNR audit ($\times 10^{-2}$, $\pm$ 95% CI, 20 seeds per cell). ρ = post/steady relearning-penalty ratio. “A6 adv.” = $(A1 - A6)/A1$ on `post_react`; positive favors RISP.

SNR	Arm	post_react	steady	ρ	A6/A9 adv.
0.5×	A1	3.778 ± 0.077	2.963 ± 0.057	1.28	
	A6	3.466 ± 0.073	2.890 ± 0.066	1.20	+8.3%
	A9	3.584 ± 0.068	3.000 ± 0.068	1.19	+5.1%
1×	A1	3.424 ± 0.095	2.459 ± 0.058	1.39	
	A6	3.072 ± 0.066	2.507 ± 0.074	1.23	+10.3%
	A9	3.210 ± 0.077	2.624 ± 0.088	1.22	+6.3%
2×	A1	2.990 ± 0.075	1.878 ± 0.062	1.59	
	A6	2.779 ± 0.117	2.273 ± 0.110	1.22	+7.0%
	A9	2.908 ± 0.121	2.369 ± 0.128	1.23	+2.7%
4×	A1	2.733 ± 0.070	1.480 ± 0.091	1.85	
	A6	3.089 ± 0.265	2.577 ± 0.217	1.20	−13.1%
	A9	3.210 ± 0.283	2.660 ± 0.250	1.21	−17.5%
8×	A1	3.286 ± 0.155	1.594 ± 0.210	2.06	
	A6	4.905 ± 0.620	4.117 ± 0.465	1.19	−49.3%
	A9	4.939 ± 0.629	4.194 ± 0.535	1.18	−50.3%

The inversion. The RISP advantage on the primary endpoint is +8.3% at 0.5×, peaks at +10.3% at the headline 1×, narrows to +7.0% at 2×, then *inverts*: −13.1% at 4× and −49.3% at 8×. The crossover sits between 2× and 4×, i.e. around 2–3× the headline SNR.

The relearning-penalty ratio isolates the mechanism. The post/steady ratio ρ tells the mechanistic story with unusual clarity. A6’s ratio is *flat* across the entire sweep: 1.20, 1.23, 1.22, 1.20, 1.19. The retention mechanism works identically at every SNR—a pinned specialist’s reactivation penalty is always the same modest $\sim 20\%$ premium over its own steady state. A1’s ratio, by contrast, climbs monotonically: 1.28, 1.39, 1.59, 1.85, 2.06. The monolith’s *relative* relearning penalty gets worse with SNR, not better. So why does the monolith win at high SNR? Because its *absolute levels* collapse: A1’s steady regret falls from 2.96 at 0.5× to 1.48 at 4×, while A6’s steady regret *deteriorates* ($2.89 \rightarrow 2.58 \rightarrow 4.12$ from 0.5× through 8×, non-monotonically). At high SNR a pooled learner extracts the strong signal from all data jointly and relearns a reactivated regime within days—faster than the probe window can penalize it—whereas each pinned specialist is confined to its own niche’s fraction of the data and carries a variance penalty calibrated for a noisier world. Pooling beats partitioning when the signal is strong enough to relearn cheaply. (A1’s own non-monotonicity at 8×—post-react 3.29 vs 2.73 at 4×, with steady CIs widening—suggests additional dynamics at extreme SNR, e.g. aggressive-fit instability, that we have not dissected; we note it rather than interpret it.)

The skyline inverts identically. The decisive control: A9, the *oracle-assigned* pinned arm, inverts in lockstep with A6 (−17.5% at 4×, −50.3% at 8×; their CIs overlap heavily at both points). The inversion therefore cannot be an artifact of RISP’s learned assignment, its pinning

heuristic, or any estimation failure in our mechanism—perfect assignment inverts too. The inversion is a property of the *regime*: partitioned, reward-independent memory is the right architecture in low-SNR, slow-relearning environments and the wrong one in high-SNR, fast-relearning environments. The boundary is part of the claim, not a caveat appended to it.

Key Point

Deployment-boundary reading. The audit gives an operational rule: RISP-style partitioned retention should be deployed only where the post-reactivation relearning cost of a pooled learner is demonstrably large—empirically, below roughly $2\text{--}3\times$ the headline SNR in this family, i.e. in environments where a monolith’s post/steady ratio is materially elevated ($\rho \gtrsim 1.4$). The monolith’s measured ρ is itself an observable diagnostic: it can be estimated on any stream *before* committing to the architecture. Financial daily data, where the E0 diagnostic already found no exploitable conditional structure at all, sits firmly outside the deployment region; the mechanism’s natural habitat is low-SNR streams with long-dormancy recurring structure.

Intuition

A useful way to compress E6: *retention is a hedge against expensive relearning*. The price of the hedge is the partitioning cost (less data per specialist, a variance penalty, a worse steady state). The value of the hedge is the relearning debt it cancels. Low SNR makes relearning slow and expensive, so the hedge is cheap relative to its payout; high SNR makes relearning nearly free, so the hedge is all premium and no payout. The post/steady ratios are precisely the two sides of this ledger, read off the data.

23 Threats to Validity

We enumerate the limits of the evidence, in roughly decreasing order of importance, with what each would take to overturn.

1. **The headline is synthetic, and the real-data result is null.** This is the central limitation and we state it without hedging: every positive number in Sections 19–22 comes from a generator we wrote. E0/E1s show that the mechanism’s precondition is absent in the real universes we tested, so the contribution is a *mechanism*, a *boundary map*, and a *falsifiable deployment gate*—not a demonstrated real-market improvement. A reader who needs the latter should treat this paper as a protocol to run on their own data, starting with E0.
2. **Small real universes.** The E0/E1s null rests on five crypto pairs and three commodities at daily frequency with one feature set and two labelers. The null is well-measured on that data ($|z| < 0.7$ across all four cells) but could fail to generalize to intraday frequencies, equities cross-sections, or richer conditioning information. The gate is cheap to rerun; we make no claim beyond the cells in Table 4.
3. **Single decision-layer family.** All arms share one decision layer (top- K portfolio from predicted returns). The decision-space dissociation literature [1, 6] cuts both ways: just as prediction nulls did not settle the decision question, our decision-layer-specific results do not settle other decision layers (e.g. mean-variance with estimated covariances, or execution-aware objectives). The within-family rankings are clean; the family is one point in a larger space.

4. **Linear specialists.** Specialists are linear in the $d = 20$ features. Retention-vs-relearning economics could shift with high-capacity nonlinear specialists, in either direction (faster relearning hurts retention’s value, as E6 shows; worse interference helps it). The GAUSE neural-expert results [14] suggest the reward-independence principle transfers to gradient-trained experts, but we have not verified that here.
5. **Task-incremental labels are given.** Arms receive the regime label stream (synthetic: true labels; E1s: causal labeler output). The harder class-incremental problem—detecting reactivations without labels—is not addressed, and the GAUSE real-data experience (label-free recovery collapsing under feature overlap) warns against assuming it is easy.
6. **Hyperparameter choices.** Each load-bearing choice is ablated where it matters: β has a $16\times$ flat plateau (Table 10), W_c is flat over $60\times$ (Table 11), λ is nearly irrelevant, pinning matters and is the mechanism itself. The probe length (15 days) and dormancy threshold (90 days) were fixed before the final run and the E3 sweep shows the effect is not knife-edged in the dormancy axis; we did not sweep probe length, and the post/steady split depends on it mechanically.
7. **Seed count and inference.** $n = 20$ seeds powers the pre-registered effects (Section 18.6) but leaves the marginal comparison A6 vs A9 unresolved (Holm $p = 0.057$) and gives only moderate resolution on variance signatures (A4). The conservative unpaired tests mean reported significances are, if anything, understated.
8. **What would change our conclusions.** Concretely: (i) an E0 run on some real universe showing a robust positive gap, followed by an E1s-style run in which arms *fail* to separate, would break the protocol’s central link; (ii) a demonstration that A2-style routers with explicit idle-expert protection match A6 would collapse the reward-independence claim to an implementation detail; (iii) an E6-style audit on a low-SNR regime showing the monolith’s ρ elevated but no RISP advantage would falsify the deployment gate. All three are runnable with the released code, and the third is the one we would run first.

Key Point

Summary of Part V. On a generator with the hypothesized structure: retention is worth 6.1%, invariance 4.4%, jointly 10.3% (38.1% of excess-over-floor), with a super-additive interaction ($I = -0.0015 \pm 0.0005$) and a reached skyline (A6 $\approx$ A10, $p = 0.72$); the effect requires capacity pressure (gap $\rightarrow 0.007$ at $K = R$, hard memory), grows then saturates with dormancy, scales with heterogeneity (gap $0.0010 \rightarrow 0.0024$), and survives every ablation with flat plateaus. On real daily crypto/commodities the precondition is absent (all E0 $|z| < 0.7$) and the arms correctly fail to separate (all E1s $p > 0.2$). At high SNR the advantage inverts (-49.3% at $8\times$), the oracle skyline inverts with it, and the monolith’s own post/steady ratio provides a measurable deployment gate. The mechanism is real, bounded, and honest about where it lives.

Part VI

Implementation Walkthrough

The preceding parts developed the theory at the blackboard. This part walks the actual code. The entire system—decision layer, synthetic market, schedules, capacity-bounded specialists, all nine allocation arms, the run loop, and the statistics—is roughly 900 lines of seeded `numpy` across three files: `risp.py` (decision layer, market, schedules, `Specialist`, arms A1–A10, `run_arm`,

statistics), `realdata.py` (loaders, causal labelers, features, block library, stitched market), and `run_experiments.py` (drivers e0–e6). There is no GPU, no deep-learning framework, and no external optimizer. The full experimental battery reruns from scratch in under an hour on a laptop, and every number quoted in the paper traces to a specific JSON file written by a specific driver function (Section 28).

Key Point

Smallness is a feature, not a budget constraint. Because the predictors are linear, the decision oracle is a sort, and the surrogate is SPO+, every mechanism claimed in the theory—interference under capacity, dormancy starvation, episode-spurious overfitting, reward-decoupled niche assignment—is visible in the code as a handful of lines, and nothing else is present that could secretly produce the effects.

24 The Decision Layer in Code

The decision set is the cardinality-constrained long-only portfolio $\mathcal{Z} = \{z \in [0, w_{\max}]^n : \mathbf{1}^\top z \leq W, \text{RISP}_{z_0} \leq k\}$ with linear utility $F(z, y) = y^\top z$. With the budget non-binding ($W \geq k w_{\max}$), the exact maximizer puts $w_{\max}$ on the k largest *positive* coordinates of the score vector. The whole decision layer is three functions:

```

1 def solve_topk(y_hat: np.ndarray, k: int, w_max: float) -> np.ndarray:
2     """Exact solution of max_z y_hat^T z over Z. 0(n log n)."""
3     n = y_hat.shape[0]
4     z = np.zeros(n)
5     idx = np.argsort(-y_hat)[:k]
6     pos = idx[y_hat[idx] > 0.0]
7     z[pos] = w_max
8     return z
9
10
11 def regret(y_hat: np.ndarray, y: np.ndarray, k: int, w_max: float) ->
float:
12     """Decision regret rho(y_hat, y) = F(z*(y), y) - F(z(y_hat), y) >= 0."""
13     z_star = solve_topk(y, k, w_max)
14     z_hat = solve_topk(y_hat, k, w_max)
15     return float(y @ z_star - y @ z_hat)
16
17
18 def spo_plus_loss_grad(w: np.ndarray, X: np.ndarray, y: np.ndarray,
19                        k: int, w_max: float):
20     """SPO+ loss and subgradient for the linear predictor y_hat = Xw.
21
22     Cast as minimization with cost c = -y, c_hat = -Xw.
23     l_SPO+(c_hat, c) = max_z (c - 2c_hat)^T z + 2c_hat^T z*(c) - c^T z*(c)
24     d_l/d c_hat = 2(z*(c) - z*(2c_hat - c)); d c_hat/d w = -X.
25     """
26     y_hat = X @ w
27     z_star = solve_topk(y, k, w_max) # argmin c^T z = argmax
        y^T z
28     # z*(2 c_hat - c) = argmin (2 c_hat - c)^T z = argmax (2 y_hat - y)
        ^T z
29     z_alt = solve_topk(2.0 * y_hat - y, k, w_max)
30     # loss value (for monitoring / variance penalty)

```

```

31     loss = float((2.0 * y_hat - y) @ z_alt - 2.0 * y_hat @ z_star + y @
32                  z_star)
33     grad_chat = 2.0 * (z_star - z_alt)
34     grad_w = -X.T @ grad_chat
    return loss, grad_w

```

Listing 1: The decision layer: exact oracle, regret, and the SPO+ subgradient (`rispy.py`).

Line-by-line.

- **solve_topk**: `np.argsort(-y_hat)[:k]` selects the k largest scores; the filter `y_hat[idx] > 0.0` drops any of them that are negative (a long-only book never holds an asset it expects to lose money on); the survivors receive weight $w_{\max}$. This is the exact $\arg \max_{z \in \mathcal{Z}} \hat{y}^\top z$, in $O(n \log n)$.
- **regret**: two oracle calls—one on the realized y (the hindsight-optimal book $z^*(y)$), one on the prediction $\hat{y}$ (the served book)—and the difference of realized utilities. This is the SPO decision regret $\rho(\hat{y}; y) \geq 0$ used as the metric *everywhere*: it is the per-day loss in `run_arm`, the gate signal in the router, the competition score in RISP, and the weight signal in both Hedge arms.
- **spo_plus_loss_grad**: the minimization-form bookkeeping. SPO+ is stated for cost *minimization*, so we set $c = -y$ and $\hat{c} = -Xw$. The loss

$$\ell_{\text{SPO}^+}(\hat{c}, c) = \max_{z \in \mathcal{Z}} (c - 2\hat{c})^\top z + 2\hat{c}^\top z^*(c) - c^\top z^*(c)$$

translates, under the sign flip, into exactly the line computing **loss**: the inner max over $(c - 2\hat{c})$ becomes a max over $(2\hat{y} - y)$ (the comment on `z_alt` spells this out), and the remaining two terms become $-2\hat{y}^\top z^* + y^\top z^*$.

- The subgradient requires *two oracle calls*: one at the true cost $(z^*(c))$, i.e. `z_star` and one at the reflected cost $2\hat{c} - c$ (i.e. `z_alt`). Then $\partial \ell / \partial \hat{c} = 2(z^*(c) - z^*(2\hat{c} - c))$ is **grad_chat**, and the chain rule through $\hat{c} = -Xw$ contributes the $-X^\top$ in **grad_w**. Note the two sign flips cancel into readable code: a head that under-ranks an asset the hindsight book holds gets pushed *toward* it.

Key Point

No Gurobi needed. Because the feasible set is a cardinality polytope with a non-binding budget and the objective is linear, the oracle is exact and costs one sort. Both the loss and the subgradient of SPO+ only ever touch the optimization problem through the oracle $z^*(\cdot)$ —**solve_topk** is called twice per gradient, never differentiated through. If the decision problem were upgraded (transaction costs, sector constraints, integer lots), the *only* change is to swap **solve_topk** for a general solver call; **regret** and **spo_plus_loss_grad** are already written against the oracle interface and would not change by a character.

25 Markets and Schedules

25.1 The synthetic generator

The synthetic market implements the structural model of Part II: per-regime invariant loadings θ_r , per-episode spurious loadings $\gamma_{r,e}$, distractor features, and regime-dependent noise.

```

1 @dataclass
2 class SynthConfig:
3     n_assets: int = 20
4     d: int = 20                                # x1,x2 invariant; x3 spurious; 16
5         distractors; const
6     k: int = 4
7     w_max: float = 0.25
8     R: int = R_DEFAULT
9     noise: float = 0.02
10    het: float = 1.0                            # episode heterogeneity scale (E4
11        sweep)
12    gamma_scale: float = 0.012
13    theta_scale: float = 0.012                # |theta_r| ~ realistic daily alpha
14        scale
15    vol_mult: tuple = (1.0, 0.75, 2.0, 3.0)    # per-regime noise
16        multiplier
17    snr_mult: float = 1.0                    # audit sweep: scales signal vs
18        noise
19
20    # ... inside SyntheticMarket ...
21    def day(self, r: int, e: int):
22        """One day's  $X, y$  in regime  $r$ , episode  $e$ ."""
23        c = self.cfg
24        X = np.ones((c.n_assets, c.d))
25        X[:, :c.d - 1] = self.rng.normal(0, 1, (c.n_assets, c.d - 1))
26        g = self.gamma_for(r, e)
27        mean = X[:, :2] @ self.theta[r] + g * X[:, 2] + self.b[r]
28        y = mean + self.rng.normal(0, c.noise * c.vol_mult[r], c.
29            n_assets)
30        return X, y

```

Listing 2: SynthConfig and the day generator (risp.py).

Each configuration field maps onto a quantity from the theory:

- **theta_scale** and the constructor loop (not shown) place the invariant loadings $\theta_r = \text{theta_scale} \cdot (\cos(2\pi r/R + 0.4), \sin(2\pi r/R + 0.4))$ on well-separated directions of the unit circle. This is what makes cross-regime interference *real*: a head dragged toward regime q 's optimum genuinely loses regime r 's.
- **gamma_scale** and **het** control the episode weather $\gamma_{r,e} \sim \mathcal{N}(0, (\text{het} \cdot \text{gamma_scale})^2)$, drawn lazily and memoized in **gamma_for** so that revisiting episode e of regime r replays the same weather. The spurious feature is column $X[:, 2]$ in the mean: an ERM learner profitably loads on it within an episode, then pays at the next episode when γ is redrawn. **het** is the dial swept in E4.
- Columns 3 through $d - 2$ are pure distractors—they enter X but never the mean—so the predictor has plenty of rope to overfit noise, and column $d - 1$ is the constant absorbing the regime intercept $\mathbf{b}[r]$.
- **vol_mult** multiplies the noise s.d. per regime ($3\times$ in crisis), so the hard regimes are hard both because they are rare *and* because they are noisy.
- **snr_mult** scales θ_r and $\gamma_{r,e}$ jointly against fixed noise; it is the dial for the E6 audit (at what SNR does retention stop mattering because relearning is instant?).

25.2 Schedules and reactivation bookkeeping

A Schedule is four aligned arrays: the regime id per day, the episode index per day, a `block_start` flag, and—only meaningful on block starts—the `dormancy` length that preceded the block.

```
1 def make_schedule(rng: np.random.Generator, R: int = R_DEFAULT,
2                 n_cycles: int = 18, block_len=(25, 60),
3                 crisis_every: int = 4, crisis_len=(15, 30)) ->
4                 Schedule:
5     """Common regimes (0,1,2) alternate in blocks; crisis (R-1) appears
6         only
7     every 'crisis_every' cycles -> long dormancy, recurring
8     reactivations."""
9     seq = []
10    cyc = 0
11    common = [0, 1, 2] if R >= 4 else list(range(R - 1))
12    while cyc < n_cycles:
13        order = rng.permutation(common)
14        for r in order:
15            L = int(rng.integers(block_len[0], block_len[1] + 1))
16            seq.append((int(r), L))
17        cyc += 1
18        if cyc % crisis_every == 0:
19            L = int(rng.integers(crisis_len[0], crisis_len[1] + 1))
20            seq.append((R - 1, L))
21    return _seq_to_schedule(seq, R)
22
23 def make_dormancy_schedule(rng: np.random.Generator, D: int,
24                           R: int = R_DEFAULT, n_react: int = 8,
25                           block_len=(25, 60), crisis_len: int = 20) ->
26                           Schedule:
27     """E3: focal regime R-1 reactivates every ~D days of dormancy
28         exactly."""
29     seq = []
30     common = [0, 1, 2] if R >= 4 else list(range(R - 1))
31     # initial training occurrences of focal regime
32     for r in [R - 1, 0, R - 1, 1, R - 1, 2]:
33         L = crisis_len if r == R - 1 else int(rng.integers(*block_len))
34         seq.append((int(r), L))
35     for _ in range(n_react):
36         filler = 0
37         while filler < D:
38             r = int(rng.choice(common))
39             L = int(rng.integers(block_len[0], block_len[1] + 1))
40             seq.append((r, L))
41             filler += L
42         seq.append((R - 1, crisis_len))
43     return _seq_to_schedule(seq, R)
```

Listing 3: Schedule builders and dormancy bookkeeping (`risp.py`).

Both builders emit a list of (regime, length) pairs and hand it to `_seq_to_schedule`, which does the bookkeeping that the reactivation metric depends on: for each block it increments the per-regime episode counter, sets `block_start[t]` on the first day, and records `dormancy[t] = t - last_seen[r]`, where `last_seen[r]` is the *final day* of the regime's previous block. The method

```

1 def reactivation_days(self, min_dormancy: int):
2     """Days that begin a block whose regime was dormant >=
3         min_dormancy."""
4     out = []
5     for t in range(self.T):
6         if self.block_start[t] and self.dormancy[t] >= min_dormancy:
7             out.append(t)
8     return out

```

is then the single source of truth for which days count as reactivations: `run_arm` masks the first `probe` (default 15) days after each such t as the post-reactivation window, and only windows in the second half of the run ($t \geq T/2$) enter the headline metric, so warm-up episodes never contaminate it. `make_schedule` gives irregular dormancy by rationing crisis to every fourth cycle; `make_dormancy_schedule` gives *controlled* dormancy of $\approx D$ days for E3, with three training occurrences of the focal regime up front so there is expertise to forget.

25.3 Causal labels on real data

On real data the regimes are not given; they are computed by labelers that must not leak day- t information into the day- t label, because the label selects which head serves the day- t decision. The L1 labeler is worth quoting in full:

```

1 def label_L1(panel: pd.DataFrame, vol_win=20, trend_win=50,
2             vol_pct_win=252) -> np.ndarray:
3     """Vol band (rolling-percentile of trailing vol) x trend sign.
4         Causal:
5         every input is shifted one day so day t uses data through t-1."""
6     px = panel.mean(axis=1) if panel.shape[1] > 1 else panel.iloc[:, 0]
7     # equal-weight index in log space to avoid scale domination
8     px = np.log(panel).mean(axis=1)
9     ret = px.diff()
10    vol = ret.rolling(vol_win).std().shift(1)
11    vol_rank = vol.rolling(vol_pct_win, min_periods=60).rank(pct=True)
12    trend = (px.shift(1) - px.shift(trend_win + 1))
13    high_vol = (vol_rank > 0.7).astype(int)
14    up = (trend > 0).astype(int)
15    lab = high_vol * 2 + (1 - up) # 0 calm-up 1 calm-down 2 vol-
    up 3 vol-down
    return lab.values

```

Listing 4: The L1 causal labeler (`realdatalabeler.py`).

The leakage prevention is the `.shift(1)` discipline, and each occurrence matters:

- `vol = ret.rolling(vol_win).std().shift(1)`: the rolling window ending at t would include the day- t return r_t —which is the prediction target. The `shift(1)` moves the window end to $t - 1$, so the volatility used to label day t is the s.d. of $r_{t-20}, \dots, r_{t-1}$.
- `vol_rank`: the percentile rank is computed over the already-shifted `vol` series, so the trailing-year band threshold (> 0.7) also sees nothing past $t - 1$. Ranking rather than thresholding in absolute units keeps the band meaningful across the secular volatility decay of the crypto sample.
- `trend = px.shift(1) - px.shift(trend_win + 1)`: the 50-day trend ends at the close of $t - 1$ (note the matching $+1$ on *both* shifts: the window is $[t - 51, t - 1]$, length 50, never touching t).

The same convention governs the features in `build_xy` (`mom5 = lp.shift(1) - lp.shift(6)`, `vol20 = r1.rolling(20).std().shift(1)`, etc.): all features are functions of information through $t - 1$, while the target $y = r1$ is the day- t return. The L2 labeler is causal by construction rather than by shifting: it is a forward-only two-state Hamilton filter whose update at day t uses `ret[t - 1]` (“yesterday’s return only,” as the inline comment says), crossed with the same shifted trend sign.

Intuition

The block-shuffle control in E0 (`block_shuffle` in `run_experiments.py`) is the converse check: it keeps the block *structure* of the labels but randomizes block *identities*. If regime-conditioned models beat block-shuffled ones on decision regret, the labels carry real conditional structure, not just autocorrelation.

26 The Specialist and Memory Models

A **Specialist** is a dictionary of at most K linear heads (one per held regime), an episode-indexed replay buffer per held regime, and a training rule (ERM or invariance). Memory pressure is implemented in `_touch`:

```

1  def _touch(self, r: int):
2      self.t += 1
3      if r not in self.heads:
4          if len(self.heads) >= self.K:
5              lru = min(self.heads, key=lambda q: self.heads[q].
6                      last_used)
7              del self.heads[lru]
8              self.buf.pop(lru, None)
9              self.heads[r] = Head(w=np.zeros(self.d))
10             self.buf[r] = {}
11             self.heads[r].last_used = self.t
12             if self.memory == "soft":
13                 for q, h in self.heads.items():
14                     if q != r:
15                         h.strength *= (1.0 - self.rho_int)
16                         h.w *= (1.0 - self.rho_int)
17             self.heads[r].strength = 1.0

```

Listing 5: Memory bookkeeping: hard LRU and soft interference (`risp.py`).

Hard model. When a new regime arrives at a full specialist ($\text{len}(\text{self.heads}) \geq K$), the least-recently-used head is deleted *together with its buffer*: `self.buf.pop(lru, None)`. This is the line that makes forgetting total in the hard model—knowledge lost is data lost, so the evicted regime must be relearned from scratch at its next reactivation, with no replay shortcut. The fresh head starts at $w = 0$, which under `solve_topk` means a near-random book (the `predict` fallback for unheld regimes is an explicit 10^{-6} -scale random tiebreaker for the same reason: zero scores would deterministically pick no assets).

Soft model. Under `memory == "soft"`, every training day on regime r multiplies every *other* held head’s weights and strength by $(1 - \rho_{\text{int}})$, with $\rho_{\text{int}} = 0.06$ by default. Untouched expertise decays geometrically toward zero at a rate set by how often its owner trains on something else—the continuous analogue of LRU eviction, matching the interference operator of Part II. The active head’s strength snaps back to 1.

Training itself lives in `learn`:

```

1  def learn(self, X, y, r: int, e: int, n_steps: int = 2):
2      """One observation: store, touch memory, take SGD steps on the
3      regime-r objective over the episode buffer."""
4      self._touch(r)
5      self.store(X, y, r, e)
6      h = self.heads[r]
7      eps = self.buf[r]
8      for _ in range(n_steps):
9          if self.mode == "erm" or len(eps) < 2:
10             # plain ERM on pooled buffer
11             keys = list(eps.keys())
12             ekey = keys[int(self.rng.integers(len(keys)))]
13             Xb, yb = ep_sample(eps[ekey], self.rng)
14             _, g = spo_plus_loss_grad(h.w, Xb, yb, self.k, self.
15                                     w_max)
16             h.w -= self.lr * g / max(1, Xb.shape[0])
17         else:
18             # invariance: Var_e[lbar_e] + beta * E_e[lbar_e]
19             losses, grads = [], []
20             for ekey in eps:
21                 Xb, yb = ep_sample(eps[ekey], self.rng)
22                 l, g = spo_plus_loss_grad(h.w, Xb, yb, self.k, self.
23                                         w_max)
24                 losses.append(l / max(1, Xb.shape[0]))
25                 grads.append(g / max(1, Xb.shape[0]))
26             E = len(losses)
27             lbar = float(np.mean(losses))
28             g_tot = np.zeros(self.d)
29             for l, g in zip(losses, grads):
30                 g_tot += (2.0 * (1 - lbar) + self.beta) * g / E
31             h.w -= self.lr * g_tot

```

Listing 6: ERM vs. invariance training over the episode buffer (rispy.py).

Line-by-line.

- The buffer `self.buf[r]` maps episode index e to a list of stored (X, y) days, capped at `buf_days = 40` days per episode and `buf_episodes = 6` episodes per regime (`store` evicts oldest-first on both axes). Episode identity is the unit the invariance objective averages over.
- *ERM branch*: sample one episode uniformly at random, draw a minibatch of up to 8 stored days from it (`ep_sample`), take one SPO+ step. Over many steps this pools episodes—exactly the estimator that cannot distinguish θ_r from the episode weather $\gamma_{r,e}$.
- *INV branch*: evaluate the per-episode mean losses $\bar{\ell}_e$ and mean gradients g_e on a fresh minibatch from *every* buffered episode, then assemble the gradient of $\text{Var}_e[\bar{\ell}_e] + \beta \mathbb{E}_e[\bar{\ell}_e]$. The identity used is

$$\nabla_w \text{Var}_e[\bar{\ell}_e] = \frac{2}{E} \sum_e (\bar{\ell}_e - \bar{\ell}) (g_e - \bar{g}) = \frac{2}{E} \sum_e (\bar{\ell}_e - \bar{\ell}) g_e,$$

where the $\bar{g}$ term vanishes because $\sum_e (\bar{\ell}_e - \bar{\ell}) = 0$; adding $\nabla_w \beta \mathbb{E}_e[\bar{\ell}_e] = \frac{\beta}{E} \sum_e g_e$ gives

$$\nabla_w [\text{Var}_e + \beta \mathbb{E}_e] = \sum_e \frac{(2(\bar{\ell}_e - \bar{\ell}) + \beta) g_e}{E},$$

which is verbatim the accumulation line `g_tot += (2.0 * (1 - lbar) + self.beta) * g / E`. Episodes whose loss is above the cross-episode mean get their gradients up-weighted; episodes below the mean can have *negative* effective weight when $\bar{\ell} - \bar{\ell}_e > \beta/2$, actively pushing the head away from solutions that win on one episode’s weather at others’ expense.

- The guard `len(eps) < 2` falls back to ERM until a second episode exists—a one-episode variance is identically zero, so the INV objective is undefined before then.

Key Point

Eviction travels with the buffer. A head that loses the LRU competition loses its replay data in the same statement, so no arm can cheat the capacity constraint by quietly re-learning from a retained buffer. Capacity K bounds *retained regimes*, in both parameters and data.

27 The Arms

All arms share one interface: `decide(X, r) -> y_hat` produces the prediction that is converted into the served book, and `observe(X, y, r, e, served_regret)` performs all internal updates. Signature-level summary:

- **MonolithArm** (A1 ERM / A7 INV): one capacity- K specialist that always learns the active regime; `observe` is a single `self.s.learn(X, y, r, e)`. This is the canonical reward-driven allocation—capacity follows activity, so dormant regimes are first in the LRU queue.
- **RecentPerfArm** (A3): the industry baseline. Capital shares are a softmax (temperature 0.02) of trailing 60-day negative regret per specialist; the served decision is the top-capital specialist’s; in `observe`, *every* specialist trains on the active regime with probability equal to its capital share, so starved specialists drift into the active regime at the capital-floor rate.
- **FixedNicheArm** (A4 random / A9, A10 oracle): a reward-independent owner map `self.owner[r]` fixed at construction (random claims, or hand-pinned $r \mapsto r \bmod N$ for the oracle skyline); `observe` trains the owner only.
- **HedgeFixedArm** (A8a): exponential weights over $2(d-1)$ fixed $\pm$ unit-direction strategies, weights decayed by realized regret; no learning. **HedgeLearnersArm** (A8b): Hedge over N learning specialists, where the current top-weight learner trains on the active regime—training follows the weight, so it is reward-coupled like the monolith, just with a softer selector.
- **RouterArm** (A2) and **RISPArm** (A5 ERM / A6 INV): quoted in full below; they are the two poles of the paper’s central contrast.

27.1 RouterArm: where reward-coupling enters

```

1  def observe(self, X, y, r, e, served_regret):
2      j = self._route(r, explore=True)
3      ex = self.experts[j]
4      rg = regret(ex.predict(X, r), y, ex.k, ex.w_max)
5      b = self.base.get(r, rg)
6      self.base[r] = 0.95 * b + 0.05 * rg

```

```

7         self.G[r, j] += self.gate_lr * (b - rg)           # reward-driven
           gate
8     ex.learn(X, y, r, e)

```

Listing 7: RouterArm.observe: the reward-driven gate (rispy.py).

Each day, the gate ε -greedily routes the active regime to expert j (10% exploration), scores that expert’s regret, and updates a running per-regime baseline b_r by EWMA. The annotated line is where reward-coupling *enters* the allocation: the gate logit $G[r, j]$ moves by `gate_lr * (b - rg)`—an advantage update, positive when the routed expert beat the regime’s recent baseline. Crucially, the update is indexed by the *active* regime r : a dormant regime emits no gate gradient at all (Observation 1 of Part III), so its routing column $G[r, :]$ is frozen while the experts it points to are being retrained—and possibly LRU-evicted—by the regimes that are active. The routed expert then trains on the day (`ex.learn`), which is the second coupling: training mass also follows activity.

27.2 RISPArm: where reward-coupling exits

```

1     def observe(self, X, y, r, e, served_regret):
2         if r in self.pinned:
3             self.specs[self.pinned[r]].learn(X, y, r, e)
4             return
5         if self.win_regime is not None and r != self.win_regime:
6             self._close_window()
7         self.win_regime = r
8         for i, sp in enumerate(self.specs):
9             self.win_quality[i] += regret(sp.predict(X, r), y,
10                                         self.k, self.w_max)
11         self.win_data.append((X, y, e))
12         self.win_days += 1
13         if self.win_days >= self.Wc:
14             self._close_window()

```

Listing 8: RISPArm.observe: window accumulation and the pinning short-circuit (rispy.py).

The first two lines are where reward-coupling *exits*: once a regime is pinned, `observe` routes the day’s data to the pinned owner and `returns`. No competition, no affinity update, no gate gradient—the owner map for that regime is permanently reward-independent from that day on, exactly the fixed-assignment regime of the theory. Until pinning, days accumulate into a competition window: every specialist is scored on the day (counterfactually—only the owner’s prediction was *served*), the day’s data is parked in `win_data`, and the window closes after $W_c = 20$ days or at a regime switch. Window closing does the actual allocation work:

```

1     def _close_window(self):
2         if self.win_days == 0 or self.win_regime is None:
3             return
4         r = self.win_regime
5         score = -self.win_quality / self.win_days \
6               + self.lam * (self.alpha[:, r] - 1.0 / self.R)
7         i_star = int(np.argmax(score))
8         # EG / Hedge update on the winner’s affinity row
9         a = self.alpha[i_star].copy()
10        a[r] *= np.exp(self.eta)
11        self.alpha[i_star] = a / a.sum()
12        # winner trains on the window’s data
13        for (X, y, e) in self.win_data:
14            self.specs[i_star].learn(X, y, r, e, n_steps=1)

```

```

15         # pinning
16         if self.pin_enabled and r not in self.pinned \
17             and self.alpha[i_star, r] >= self.pin_thresh:
18             self.pinned[r] = i_star
19         self.win_quality = np.zeros(self.N)
20         self.win_days = 0
21         self.win_data = []

```

Listing 9: RISPArm._close_window: EG update, winner-trains, pinning (risp.py).

The winner is the specialist with the best (lowest) mean window regret plus a niche bonus $\lambda(\alpha_{i,r} - 1/R)$ that rewards already-declared affinity—the incumbency term that prevents ownership thrash. The winner’s affinity row gets a multiplicative EG/Hedge update ($\alpha_{i*,r} *= e^\eta$, then renormalize over regimes; since rows are simplexes over regimes, boosting r implicitly taxes i^* ’s claim on every other regime—winning here means withdrawing elsewhere). *Only the winner* trains, on the whole window’s parked data, with `n_steps=1` per day to keep total gradient work comparable to the streaming arms. When the winner’s affinity crosses `pin_thresh = 0.95`, the regime is pinned and the `observe` short-circuit takes over forever.

Intuition

Read the two `observe` methods side by side. The router updates its allocation *every day a regime is active and never otherwise*—it is reward-coupled by construction, and dormancy starves it. RISP uses realized regret only as a *transient* signal to discover who should own what; the moment ownership is settled, the signal is disconnected. The allocation converges *to* reward-independence. The E5(d) ablation (`pinned` vs. `never`) isolates exactly the value of that disconnection.

28 Reproducibility Map

Every quantitative claim in the paper traces to a JSON file in `results/`, written by one driver in `run_experiments.py`.

Seeding discipline. Within a seed s , three independent RNG streams are used, and they are constructed so that *all arms see identical data*:

- *Schedule stream*: `np.random.default_rng(1000 + s)` builds the schedule once per seed (E3 uses $3000 + s$, E1s uses $2000 + s$), before any arm runs.
- *Market stream*: the market is *reconstructed per arm* with the same seed—`SyntheticMarket(cfg, seed=5000 + s)` inside the arm loop ($7000 + s$ in E3)—so every arm replays the identical sequence of (X_t, y_t) days. Comparisons across arms within a seed are therefore paired, not merely matched in distribution.
- *Arm stream*: each arm receives a fresh `np.random.default_rng(99 * s + 7)` ($13s + 5$ in E3, $55s + 3$ in E1s). Multi-specialist arms then spawn one sub-RNG per specialist via `np.random.default_rng(rng.integers(2**31))` in their constructors, so each specialist owns a private stream derived from the order of construction-time draws.

Rerunning. From the `code/` directory:

```

python run_experiments.py e1          # headline table, 20 seeds
python run_experiments.py e1 --seeds 5 # quick smoke run
python run_experiments.py e0          # real-data diagnostic (needs
    GAUSE/data)

```

Paper artifact	Driver	Results JSON
Real-data structure diagnostic (regime labels vs. block-shuffled controls, crypto + commodities)	e0()	e0_structure_diagnostic.json
Headline 2×2 dissociation table; Welch/Holm p -values; overall / post-reactivation / steady-state regret, all arms	e1()	e1_synth.json
Stitched-crypto replication of the headline table	e1s()	e1s_stitched_crypto.json
Capacity sweep ($K \in \{1, 2, 3, 4\}$, hard vs. soft memory)	e2()	e2_K{K}_{mem}.json, e2_capacity_sweep.json
Dormancy sweep ($D \in \{21, 63, 126, 252, 504\}$)	e3()	e3_dormancy_sweep.json
Episode-heterogeneity sweep ($\text{het} \in \{0, 0.5, 1, 1.5, 2\}$, ERM vs. INV at fixed allocation)	e4()	e4_het{h}.json, e4_heterogeneity_sweep.json
Ablations: β sweep (incl. variance-only $\beta=0$), W_c sweep, λ on/off, pinned vs. never-pinned	e5()	e5_ablations.json
SNR audit (break-even re-learning speed)	e6()	e6_snr_audit.json

Table 13: Map from paper artifacts to experiment drivers and output files.

```
for e in e1s e2 e3 e4 e5 e6; do python run_experiments.py $e; done
```

The only external dependencies are `numpy`, `scipy` (Welch tests), and `pandas` (real-data loaders); `e0/e1s` additionally require the Bybit and FRED CSVs under `GAUSE/data/`. Each driver prints the absolute path of every JSON it saves.

Important

The one nondeterminism trap. All randomness flows from explicit seeds, but the *construction-time draw order* is part of the specification. Each arm’s top-level RNG is freshly seeded ($99*s + 7$ for all arms in E1), so adding or removing arms from the run list is safe; what is *not* safe is changing what an arm does with that RNG at construction. Every sub-specialist seed comes from a sequential `rng.integers(2**31)` call in the constructor, so reordering the arm factories’ internals, changing N , or inserting an extra draw in any `__init__` silently shifts every downstream stream of that arm—exploration choices, minibatch sampling, tie-breaking predictions—and the rerun will no longer bit-match an archived JSON even though nothing statistically meaningful changed. When validating against archived results, leave `ARM_FACTORIES` and all arm constructors untouched; when changing them deliberately, regenerate *all* JSONs and re-archive.

Part VII

Conclusion

Summary of the Deep Dive

The mathematics of RISP reduces to one decomposition and two dichotomies. The decomposition: post-reactivation decision regret splits into an invariance gap—mean plus variance of episode risks, controlled by Cantelli’s inequality and priced for finite episodes by Hoeffding’s—and a forgetting deficit—eviction probability times relearning cost. The first dichotomy: the forgetting deficit is identically zero for reward-independent covering assignments and tends exponentially to its maximum for reward-driven ones, because a dormant regime emits no signal with which a reward-driven allocator could protect it; this is a property of *classes* of mechanisms, confirmed experimentally as a class boundary. The second dichotomy: the invariance gap is governed by the training objective alone, is provably reduced by variance penalization across episodes when episodes are heterogeneous draws of a stable decision structure, and is *inert*—measurably, predictably—when the allocation axis has already discarded the head it trained. The super-additivity of the two fixes is therefore not a curiosity but the signature of the account: each mechanism is the other’s precondition.

Around these results sit the honesty obligations this research program treats as part of the mathematics: the plug-in caveat on every generalization bound applied to a trained head; the explicit looseness of the Pinsker bridge constant; the refusal to claim convergence theorems for coupled winner-take-all dynamics that we can only demonstrate; the estimation arithmetic showing that six crisis episodes cannot certify crisis-grade invariance without environment enlargement; and the three pre-registered expectations our own experiments refuted—reported here with the same care as the confirmations, because a theory of honest retention should itself retain what it got wrong.

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